

MANAGEMENT OF COMMON NEONATAL PROBLEMS IN ASELA TEACHING AND REFERAL HOSPITAL

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List of Acronyms:

AGA Appropriate for gestational age
BP Blood pressure
BPD Broncho pulmonary dysplasia
BW Birth weight
CHD Congenital heart disease
CMV Cytomegalovirus
CPAP Continuous positive airway pressure
CVP Central venous pressure
DDH Developmental dysplasia of the hips
DIC Disseminated intravascular coagulation
EBM Expressed Breast Milk
ECG Electrocardiogram
EDD Estimated date of delivery
ELBW Extremely low birth-weight
FFP Fresh frozen plasma
FHB Fetal heart beat
HIE Hypoxic ischemic encephalopathy
HMD Hyaline membrane disease
HR Heart rate
ICP Intracranial pressure
IPPV Intermittent positive pressure ventilation
IUGR Intrauterine growth restriction
IVH Intraventricular hemorrhage
KCL Potassium chloride
LBW Low birth weight
LFT Liver function test
LGA Large for gestational age
LMP Last menstrual period
LP Lumbar puncture
MAS Meconium aspiration syndrome
NEC Necrotizing enterocolitis
NGT Naso-gastric tube
NICU Newborn intensive care unit
OR Operating room
PaCO₂ Partial pressure arterial carbon dioxide
PaO₂ Partial pressure arterial oxygen
PDA Patent ductus arteriosus
PIP peak inspiratory pressure
PPHN Persistent pulmonary hypertension of the newborn
PPROM Prelabour premature rupture of the membranes
PROM Prolonged rupture of membranes
PSV Pressure support ventilation
PVH Periventricular hemorrhage
RDS Respiratory distress syndrome

ROP Retinopathy of prematurity
RV Residual volume
SpO2 Oxygen saturation
SGA Small for gestational age
SVD Spontaneous vaginal delivery
TGA Transposition of the great arteries
TOF Tracheal esophageal fistula
TTN Transient tachypnoea of the newborn
UVC Umbilical venous catheter
VLBW Very low birth weight

Anemia in the Newborns

I. Hematologic Physiology of the Newborn.

A. Normal development: The physiologic anemia of infancy

1. In utero, the fetal aortic oxygen saturation is 45%, erythropoietin levels are high, and RBC production is rapid. The fetal liver is the major site of erythropoietin production.

2. After birth, the oxygen saturation is 95%, and erythropoietin is undetectable. RBC production by day 7 is <1/10th the level in utero. Reticulocyte counts are low, and the hemoglobin level falls; i.e.: Erythropoietin decreases after birth as a result of increased tissue oxygenation due to the onset of breathing and closure of the ductus arteriosus, and a reduced production of erythropoietin (EPO).

3. Despite dropping hemoglobin levels, the ratio of hemoglobin A to hemoglobin F increases, and the levels of 2, 3-diphosphoglycerate (2, 3-DPG) (which interacts with hemoglobin A to decrease its affinity for oxygen, thereby enhancing oxygen release to the tissues) are high. As a result, oxygen delivery to the tissues actually increases. This physiologic “anemia” is not a functional anemia in that oxygen delivery to the tissues is adequate. Iron from degraded RBCs is stored.

4. At 8 to 12 weeks, hemoglobin levels reach their nadir (11 g/dL), oxygen delivery to the tissues is impaired, renal erythropoietin production is stimulated, and RBC production increases.

5. During this period of active erythropoiesis, iron stores are rapidly utilized. Iron stores are sufficient for 15 to 20 weeks in term infants. After this time, the hemoglobin level decreases if iron is not supplied.

In preterm infants who are already born with a lower hematocrit, this decline, referred to as anemia of prematurity (AOP), occurs earlier and is more pronounced in its severity than the anemia seen in term infants.

B. Anemia of prematurity is an exaggeration of the normal physiologic anemia.

Anemia usually is defined by a Hb or Hct value that is more than two standard deviations below the mean for age.

1. In preterm infants, anemia of prematurity is common occurrence between 4- 8 weeks. However a nadir could be seen as early as 2 weeks. Hence weekly PCV screen till discharge is important for all infants < 1500g. A “physiologic” decrease in hemoglobin content is noticed at 8-12 wk in term infants (hemoglobin, 11 g/dL) and at approximately 6 wk in premature infants (7-10 g/dL).

The primary cause of anemia of prematurity (AOP) is the impaired ability to increase serum erythropoietin (EPO) appropriately in the setting of anemia and decreased tissue availability of oxygen. Circulating and bone marrow red cell progenitors respond to EPO, if present, indicating that the impaired erythropoiesis is due to lack of EPO, not a failure to respond to the hormone.

2. The hemoglobin nadir is reached earlier than in the term infant because of the following:

The physiologic effects of the transition from fetal (low Pao₂ and hemoglobin saturation) to neonatal life (high Pao₂ and hemoglobin saturation) contribute to anemia of prematurity.

a. RBC survival is decreased in comparison with the term infant. The reduced red cell life span contributes to the severity of anemia. Increased susceptibility to oxidant injury may contribute to shortened red cell survival in the neonate.

b. There is a relatively more rapid rate of growth in premature babies than in term infants. For example, a premature infant gaining 150 g per week requires approximately a 12 mL per week increase in total blood volume.

c. Many preterm infants have reduced red cell mass and iron stores because of iatrogenic phlebotomy for laboratory tests. This has been somewhat ameliorated with the use of micro techniques.

d. Vitamin E deficiency is common in small premature infants, unless the vitamin is supplied exogenously.

3. The hemoglobin nadir in premature babies is lower than in term infants because erythropoietin is produced by the term infant at a hemoglobin level of 10 to 11 g/dL but is produced by the premature infant at hemoglobin level of 7 to 9 g/dL.

4. Iron administration before the age of 10 to 14 weeks does not increase the nadir of the hemoglobin level or diminish its rate of reduction. However, this iron is stored for later use.

Although it is not involved in its pathogenesis, iron depletion may impair recovery from AOP. Because of their rapid growth rate, preterm infants have increased utilization and depletion of iron stores and, as noted above, blood loss from phlebotomy. The administration of iron does not inhibit the fall in hemoglobin concentration due to AOP. However, in term infants, it reduces the incidence of iron deficiency anemia in the first year of life.

5. Once the nadir is reached, RBC production is stimulated, and iron stores are rapidly depleted because less iron is stored in the premature infant than in the term infant.

II. Etiology of Anemia in the Neonate

A. Blood loss is manifested by a decreased or normal Hct, increased or normal reticulocyte count, and a normal bilirubin level (unless the hemorrhage is retained). If blood loss is recent (e.g., at delivery), the Hct and reticulocyte count may be normal, and the infant may be in shock. The Hct will fall later because of hemodilution.

An infant with acute blood loss may not be anemic if blood sampling is done soon enough after the acute event that hemodilution has not yet occurred. Anemia usually develops within 3 to 4 hours after blood loss; repeat testing 6 to 12 hours after the event should reveal the true extent of the loss. In acute blood loss, the infant might exhibit signs and symptoms of hypovolemia and hypoxemia (e.g., tachycardia, tachypnea, and hypotension). The RBCs should be morphologically normal.

If the bleeding is chronic, the Hct will be low, the reticulocyte count up, and the baby normovolemic.

1. Obstetric causes of blood loss, including the following malformations of placenta and cord:

- ✓ Abruptio placentae
- ✓ Placenta previa
- ✓ Incision of placenta at cesarean section
- ✓ Rupture of anomalous vessels (e.g., vasa previa, velamentous insertion of cord, or rupture of communicating placental vessels in a multi lobed placenta)
- ✓ Rupture of cord (more common in short cords and in dysmature cords) tearing or cutting of the umbilical cord during delivery

2. Occult blood loss

- ✓ **Fetomaternal bleeding** may be chronic or acute. It occurs in 8% of all pregnancies, and in 1% of pregnancies, the volume may be as large as 40 mL. The diagnosis of this problem is by Kleihauer-Betke stain of maternal smear for fetal cells. Chronic fetal-to-maternal transfusion is suggested by a reticulocyte count >10%.
- ✓ **Fetoplacental bleeding**
- ✓ **Twin-to-twin transfusion:** In twin-twin transfusion syndrome (TTTS), an artery from one twin acutely or chronically delivers blood that is drained into the vein of the other. The latter develops polyhydramnios, plethoric and large for dates, and the former has oligohydramnios, anemic and small. TTTS is more common in monozygotic twins and affects up to 30% of monochorionic twins. Maternal polyhydramnios in a twin pregnancy suggests TTTS.
- ✓ **Twin anemia polycythemia sequence (TAPS)**, an uncommon form of chronic inter-twin transfusion between monochorionic twins characterized by large inter-twin hemoglobin differences in the absence of amniotic fluid discordance.

3. Bleeding in the neonatal period may be due to the following causes:

- ✓ **Massive cephalohematoma, subgaleal hemorrhage, Intracranial bleeding.**
- ✓ **Retroperitoneal bleeding**
- ✓ **Ruptured liver or spleen**
- ✓ **Adrenal or renal hemorrhage**
- ✓ **Gastrointestinal bleeding** (maternal blood swallowed from delivery or breast should be ruled out by the Apt test)
- ✓ **Bleeding from umbilicus**

4. Iatrogenic causes. Excessive blood loss may result from blood sampling with inadequate replacement.

B. Hemolysis is manifested by a decreased Hct, increased reticulocyte count, and an increased bilirubin level.

1. Immune hemolysis

- ✓ **Rh incompatibility**
- ✓ **ABO incompatibility**
- ✓ **Minor blood group incompatibility** (e.g., c, E, Kell, Duffy)

Hemolytic Disease of the Fetus and Newborn

Hemolytic disease of the fetus and newborn (**HDFN**), also known as **erythroblastosis fetalis**; Clinically significant disease is associated primarily with incompatibility of **ABO blood groups** and the **RhD antigen**. Less frequently, hemolytic disease may be caused by differences in other antigens of the Rh system or by other RBC antigens which are minor bloodgroup incompatibilities, such as Kell, Duffy, and MNS blood groups.

When Rh-positive blood is infused into an unsensitized Rh-negative woman, antibody formation against the mismatched Rh antigen is induced in the recipient. This can occur through transfusion, but the typical scenario is when small quantities (usually >1 mL) of Rh-positive fetal blood, inherited from an Rh-positive father, enter the maternal circulation during pregnancy, through spontaneous or induced abortion, or at delivery. Once sensitization has occurred, considerably smaller doses of antigen can stimulate an increase in antibody titer. Initially, a rise in immunoglobulin (Ig) M antibody occurs, which is later replaced by IgG antibody. Unlike IgM antibodies, IgG readily crosses the placenta to cause hemolytic manifestations.

Although ABO incompatibility is the most common cause of HDFN, this form is usually much milder than Rh. Approximately 20% of live births are at theoretical risk for immune-mediated hemolysis based on ABO mismatch, most often the mother being group O and the infant either group A or B. Less often, the mother will be group A and the infant group B, or vice versa. However, clinical manifestations of hemolysis develop in only 1–10% of at-risk infants, primarily because naturally occurring maternal antibodies against ABO blood group antigens are almost exclusively IgM and therefore do not cross the placenta. Some group O mothers will produce IgG antibodies against blood group A or B antigens, and these can cross the placenta and cause immune-mediated hemolysis. A 2nd factor that accounts for the lower-than-predicted incidence of severe ABO hemolytic disease is the relatively low antigen frequency and expression on the RBC of the fetus and newborn infant. With few strong binding sites available for the maternal antibodies to bind, there is less hemolysis.

When there is both ABO and RH setup, the Rh-negative mother is partially protected against sensitization due to rapid removal of the fetal Rh-positive cells by maternal

iso-hemagglutinins (preexisting IgM anti-A or anti-B antibodies that do not cross the placenta).

Hemolytic Disease of the Fetus and Newborn (From: Nelson 21st Ed.)

	Rh	ABO	KELL
BLOOD GROUPS			
Mother	Rh-negative	O (occasionally B)	K1-negative
Infant	Rh-positive (D is most common)	A (sometimes B)	K1-positive
CLINICAL FEATURES			
Occurrence in firstborn	5%	40–50%	Rare
Severity in subsequent pregnancies:	Predictable	Difficult to predict	Somewhat predictable
Stillbirth/hydrops	Frequent (less with Rh-immunoglobulin use)	Rare	10%
Severe anemia	Frequent	Rare	Frequent
Jaundice	Prominent, severe	Mild-moderate	Mild
LABORATORY TESTS			
Direct antiglobulin test (infant)	Positive	Positive or negative	Positive or negative
Reticulocyte count	Elevated	Elevated	Variable
Red blood cell (RBC) antibodies (mother)	Usually detectable Antibody titers may help predict severity of fetal disease.	May not be detectable Antibody titers may not correlate with fetal disease.	Usually detectable Antibody titers may not correlate with fetal disease; fetus can be affected at titers lower than for Rh-mediated hemolysis.

2. Hereditary RBC disorders

- ✓ **RBC membrane defects** such as spherocytosis, elliptocytosis, or stomatocytosis
- ✓ **Metabolic defects**—glucose-6-phosphate dehydrogenase (G6PD) deficiency, pyruvate-kinase deficiency,
- ✓ **Hemoglobinopathies:** α - and γ -Thalassemia syndromes

3. Acquired hemolysis

- ✓ **Infection**—bacterial or viral
- ✓ **Disseminated intravascular coagulation**
- ✓ **Vitamin E deficiency** and other nutritional anemias
- ✓ **Microangiopathic hemolytic anemia**

C. Diminished RBC production is manifested by a decreased Hct, decreased reticulocyte count, and normal bilirubin level.

- ✓ **Physiologic anemia and anemia of prematurity**

- ✓ **Infection (rubella, CMV, parvovirus B19)**
- ✓ **Diamond-Blackfan syndrome**
- ✓ **Congenital leukemia**
- ✓ **Osteopetrosis**, leading to inadequate erythropoiesis
- ✓ **Drug-induced suppression of RBC production**

III. DIAGNOSTIC APPROACH TO ANEMIA IN THE NEWBORN

A. The **family history** should include questions about anemia, jaundice, gallstones, and splenectomy.

B. The **obstetric history** should be evaluated.

C. The **physical examination** may reveal an associated abnormality and provide clues to the origin of the anemia.

1. Acute blood loss leads to shock, with cyanosis, poor perfusion, and acidosis.

2. Chronic blood loss produces pallor, but the infant may exhibit only mild symptoms of respiratory distress or irritability.

3. Chronic hemolysis is associated with pallor, jaundice, and hepatosplenomegaly.

Characteristics of Anemia in the Newborn (from: Eichenwald. E.C., Hansen. AR., Martin. CR., Stark. AR., Cloherty and Starks Manual of Neonatal Care; 8th edition, 2016.)

Reticulocytes	Bilirubin	Coombs Test	RBC Morphology	Diagnostic Possibilities
Normal or ↓	Normal	Negative	Normal	Physiologic anemia of infancy or prematurity; congenital hypoplastic anemia; other causes of decreased production
Normal or ↓	Normal	Negative	Normal	Acute hemorrhage (fetomaternal, placental, umbilical cord, or internal hemorrhage)
Normal or ↑	↑	Positive	Hypochromic microcytes	Chronic fetomaternal hemorrhage
↑	↑		Spherocytes	Immune hemolysis (blood group incompatibility or maternal autoantibody)
Normal or ↑	↑	Negative	Spherocytes	Hereditary spherocytosis
			Elliptocytes	Hereditary elliptocytosis
			Hypochromic microcytes	Hereditary elliptocytosis
			Hypochromic microcytes	α- or γ-Thalassemia syndrome
			Spiculated RBCs	Pyruvate-kinase deficiency
			Schistocytes and RBC fragments	DIC; other microangiopathic processes
			Bite cells (Heinz bodies with Supravital stain)	Glucose-6-phosphate dehydrogenase deficiency
			Normal	Infections; enclosed hemorrhage (cephalohematoma)

D. Complete blood cell count.

E. Reticulocyte count (elevated with chronic blood loss and hemolysis, depressed with infection and production defect)

F. Blood smear

G. Coombs test and bilirubin level

H. Apt test on gastrointestinal blood of uncertain origin

I. Kleihauer-Betke preparation of the mother's blood. A 50-mL loss of fetal blood into the maternal circulation will show up as 1% fetal cells in the maternal circulation.

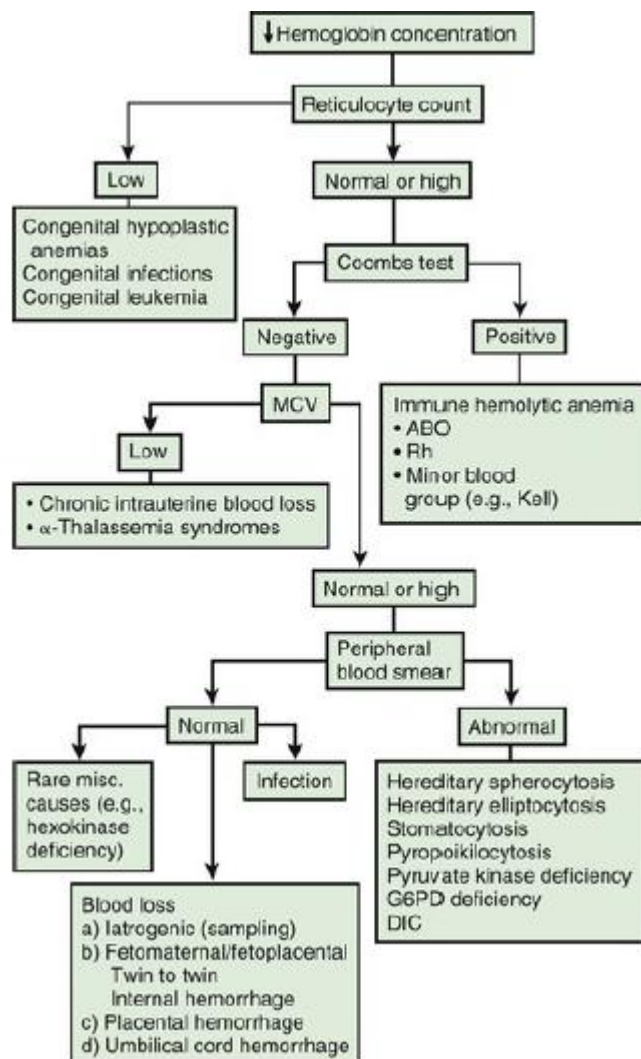
J. Ultrasound of abdomen and head

K. Parental testing—complete blood cell count, smear, and RBC indices are useful screening studies. Osmotic fragility testing and RBC enzyme levels (e.g., G6PD, pyruvate kinase) may be helpful in selected cases.

L. Studies for infection: TORCH

M. Bone marrow (rarely used except in cases of bone marrow failure from hypoplasia or tumor)

Diagnostic algorithm showing the approach to anemia in newborn infants:



DIC, Disseminated intravascular coagulation; G6PD, glucose-6-phosphate dehydrogenase; MCV, mean corpuscular volume.
(From: Blanchette VS, Zipursky A: Assessment of anemia in newborn infants, *Clin Perinatol* 11:489–510, 1984.)

IV. THERAPY

A. Transfusion.

1. Indications for transfusion. The decision to transfuse must be made in consideration of the infant's condition and physiologic needs.

a. Infants with significant respiratory disease or congenital heart disease (e.g., large left-to-right shunt) may need their Hct maintained above 40%. Transfusion with adult RBCs provides the added benefit of lowered hemoglobin oxygen affinity, which augments oxygen delivery to tissues. Blood should be fresh (3 to 7 days old) to ensure adequate 2,3-DPG levels.

Suggested Transfusion Thresholds of Neonates including Infants with Anemia of Prematurity

Postnatal Age	Respiratory Support	No Respiratory Support
Week 1	11.5 (35)	10.0 (30)
Week 2	10.0 (30)	8.5 (25)
Week 3 and older	8.5 (25)	7.5 (23)

(From: 1. Nelson 21st Ed.

2. Whyte R, Kirpalani H. Low versus high haemoglobin concentration threshold for blood transfusion for preventing morbidity and mortality in very low birth weight infants. Cochrane Database Syst Rev 2011;(11):CD000512.)

Infants with significant respiratory disease or congenital heart disease (e.g., large left-to-right shunt) may need their Hct maintained above **40%**.

c. Infants with ABO incompatibility who do not have an exchange transfusion may have protracted hemolysis and may require a transfusion several weeks after birth. This may be ameliorated with the use of intravenous immunoglobulin (IVIG).

d. Premature babies may be quite comfortable with hemoglobin levels of 6.5 to 7.0 mg/dL. The level itself is not an indication for transfusion. Growing premature infants may manifest a need for transfusion by exhibiting poor weight gain, apnea, tachypnea, or poor feeding. Sick infants (e.g., with sepsis, pneumonia, or bronchopulmonary dysplasia) may require increased oxygen-carrying capacities and therefore need transfusion.

Transfusion Guidelines for Premature Infants (From Bishara N, Ohls RK. Current controversies in the management of the anemia of prematurity. *Semin Perinatol* 2009;33(1):29-34.)

1. Asymptomatic infants with Hct $\leq 18\%$ (hemoglobin ≤ 6 g/dL) and reticulocytes $< 100,000$ cells/ μ L ($< 2\%$)

2. Infants with Hct $\leq 20\%$ (hemoglobin ≤ 7 g/dL) on supplemental oxygen who are not requiring mechanical ventilation but have one or more of the following:

a. ≥ 24 hours of tachycardia (heart rate > 180 bpm) or tachypnea (respiratory rate > 80 breaths per minute)

b. A doubling oxygen requirement from the previous 48 hours

c. Acute metabolic acidosis (pH < 7.20) or lactate ≥ 2.5 mEq/L

d. Weight gain of < 10 g/kg/day for 4 days while receiving ≥ 120 kcal/kg/day

e. If the infant will undergo major surgery within 72 hours

3. Infants with Hct $\leq 25\%$ (hemoglobin ≤ 8 g/dL) requiring minimal mechanical ventilation, defined as MAP ≤ 8 cm H₂O by CPAP or conventional ventilation, or MAP < 14 on high-frequency ventilation, and/or FiO₂ ≤ 0.40
4. Infants with Hct $\leq 30\%$ (hemoglobin ≤ 10 g/dL) requiring moderate or significant mechanical ventilation, defined as MAP > 8 cm H₂O on conventional ventilation, or MAP > 14 on high-frequency ventilation, and/or FiO₂ > 0.40
5. A transfusion should be considered if acute blood loss of $\geq 10\%$ associated with symptoms of decreased oxygen delivery occurs, or if significant hemorrhage of $\geq 20\%$ total blood volume occurs.

Hct, hematocrit; bpm, beats per minute; MAP, mean airway pressure; CPAP, continuous positive airway pressure.

Transfusion guidelines for patients aged < 4 months:

Hct $< 20\%$ with low reticulocyte count and symptoms of anemia ^a
Hct $< 30\%$ with an infant:
On $< 35\%$ oxygen by hood
On continuous positive airway pressure and/or intermittent mandatory ventilation with mechanical ventilation with mean airway pressure < 6 cm H ₂ O
With significant apnea or bradycardia ^b
With significant tachycardia or tachypnea ^c
With poor weight gain ^d
Hct $< 35\%$ with an infant:
On $> 35\%$ oxygen by hood
On continuous positive airway pressure/intermittent mandatory ventilation with mean airway pressure ≥ 6 –8 cm H ₂ O
Hct $< 45\%$ with an infant:
On ECMO
With congenital cyanotic heart disease

a Tachycardia, tachypnea, poor feeding

b More than six episodes in 12 h or two episodes in 24 h requiring bag and mask ventilation while on methylxanthines

c Heart rate > 180 beats/min for 24 h, respiratory rate > 80 breaths per min for 24 h

d Gain of < 10 g/day observed over 4 days while receiving ≥ 100 kcal/kg/day

(From: Sesok-Pizzini DA., Neonatal Transfusion Practices, Springer International Publishing Switzerland 2017.)

Complications of transfusion:

- a. Acute Hemolytic reaction:
 - a. Due to incompatibility of donor RBC with patient plasma: Rare in neonates
 - b. Tachycardia, hypotension, fever, hematuria
 - c. Management: - Stop transfusion
 - d. Fluids and furosemide as necessary
 - e. Vasopressors for hypotension
- b. Volume overload – Monitor vitals, stop transfusion. Fluid restriction, Furosemide, Partial exchange.
- c. Hypocalcemia, Hypothermia
- d. Transfusion-associated acute lung injury (TRALI).
- e. Hyperkalemia – In exchange transfusion use fresh RBC < 5 days
- f. NEC – anemia itself rather than transfusion is a reported cause of NEC

2. Blood products, methods of transfusion and Volume of transfusion:

a. Packed RBCs.

The average newborn blood volume is 80 mL/kg; the Hct of packed RBCs is 60% to 80% and should be checked before transfusion. We generally transfuse 15 to 20 mL/kg; larger volumes may need to be divided in more than one aliquots.

Packed RBC transfusion (10-20 mL/kg) is given at a rate of 2-3 mL/kg/hr. Transfusions of 10 mL/kg increase hb by 2g/dl. (Assuming PC hct to be 60%).

Hemorrhage should be treated with whole blood if available; alternatively, fluid resuscitation is initiated, followed by packed RBC transfusion.

b. Isovolemic transfusion with high Hct-packed RBCs may be required for severely anemic infants; when routine transfusion of the volume of packed RBCs necessary to correct the anemia would result in circulatory overload.

In case of large volume transfusions for a severely anemic or hydropic infant **partial exchange must be done:**

$$\frac{\text{Weight in kilogram} \times \text{blood volume per kilogram} \times (\text{Hct desired} - \text{Hct observed})}{\text{Hct of blood to be given}} = \text{volume of transfusion}$$

c. Irradiated RBCs are recommended in premature infants weighing <1,200 g. Premature infants may be unable to reject foreign lymphocytes in transfused blood. **We use irradiated blood for all neonatal transfusions.**

All neonates must be given leucoreduced blood

d. Avoid transfusion from relatives (Directed donor transfusion)

e. Because of concern for multiple exposure risk associated with repeated transfusions in ELBW infants, **we recommend transfusing stored RBCs from a single unit reserved for an infant.**

B. Prophylaxis

1. Term infants.

breastfed infants should be started on iron supplementation at the age of 4 months. Nonbreastfed infants should be sent home from the hospital on iron-fortified formula (2 mg/kg/day).

2. Premature infants (preventing or ameliorating the anemia of prematurity).

a. Iron supplementation in the preterm infant introduced between 4 and 6 weeks of age at the onset of reticulocytosis, leads to higher hemoglobin levels, improves iron stores, and lowers the risk of iron deficiency anemia after the first 6 months of life. Early (up to 3 weeks of age) versus late (4 weeks to 60 days) commencement of iron supplementation did not result in differences in cognitive outcome, but an increased rate of abnormal neurologic exam at 5 years of age was noted in the late iron group.

It remains unclear whether iron supplementation in preterm and low birth weight infants has long-term benefits in terms of neurodevelopmental outcome and growth.

We routinely supplement iron in premature infants at a dose of 2 to 4 mg of elemental iron/kg/day once full enteral feeding is achieved.

b. Vitamin E (5 to 25 IU of water-soluble form) is given daily until the baby is 38 to 40 weeks' postconceptional age (this is usually stopped at discharge from the hospital).

3. Delayed cord clamping (30-180 sec or after cessation of cord pulsation) may be beneficial in otherwise well newborns in preventing anemia in full-term infants, with effects extending beyond the neonatal period. The benefits of delayed cord clamping persist for 2-6 mo as improved hematocrit, iron status as measured by ferritin concentration and stored iron, and a clinically important reduction in the risk of anemia in infancy. Late clamping may result in delivery of an extra 20-40 mL of blood and 30-35 mg of iron to the newborn. Polycythemia is a risk with delayed clamping but is often asymptomatic.

f. Recombinant human erythropoietin (rh-EPO): Administration of recombinant human erythropoietin (EPO) appears to have limited efficacy in decreasing the number of blood donors to which the infant is exposed. It is a costly intervention, and the available evidence suggests limited clinical benefit. As a result, we recommend **not** to routinely administer EPO to preterm infants.

A more effective approach to reduce the number of donor exposures for the preterm infant than routine administration of EPO is combination of limiting the amount of blood drawn, a restrictive policy for red cell transfusions, and the use of satellite packs, which allow for repeated transfusions from the same donor to the individual infant.

Apnea in the Newborn

Definition: Apnea is defined as the cessation of airflow or cessation of breathing for longer than 20 sec or of any duration associated with cyanosis/bradycardia <100/min /poor tone.

Incidence of apnea increases with decreasing gestational age. Essentially, all infants <28 weeks' gestational age have apnea. As many as 25% of all premature infants who weigh <1,800 g (~34 weeks' gestational age) have at least one apneic episode.

Classification of apnea is based on whether absent airflow is accompanied by continued inspiratory efforts and upper airway obstruction. Most spells in preterm infants are mixed.

1. Central apnea occurs when inspiratory efforts are absent. it is caused by decreased central nervous system (CNS) stimuli to respiratory muscles, both airflow and chest wall motion are absent (no airflow, no respiratory efforts).
2. Obstructive apnea occurs when inspiratory efforts persist in the presence of airway obstruction, usually at the pharyngeal level. It is characterized by absence of airflow but persistent chest wall motion (no airflow, despite respiratory efforts).
3. Mixed apnea occurs (50-75% of cases), when airway obstruction with inspiratory efforts precedes or follows central apnea.

Short episodes of apnea are usually central, where as prolonged ones are often mixed. Apnea depends on the sleep state; its frequency increases during active (rapid eye movement) sleep.

Onset: Apneic spells generally begin at 1 or 2 days after birth; if they do not occur during the first 7 days, they are unlikely to occur later.

Duration: Apneic spells persist for variable periods postnatally and usually cease by 36 to 37 weeks' postmenstrual age (PMA) in infants born at 28 weeks' gestation or more.

The onset of apnea in a previously well preterm neonate after the 2nd wk of life or in a term infant at any time is a critical event that warrants prompt investigation. Recurrent apnea of prematurity may be treated with caffeine or theophylline.

Idiopathic apnea of prematurity occurs in the absence of identifiable predisposing diseases.

The incidence of idiopathic apnea of prematurity varies inversely with gestational age. The onset of idiopathic apnea can be during the 1st 1-2 wk after birth but is often delayed if there is RDS or other causes of respiratory distress.

Although apnea of prematurity is the most common cause of apnea in preterm infants, it is a diagnosis of exclusion.

Potential Causes of Neonatal Apnea (From: Nelson Textbook of Pediatrics.)

Idiopathic	Apnea of prematurity
Central nervous system	Intraventricular hemorrhage, seizures, hypoxic injury, herniation, drugs – depressant, maternal narcotics and magnesium sulphate,
Respiratory	Pneumonia, obstructive airway lesions, upper airway collapse, atelectasis, extreme prematurity, laryngeal reflex, phrenic nerve paralysis, pneumothorax, hypoxia, RDS, nasal occlusion caused by phototherapy eye patches, tracheal occlusion caused by neck flexion
Infectious	Sepsis, meningitis, pertussis
Gastrointestinal	Necrotizing enterocolitis, GERD,
Metabolic	↓ Glucose, ↓ calcium, ↓/↑ sodium, ↑ ammonia, ↑ organic acids, ↑ ambient temperature, hypothermia
Cardiovascular	Hypotension, hypertension, heart failure, anemia, hypovolemia, PDA
Haematological	Anemia, polycythemia
Other	Immaturity of respiratory center, sleep state

Monitoring and evaluation

All babies less than 34 weeks gestation should be monitored for at least in the first week of life or till absence of apneic episodes for at least 7 days. Babies >34 weeks gestation should be monitored if they are sick.

Though various apnea monitors (sensor mattress, pressure sensitive capsule, thoracic impedance based monitors, respiratory inductive plethysmography, magnetometers) are available, **Pulse oximeter** is an effective way of monitoring for apnea. Though it does not detect chest wall movement; it detects the most important clinical consequences of apnea i.e. hypoxia and bradycardia.

Evaluation of an Infant with Apnea (from: Eichenwald. E.C., Hansen. AR., Martin. CR., Stark. AR., Cloherty and Starks Manual of Neonatal Care; 8th edition, 2016.)

Potential Cause	Associated History or Signs	Evaluation
Infection	Feeding intolerance, lethargy, temperature instability	Complete blood count, cultures, if appropriate
Impaired oxygenation	Desaturation, tachypnea, respiratory distress	Continuous oxygen saturation monitoring, arterial blood gas measurement, chest x-ray examination
Metabolic disorders	Jitteriness, poor feeding, lethargy, CNS depression, irritability	Glucose, calcium, electrolytes
Drugs	CNS depression, hypotonia, maternal history	Magnesium; screen for toxic substances in urine
Temperature instability	Lethargy	Monitor temperature of patient and environment.
Intracranial pathology	Abnormal neurologic examination, seizures	Cranial ultrasonographic examination

Differential diagnosis:

1. Periodic breathing: About 30-45% of preterm babies exhibit a periodic breathing pattern characterized by 3 or more respiratory pauses of 3-10 seconds. It is not associated with bradycardia or desaturations. Periodic breathing is a normal event, reflective of immaturity of respiratory control system in these infants and does not merit any treatment.

2. Subtle seizures: Apnea is an uncommon presentation of a neonatal seizure in preterm infants. Sudden alteration in muscle tone, twitching movements, vacant stare and up rolling of eyes suggests a seizure. Tachycardia preceding/ accompanying an apneic attack usually suggests seizure activity.

TREATMENT

A. General measures

I. General Measures

1. Avoid reflexes that trigger apnea. Careful suctioning
2. Withheld oral feeding
3. **Positions of extreme flexion** or extension of the neck should be avoided to reduce the likelihood of airway obstruction. Prone positioning stabilizes the chest wall and may reduce apnea.
4. Warmth and temperature regulation.
5. Oxygen as needed

II. Specific Measures:

1. **Correction of metabolic disturbances** like hypoglycaemia/hypocalcemia/acidosis
2. **Correction of Anemia:** Packed cell transfusion if PCV <31 % and recurrent apneas
3. **Antibiotics** for suspected infection
4. **Methyl xanthines** is indicated in preterm infants < 34 weeks if recurrent apnea (> 2 apneas/ h or > 3 in 2 consecutive hours).

Methylxanthines are competitive inhibitors of adenosine receptors. Because adenosine is an inhibitory neuromodulator of respiratory drive, blockage of its receptors by blocking both its inhibitory A1 receptor and its excitatory A2A receptors by methylxanthines results in increased ventilatory responsiveness to carbon dioxide or increased carbon dioxide sensitivity, reversal of central hypoxic depression of breathing, decreased periodic breathing, enhanced force of diaphragmatic contraction, and improved pharyngeal muscle tone.

Caffeine:

Caffeine: a methylxanthine, reduces apneic spells and need for mechanical ventilation. Survival without neurodevelopmental disability is improved and reduces rate of BPD/CLD. It is prophylactically started in ELBW and extreme preterm neonates and also given periextubation in VLBW infants.

- ✓ We use a loading dose of 20 mg/kg of caffeine citrate (10 mg/kg caffeine base) orally or intravenously >30 minutes, followed by maintenance doses of 5-8mg/kg

of caffeine citrate (2.5-5 mg/kg of caffeine base) in one daily dose beginning 24 hours after the loading dose.

- ✓ If apnea continues at the lower range of maintenance doses, we give an additional loading dose of 10 mg/kg caffeine citrate and increase the maintenance dose by 20%.
- ✓ Caffeine serum levels of 5 to 20 µg/mL are considered therapeutic.
- ✓ Caffeine Prophylaxis: < 28 weeks or <1000g.
- ✓ Caffeine is generally discontinued at 33 to 34 weeks' PMA if no apneic spells have occurred for 7 days, 48 hours after extubation, if started empirically in ELBW / extreme preterm, it is generally discontinued at 34-36 wk of corrected age if no apneic spells have occurred in last 7 days. In some ELBW infant's caffeine may have to continued even at discharge.

The effect of caffeine likely remains for approximately 1 week after it has been discontinued. We continue monitoring until no apnea has been detected for at least 5 - 7 days after that period. Caffeine appears to be less toxic than theophylline and is well tolerated.

Aminophylline can also be used with loading dose of 5 - 8 mg/kg, followed by 2 mg/kg /dose TID. However, the therapeutic index of aminophylline is narrower.

Caffeine versus theophylline: Caffeine has therapeutic advantages over theophylline including its longer half-life, ranging from 65 to 100 hours. As a result, caffeine can be administered once daily instead of the more frequent dosing required for theophylline. In addition, enteral absorption of caffeine is more reliable, and the wide therapeutic index of caffeine minimizes side effects. Finally, monitoring serum levels is generally required in patients treated with theophylline because of the smaller margin of safety and greater variability of absorption, whereas caffeine levels are usually only measured if there are signs of toxicity.

Adverse effects of methylxanthine treatment include feeding intolerance and tachycardia. Theophylline can cause gastroesophageal reflux, perhaps because of delayed gastric emptying.

5. Ventilation: If the apneas are very frequent mandating immediate respiratory support: despite methylxanthines:

Nasal continuous positive airway pressure (CPAP) at moderate levels (4 to 6 cm H₂O) can reduce the number of mixed and obstructive apneic spells.

Nasal intermittent positive pressure ventilation (NIPPV) may reduce extubation failure due to apnea following mechanical ventilation.

NIPPV PIP – 12- 15 cm H₂O, PEEP - 5 , Rate 20-30 / min, FiO₂ 0.21 for Normal lungs

SIMV with minimal settings if all other interventions unsuccessful PIP –

12- 13 cm H₂O, PEEP - 5 , Rate 20-30 / min, FiO₂ 0.21

Bleeding in the Newborn

I. Etiology

A. Deficient clotting factors

1. Transitory deficiencies of the procoagulant vitamin K-dependent factors II, VII, IX, and X leading to HDN.

- ✓ **lack of administration of vitamin K**
- ✓ Liver disease may interfere with the production of clotting factors by the liver.
- ✓ **Transplacental exposure to certain drugs can cause bleeding in the first 24 hours of life:** Phenytoin (Dilantin), phenobarbital, and salicylates interfere with the effect of vitamin K on clotting factor synthesis. And Warfarin and related compounds given to the mother interfere with the synthesis of vitamin K-dependent clotting factors.

2. Disturbances of clotting: Disseminated intravascular coagulation (DIC) may be due to infection, shock, anoxia, necrotizing enterocolitis (NEC), renal vein thrombosis (RVT), or the use of vascular catheters.

3. Inherited abnormalities of clotting factors

a. X-linked; Hemophilias:

- ✓ **Factor VIII** levels are decreased in the newborn with hemophilia A (1 in 5,000 males).
- ✓ **Hemophilia B, or Christmas disease**, is due to a deficiency of factor IX (1 in 25,000 males).

b. Autosomal dominant (expressed in boys and girls with one parent affected)

1. Von Willebrand disease (VWD)
2. Dysfibrinogenemia (very rare) is due to fibrinogen structural mutations.

B. Platelet problems

1. Qualitative disorders include hereditary conditions (e.g., storage pool defects, Glanzmann thrombasthenia, Bernard-Soulier syndrome, platelet-type VWD) and transient disorders that result from maternal use of antiplatelet agents.

2. Quantitative disorders include the following:

- ✓ Immune thrombocytopenia (maternal idiopathic thrombocytopenic purpura [ITP] or neonatal alloimmune thrombocytopenia [NAIT])
- ✓ Maternal preeclampsia or Hemolysis, elevated liver enzymes, and low platelets (HELLP) syndrome or severe uteroplacental vascular insufficiency
- ✓ DIC
- ✓ Inherited marrow failure syndromes, including Fanconi anemia and congenital amegakaryocytic thrombocytopenia
- ✓ Congenital leukemia
- ✓ Inherited thrombocytopenia syndromes, including gray-platelet syndrome and the macrothrombocytopenias
- ✓ Consumption of platelets, i.e., catheter-related thrombosis, RVT, NEC, or vascular anomalies, such as Kasabach-Merritt phenomenon (KMP) from kaposiform hemangioendothelioma or tufted angioma.

C. Other potential causes of bleeding

Trauma:

- ✓ Rupture of spleen or liver associated with breech delivery
- ✓ Retroperitoneal or intraperitoneal bleeding may present as scrotal ecchymosis.
- ✓ Subdural hematoma, cephalohematoma, or subgaleal hemorrhage (the latter may be associated with vacuum extraction).

II. Diagnostic workup of the bleeding infant

A. History:

Family history of excessive bleeding or clotting

Maternal:

- ✓ Maternal medications (e.g., aspirin, anticonvulsants, anticoagulants)
- ✓ Pregnancy and birth history
- ✓ Maternal history of a prior infant with a bleeding disorder

Baby:

- ✓ Mode of delivery
- ✓ Vitamin K administration
- ✓ Illness, medication, anomalies, or procedures performed on the infant

B. In the examination Determine whether the baby is sick or well:

- ✓ **Sick infant.** Consider DIC, viral or bacterial infection, or liver disease. Hypoxic/ischemic injury may lead to DIC.
- ✓ **Well infant.** Consider vitamin K deficiency, isolated clotting factor deficiencies, or immune thrombocytopenia. Maternal blood in the infant's GI tract will not cause symptoms in the infant.
- ✓ Petechiae, small superficial ecchymosis, or mucosal bleeding suggests a platelet problem or von Willebrand disease.
- ✓ Large bruises suggest deficiency of clotting factors, DIC, liver disease, or vitamin K deficiency.
- ✓ Enlarged spleen suggests possible congenital infection or erythroblastosis.
- ✓ Jaundice suggests infection, liver disease, or resorption of a large hematoma.
- ✓ Abnormal retinal findings suggest infection.

Differential Diagnosis of Bleeding in the Neonate (from: Eichenwald. E.C., Hansen. AR., Martin. CR., Stark. AR., Cloherty and Starks Manual of Neonatal Care; 8th edition, 2016.)

Clinical Evaluation	Laboratory Studies			Likely Diagnosis
	Platelets	PT	PTT	
"Sick"	D-	I+	I+	DIC
	D-	N	N	Platelet consumption (infection, necrotizing enterocolitis, renal vein thrombosis, KMP)
	N	I+	I+	Liver disease
	N	N	N	Compromised vascular integrity associated with hypoxia, prematurity, acidosis, hyperosmolality
"Healthy"	D-	N	N	Immune thrombocytopenia, occult infection, thrombosis, bone marrow hypoplasia (rare), or bone marrow infiltrative disease
	N	I+	I+	Hemorrhagic disease of newborn (vitamin K deficiency)
	N	N	I+	Hereditary clotting factor deficiencies
	N	N	N	Bleeding due to local factors (trauma, anatomic abnormalities), qualitative platelet abnormalities (rare), factor XIII deficiency (rare), von Willebrand disease

PT, prothrombin time; PTT, partial thromboplastin time; D-, decreased; I+, increased; DIC, disseminated intravascular coagulation; N, normal; KMP, Kasabach-Merritt phenomenon.

C. Laboratory tests

Venipuncture blood draw is the method of choice if cord blood samples are not obtained.

Normal Values for Laboratory Screening Tests in the Neonate (From: Eichenwald. E.C., Hansen. AR., Martin. CR., Stark. AR., Cloherty and Starks Manual of Neonatal Care; 8th edition, 2016.)

Laboratory Test	Premature Infant Having Received Vitamin K	Term Infant Having Received Vitamin K	Child 1-2 Months of Age
Platelet count/ μ L	150,000-400,000	150,000-400,000	150,000-400,000
PT (seconds)*	14-22	13-20	12-14
PTT (seconds)*	35-55	30-45	25-35
Fibrinogen (mg/dL)	150-300	150-300	150-300

1. The Apt test is used to rule out maternal blood. If the infant is well and only "GI bleeding" is noted, an Apt test is performed on gastric aspirate or stool to rule out the presence of maternal blood swallowed during labor or delivery or from a bleeding breast.

- ✓ **Procedure.** Mix one-part bloody stool or vomitus with five parts water; centrifuge it and separate the clear pink supernatant (hemolysate); add 1 mL of sodium hydroxide 1% (0.25 M) to 4 mL of hemolysate.
- ✓ **Result.** Hemoglobin A (HbA) changes from pink to yellow brown (maternal blood); hemoglobin F (HbF) stays pink (fetal blood).

2. Peripheral blood smear is used to assess the number, size, and granulation of platelets and the presence of fragmented red blood cells (RBCs) as seen in DIC. Large platelets reflect either a congenital macrothrombocytopenia or young platelets, suggesting an immunemediated or destructive thrombocytopenia.

3. Platelet count

Significant bleeding from thrombocytopenia **is a greater risk with platelet counts $\leq 20,000$ to $30,000/\text{mm}^3$** ; however, bleeding with platelet counts up to $50,000$ platelets/ mm^3 may be seen with neonatal alloimmune thrombocytopenia.

4. PT is a test of the “extrinsic” clotting system, integrating activation of factor X by factor VII and tissue factor.

Factor Xa, with factor Va as a cofactor, activates prothrombin (factor II) to form thrombin. Thrombin cleaves fibrinogen to fibrin.

5. PTT is a test of the “intrinsic” clotting system and of the activation of factor X by factors XII, XI, IX, and VIII as well as the downstream factors of the common coagulation pathway (factor V, prothrombin, and fibrinogen).

6. Fibrinogen can be measured on the same sample used for PT and PTT. It may be decreased in liver disease and consumptive states. The usual functional assay is low in dysfibrinogenemia.

7. D-Dimer assays measure degradation products of fibrin found in the plasma.

Levels are increased in patients with liver disease who have problems clearing fibrin split products, thromboembolism, and DIC.

8. Specific factor assays and von Willebrand panel for patients with positive family history **can be measured in cord blood or by venipuncture after birth**. Age-specific norms must be used.

Other tests:

- ✓ Blood grouping, cross matching, CBC with reticulocyte count.
- ✓ Septic profile
- ✓ Liver Function Tests
- ✓ Ultrasound of the brain for IVH

IV. Treatment of bleeding:

In general:

- ✓ If PT APTT altered
- Inj Vit K 1 mg stat X 3 d
- FFP 10 ml/kg if bleeding persists for 4-6 hours after Vit K
- ✓ Platelet Transfusion 10-20 ml/kg if
- Platelet count $< 50,000$ and bleeding
- Platelet count $< 30,000$

- ✓ Supportive Treatment
- ✓ Treatment of underlying cause
- ✓ Packed cell transfusion if blood loss is significant

A. Replacement therapies

1. Vitamin K1 (Aquamephyton). Vitamine K; 1 – 5mg im/iv. correction of bleeding using vitamin K can take 12 to 48 hours.

2. FFP and cryoprecipitate. FFP (10 mL/kg) is given intravenously for active bleeding and is repeated every 8 to 12 hours as needed. FFP replaces all the clotting factors; however, 10 mL/kg of FFP will transiently raise the factor levels approximately to 20% of adult control, so specific factor deficiencies should be treated with factor concentrate when available. Cryoprecipitate contains only factor VIII, VWF, fibrinogen, and factor XIII. Cryoprecipitate is the most practical source of fibrinogen or factor XIII for neonates until a specific diagnosis is made.

3. Platelets. In the absence of platelet destruction (such as DIC, immune destruction, or sepsis), 1 unit of random donor platelets should raise the platelet count by 50,000 to 100,000/mm³ in a neonate.

The platelet count will drop over 3 to 5 days unless platelet production increases.

Guidelines for platelet transfusion in neonates: (From: St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)

Platelet count	Action
< 30,000/ cu mm	Transfuse all
30000-49000/cumm	Transfuse if: ▪ BW <1,500 g and ≤7 days old ▪ Clinically unstable ▪ Recent diagnosis of NEC ▪ Concurrent coagulopathy ▪ Previous major hemorrhage (i.e., grade 3 or 4 IVH) ▪ Prior to surgical procedure ▪ Postoperative period (72 hours)
50,000 to 100,000/mm ³	Transfuse if: ▪ Active bleeding ▪ Neonatal alloimmune thrombocytopenia (NAIT) with intracranial bleed ▪ Before or after neurosurgical procedures

NB: 5- 10 ml/kg increases platelet count by 50,000 to 100,000/mm³.

4. Fresh whole blood. Initial transfusion may be 10 mL/kg but should be tailored to the clinical situation. Reconstituted components (FFP, packed red blood cell [PRBC], cryoprecipitate, and platelets) are more flexible and readily dosed than fresh whole blood.

5. Clotting factor concentrates. Factor concentrates are available for factors VIII, IX, VII, and XIII.

If severe VWD is considered, a VWF-containing, plasma-derived factor VIII concentrate should be used.

B. Treatment of specific disorders

1. DIC. The infant typically appears ill and may have petechiae, GI hemorrhage, oozing from venipuncture sites, signs of infection, asphyxia, or hypoxia. The platelet count is decreased; PT and PTT are increased. Fibrinogen is decreased, and D-dimers are increased. Fragmented RBCs are seen on the blood smear.

Treatment involves the following steps:

a. Identify and treat the underlying cause (e.g., sepsis, NEC, herpes). This is **always** the most important factor in treatment of DIC.

b. Confirm that vitamin K1 has been given.

c. Administer platelets and FFP as needed to keep the platelet count $\geq 50,000/\text{mL}$ and to control bleeding.

FFP contains anticoagulant proteins, which may slow down or stop ongoing consumption.

d. For persistent bleeding consider the following:

- ✓ Exchange transfusion with fresh citrated whole blood or reconstituted whole blood (PRBCs, platelets, FFP)
- ✓ Continued transfusion with platelets, PRBCs, and FFP as needed particularly if exchange is not possible.
- ✓ Administration of cryoprecipitate (1 to 2 units per 10 kg) for hypofibrinogenemia

e. For consumptive coagulopathy secondary to large vessel thrombosis without concurrent bleeding, consider treatment with unfractionated heparin (UFH) infusion. Administer platelets and FFP after heparin initiation to maintain platelet counts $\geq 50,000/\text{mL}$.

2. Hemorrhagic disease of the newborn (HDN) occurs in 1 out of every 200 to 400 neonates not given vitamin K prophylaxis.

Hemorrhagic Disease of the Newborn (From Nelson 21st Ed.)

Age	Early-onset disease	Classic disease	Late-onset disease
	0-24 hr	2-7 days	1-6 mo
Site of hemorrhage	Cephalohematoma Subgaleal Intracranial Gastrointestinal Umbilicus Intraabdominal	Gastrointestinal Ear-nose-throat-mucosal Intracranial Circumcision Cutaneous Gastrointestinal Injection sites	Intracranial Gastrointestinal Cutaneous Ear-nose-throat-mucosal Injection sites Thoracic
Etiology/risks	Maternal drugs (phenobarbital, phenytoin, warfarin, rifampin, isoniazid) that interfere with vitamin K Inherited coagulopathy	Vitamin K deficiency Breastfeeding	Cholestasis—malabsorption of vitamin K (biliary atresia, cystic fibrosis, hepatitis) Abetalipoprotein deficiency Idiopathic in Asian breastfed infants Warfarin ingestion
Prevention	Possibly, administrations of vitamin K to infant at birth or to mother (20 mg) before birth Avoid high-risk medications	Prevented by parenteral vitamin K at birth Oral vitamin K regimens require repeated dosing over time	Prevented by parenteral and high-dose oral vitamin K during periods of malabsorption or cholestasis
Treatment	there is active bleeding, 10 mL/kg of FFP and an IV dose of 1 mg of vitamin K are given.	there is active bleeding, 10 mL/kg of FFP and an IV dose of 1 mg of vitamin K are given.	
Incidence	Very rare	$\approx 2\%$ if infant not given vitamin K	Dependent on primary disease

Care of Preterm and LBW babies

Classification of Newborns:

1. Based on the gestational age, a newborn could be classified in to:

- ✓ **Preterm:** less than 37 Completed weeks
- ✓ **Term:** 37- 42 weeks
- ✓ **Post term:** more than 42 weeks

Gestational age could be estimated by one of the following methods

- ✚ On the bases of the first day of the last menstrual period
- ✚ Ultrasound estimation: gestational age estimate during the first trimester is ultrasonography can be accurate within +/- 5-7 days.
- ✚ Based on Ballard score. The new Ballard has two components, neuromuscular and physical maturity scoring and the accuracy is within a range of +/- two weeks.

2. Classifications of the newborn based on the birth weight:

- ✓ **Macrosomia:** birth weight of 4000 gram and above
- ✓ **Normal weight:** 2500 – 3999 grams
- ✓ **Low birth weight:** 1500 – 2499 grams
- ✓ **Very low birth weight:** 1000 – 1499 grams
- ✓ **Extremely low birth weight:** less than 1000 grams

3. A newborn can also be classified with respect to birth weight and gestational age as follows:

- ✓ **Appropriate for gestational age (AGA)** if the birth weight is between 10-90%
- ✓ **Large for gestational age (LGA)** if birth weight is greater than 90%
- ✓ **Small for gestational age (SGA)** if birth weight is less than 10%

Identifiable Risk Factors for Preterm Birth (From: Nelson Textbook of Pediatrics 21st Ed.)

Fetal:

- ✓ Fetal Distress, Multiple Gestation, Erythroblastosis, Nonimmune Hydrops

Placental

- ✓ Placental Dysfunction, Placenta Previa, Placental Abruption

Uterine

- ✓ Bicornuate Uterus, Incompetent Cervix (Premature Dilation)

Maternal

- ✓ Previous Preterm Birth, Preeclampsia, Black Race,
- ✓ Chronic Medical Illness (Cyanotic Heart Disease, Renal Disease, Thyroid Disease)
- ✓ Short Interpregnancy Interval

- ✓ Infection (Listeria Monocytogenes, Group B Streptococcus, Urinary Tract Infection, Bacterial Vaginosis, Chorioamnionitis)
- ✓ Obesity, Drug Abuse (Cocaine)
- ✓ Young or Advanced Maternal Age

Other

- ✓ Premature Rupture of Membranes, Polyhydramnios, Iatrogenic, Trauma
- ✓ Assisted Reproductive Technology

Neonatal Morbidities Associated with Prematurity (From: Nelson Textbook of Pediatrics 21st Ed.)

Respiratory

- ✓ Respiratory Distress Syndrome (Hyaline Membrane Disease)
- ✓ Bronchopulmonary Dysplasia*
- ✓ Pneumothorax, Pneumomediastinum; Interstitial Emphysema
- ✓ Congenital Pneumonia
- ✓ Apnea

Cardiovascular

- ✓ Patent Ductus Arteriosus, Hypotension, Bradycardia (With Apnea)

Hematologic

- ✓ Anemia (Early or Late Onset)

Gastrointestinal

- ✓ Poor Gastrointestinal Function—Poor Motility
- ✓ Necrotizing Enterocolitis*
- ✓ Hyperbilirubinemia—Direct and Indirect
- ✓ Spontaneous Gastrointestinal Isolated Perforation

Metabolic-Endocrine

- ✓ Hypocalcemia, Hypoglycemia/Hyperglycemia, Metabolic Acidosis
- ✓ Hypothermia
- ✓ Osteopenia

Central Nervous System

- ✓ Intraventricular Hemorrhage*, Periventricular Leukomalacia*
- ✓ Seizures, Hypotonia
- ✓ Retinopathy of Prematurity*, Deafness

Renal

- ✓ Hyponatremia, Hypernatremia, Hyperkalemia
- ✓ Renal Tubular Acidosis, Renal Glycosuria
- ✓ Edema

Other

- ✓ Infections* (Congenital, Perinatal, Nosocomial: Bacterial, Viral, Fungal, Protozoal)

* Major Neonatal Morbidities.

History:

- ✓ Determine gestational age.
- ✓ Antenatal steroids:
 - If GA is 24-34 weeks, determine if the mother has received adequate antenatal steroids.
 - Betamethasone 12 mg 24 hours apart 2 doses. Or Dexamethasone 6 mg 12hours apart 4 doses.
- ✓ Record all risk factors.
- ✓ Determine risk factors for sepsis – PPROM, UTI in last 1-month, intrapartum fever. (other signs of chorioamnionitis are elevated WBC > 15000/ mm³, uterine tenderness, foul smelling vaginal discharge /liquor).
- ✓ The risk of sepsis is higher in PPROM and unexplained preterm.
- ✓ Determine cause of prematurity – PPROM / Iatrogenic / unexplained preterm.
- ✓ Determine cause for LBW – Calculate the mothers prepregnant BMI

Care in delivery room and Nursery Care:

- ✓ At birth, the general measures needed to clear the airway, initiate breathing, care for the umbilical cord and eyes, and administer vitamin K are the same for premature infants as for those of normal weight and maturity.
- ✓ Additional considerations are the need for (1) thermal control and monitoring of the heart rate and respiration, (2) oxygen therapy, and (3) special attention to the details of fluid requirements and nutrition.
- ✓ Resuscitate as per guidelines.
- ✓ Increase the temperature of the delivery room to > 25 C.
- ✓ Delayed cord clamping in infants (>32 weeks) who cry immediately after birth.
- ✓ Skin to skin contact for those > 32 weeks who are stable.
- ✓ Warmth: Keep the baby under radiant warmer, Wrap in prewarmed, sterile sheets, Dry the baby thoroughly.
- ✓ Delivery room CPAP if in distress.
- ✓ Examine placenta, send umbilical cord and placenta for histopathology if required.
- ✓ Look for cause of prematurity, Perform shake test in preterm babies < 34 weeks.

Transport:

- ✓ Transfer the baby well wrapped in pre-warmed clothes if the baby is stable.
- ✓ During the transfer observe the baby for
 - ✚ Hypothermia
 - ✚ Respiratory distress
 - ✚ Apnea
 - ✚ Cyanosis

Neonatal Intensive Care Unit (NICU):

On admission:

- ✓ Receive the baby on sterile sheet.
- ✓ Weigh the baby.
- ✓ Record temperature.
- ✓ Check whether baby has received Inj. Vit. K.
- ✓ Check blood sugar level.
- ✓ Check HR, RT, SaO₂ and blood pressure. If baby has respiratory distress, provide CPAP.
- ✓ After initial stabilization, record anthropometry and calculate gestational age according to New Ballard score / Parkin score and classify as AGA, SGA, LGA as per intra uterine growth (Lubchenco's) charts. Calculate **Ponderal index** (PI) in SGA babies.

PI = Weight in gm ÷ (length in cm)³ x 100

- ✚ < 2 = Asymmetrical SGA
- ✚ > 2 = Symmetrical SGA

Potential Adverse Reactions to Drugs Administered to Premature Infants (From: Nelson Textbook of Pediatrics 21st Ed.)

DRUG	REACTION(S)
Oxygen	Retinopathy of prematurity, bronchopulmonary dysplasia
Sulfisoxazole	Kernicterus
Chloramphenicol	Gray baby syndrome—shock, bone marrow suppression
Vitamin K analogs	Jaundice
Novobiocin	Jaundice
Hexachlorophene	Encephalopathy
Benzyl alcohol	Acidosis, collapse, intraventricular bleeding
Intravenous vitamin E	Ascites, shock
Phenolic detergents	Jaundice
NaHCO ₃	Intraventricular hemorrhage
Amphotericin	Anuric renal failure, hypokalemia, hypomagnesemia
Reserpine	Nasal stuffiness
Indomethacin	Oliguria, hyponatremia, intestinal perforation
Cisapride	Prolonged QTc interval
Tetracycline	Enamel hypoplasia
Tolazoline	Hypotension, gastrointestinal bleeding
Calcium salts	Subcutaneous necrosis
Aminoglycosides	Deafness, renal toxicity
Enteric gentamicin	Resistant bacteria
Prostaglandins	Seizures, diarrhea, apnea, hyperostosis, pyloric stenosis
Phenobarbital	Altered state, drowsiness
Morphine	Hypotension, urine retention, withdrawal
Pancuronium	Edema, hypovolemia, hypotension, tachycardia, vecuronium contractions, prolonged hypotonia
Iodine antiseptics	Hypothyroidism, goiter
Fentanyl	Seizures, chest wall rigidity, withdrawal
Dexamethasone	Gastrointestinal bleeding, hypertension, infection, hyperglycemia, cardiomyopathy, reduced growth
Furosemide	Deafness, hyponatremia, hypokalemia, hypochloremia, nephrocalcinosis, biliary stones
Heparin (<i>not</i> low-dose prophylactic use)	Bleeding, intraventricular hemorrhage, thrombocytopenia
Erythromycin	Pyloric stenosis

Investigations:

- ✓ RBS, CBC
- ✓ If symptomatic and has risk for sepsis send blood culture, CRP
- ✓ Serum creatinine.
- ✓ Serum electrolytes
- ✓ Serum bilirubin if clinical jaundice
- ✓ Chest X-ray if symptomatic
- ✓ Cranial ultrasound if clinically indicated.

Management:

- ✓ Start IV fluids for babies with gestational age < 30 weeks/ birth weight < 1200 g
- ✓ Temperature, Respiratory distress, jaundice, pain management as per protocol
- ✓ Kangaroo mother care is the most important aspect of developmentally supportive care (**DSC**). Other aspects of **DSC** are: Nesting, frequent change of position, limiting light and noise exposure.
- ✓ Central lines: Insert UVC in all neonates < 1000 g and in sick LBW neonates requiring ventilatory care / Inotropes. Insert UAC in sick LBW neonates requiring frequent ABG. If UAC is not possible then a peripheral arterial line can be inserted.
- ✓ Prevention of infection.
- ✓ In hospital transfers should be done in thermoneutral range either using the transport incubator if the baby is sick or < 1500 g or with embrace bag.
- ✓ **Nutrition Management:**

Feeding of preterm and low birth weight infants (specialty G/A < 34wks)

To attain fast growth and prevent tissue loss, preterm need relatively high nutrients.

Protein and energy requirements of premature newborns:

Body weight (gm)	Protein (gm/kg/d)	Energy (kcal/kg/d)	Protein/Energy (gm/100kcal)
500 – 700	4.0	105	3.8
700 – 900	4.0	108	3.7
900 – 1200	4.0	119	3.4
1200 – 1500	3.9	125	3.1
1500 – 1800	3.6	128	2.8
1800 – 2200	3.4	131	2.6

(From: Ziegler EE., Protein requirement of preterm; Journal of Ped. Gastr. Enterology; Nut. 45, 170; 2007.)

Source of nutrition for preterm:

- ✓ Human milk (breast milk): Is the primary nutritional source for premature infant
- ✓ Premature milk is specially adapted to preterm
- ✓ Well tolerated
- ✓ Improved gastric emptying
- ✓ Reduced NEC and infections
- ✓ Possibly better neurocognitive development

Maturation of oral feeding skills across various gestational periods (From: St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)

Gestational age	Maturation of oral feeding skill	Relevance to feeding the preterm neonate
29 – 30 weeks	Organized peristalsis Sucking bursts develop No coordination between suck/swallow and breathing	<30 weeks and < 1200g – feeding preferably with-held until stabilized. Start intravenous fluids.
32 – 34 weeks	Slightly mature sucking pattern Co-ordination between sucking, swallowing, breathing	Feeding by palada
>34 weeks	More mature sucking pattern More co -ordination between sucking, swallowing, breathing	Can take breast feeds

When to start feeding in preterm newborns

- ✓ Feeding should not be delayed once baby is stabilized, usually starting from 1st day.
- ✓ The 1st feed is called priming or trophic feeding
- ✓ It is intended not to provide nutrients to support growth, rather is to facilitate maturation and keep integrity of preterm gut
- ✓ Early priming facilitates tolerance and early achievement of full feeding

How to advance daily to full feeding:

The goal is to achieve – Volume: 140 – 150 ml/kg/day & Calorie: 110 – 120 kcal/kg/day.

- Infants weighing < 1.5 kg at birth are at the highest risk of feeding problems and necrotizing enterocolitis. The smaller the infant, the higher the risk Initial volume to start is 2ml/kg/feed as trophic.

- Starting on the first day, give 10 ml/kg per day of enteral feeds, preferably expressed breast milk, with the remaining fluid requirement at 50 ml/kg per day met by IV fluids.

The frequency of giving per day varies based on birth weight and or baby's tolerance

The guide to proceed the next feed is baby's tolerance

- ✓ Usually is three or four times per day
- ✓ Advancement per day shouldn't exceed 20ml/kg/day
- ✓ After full volume feeding is achieved, frequency of feeding is every 2 - 3 hourly
- ✓ If the infant is well and active and not receiving IV fluids, give 2–4 ml of expressed breast milk every 2 h through a nasogastric tube, depending on the weight of the infant

For stable babies (From: St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)

- ✓ >34 wks / >1800 g – Breastfeeds/palladai (cup feeding)
- ✓ 32-34 wks / 1500-1800 g – palladai (cup feeding)
- ✓ 30-32 wks / 1200-1500 g –gavage.
- ✓ < 30 wks / < 1200 g – gavage (tube)
- ✓ Infants < 30weeks and < 1200g must be started on IV fluids

1. **Recommendation for initiation and advancement of preterm feeding for (24 – 32wk)** (From: Federal Ministry of Health of Ethiopia, Neonatal Intensive Care Unit (NICU) Training Participants' Manual, November 2014, Addis Ababa.)

Gestational age (weeks)	Volume of initial Feed (ml/kg)	Rate of feeding		
		frequency	Advancement	
24 – 26	2	6 – 8 hours	Non for 5 – 7 days	10 -15 ml/k/d
26 – 28	2	6 – 8 hours	Non for 3 – 5 days	10 – 20 ml/k/d
28 – 32	2	6 – 8 hours	As tolerated	Aim full feed at 7 postnatal day

Initial volumes of feeding for preterm neonates & advancement of feed volumes

(for infants who are AGA and normal dopplers) (From: St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)

Gestational age	Weight(grams)	Initial feeds	Progression of feeds
<28 weeks	<1000	12 ml/kg/day	20 cc/kg/d
28 –32weeks	1000–1500	20 ml/Kg /day	Increase by 35 ml/kg/day. Aim to reach full feeds by 7 days.
32- 34 weeks	1500 – 1800	60ml/kg/day by paladai. Allow breast feeds	Increase by 30 ml/kg/day – to attain full feeds
> 34 weeks	>1800	Breast feeds from birth (or) 60ml/kg/day by paladai	Increase by 30 ml/kg/day – to attain full feeds

NG Tube Feeding

- ✓ Use bolus over 20 – 25 minutes
- ✓ Avoid injecting the milk rapidly. Results gastric distension and intolerance
- ✓ Avoid continuous pouring of the milk. Enhances bacterial colonization

Further, follow up of feeding

- Check whether the previously administered milk is emptied before next bolus is given by sucking out gastric content.

• Indications for stopping/withholding feeds:

- ✓ Signs of feed intolerance
- ✓ Hemodynamic instability
- ✓ Suspect NEC
- ✓ Voluminous gastric residuals i, e; >50% of previous feed volume if ≥6ml/feed, if feed volume <6ml/feed - 2 episodes of >50% in a 6 hr period or >100%
- ✓ Apneic episode if suspecting GERD/seizures
- ✓ Withheld until all the above signs resolve.

• Definition of feed intolerance is presence of *any one* of the following:

- ✓ Abdominal distension ≥2cm from the previous measurement
- ✓ Vomiting ≥2 episodes in the past 6 hrs
- ✓ Gastric aspirate blood stained
- ✓ >2 episodes of voluminous gastric aspirates in a 6 hr period

Assessing for adequacy of preterm feeding:

- ✓ Check for adequate urine passage. Usually passage of more than 6 times per day is fair
- ✓ Check weight daily. There will be initial about 15% loss of birth weight. In the 1st week.
 - ✚ Accepted weight gain in preterm is 15 – 20gm/kg/day until 2kg is reached then in average weight gain is 20 – 30gm/kg/day.
 - ✚ Once the baby achieved weight of 2 – 2.5 kg, feeding can be switched to full breast milk or preterm formula.

Addition of vitamins and minerals

- ✓ Preterm is particularly at risk for Iron deficiency anaemia and vitamin D deficiency
- ✓ Start Iron drops. Elemental Iron 3mg/kg/day starting from 3rd – 4th weeks of life
- ✓ Vitamin D 150 – 400IU. Giving 2 ml/day of cod liver oil (seven seas without vitamin) can provide.

Approach to a baby who has inadequate weight gain:

- ✓ Assess the baby for any illness
- ✓ Ensure normal temperature at all times.
- ✓ Increase feeds gradually upto 180- 200 ml/kg/day, provided there are no contraindications like CLD, PDA.
- ✓ Provide hind milk, Discard the first 5 ml of milk.
- ✓ Use Human milk fortifier
- ✓ Increase the duration of KMC
- ✓ If PCV is low, consider transfusion

Discharge routine and Discharge criteria:

- ✓ Weight of 1400 g, Gestation age 34 weeks, Weight gain for 3 consecutive days.
- ✓ Baby feeding well by cup / direct feeds
- ✓ Maintains temperature.
- ✓ No evidence of infection.
- ✓ Mother is confident of KMC and care of baby
- ✓ Enroll the baby for high risk follow up clinic if birth wt < 1800 gm / GA < 34 wks.



Hyperbilirubinemia in the Neonatal

I. Introduction.

Almost all newborn infants develop a total serum or plasma bilirubin (TB) level greater than 2 mg/dL. As the TB increases, it produces neonatal jaundice [jaundice (French *jaune*, yellow) or icterus (Greek *ikteros*)]. Newborns appear jaundiced when serum bilirubin level is greater than 5-7 mg/dl.

Neonates with severe hyperbilirubinemia (defined as a TB >25 mg/dL [428 micromol/L]) are at risk for bilirubin-induced neurologic dysfunction (BIND), which occurs when bilirubin crosses the blood-brain barrier and binds to brain tissue.

Jaundice is observed during the 1st wk after birth in approximately 60% of term infants and 80% of preterm infants. The yellow color usually results from the accumulation of unconjugated, nonpolar, lipid-soluble bilirubin pigment in the skin. This unconjugated bilirubin (designated **indirect-acting** by nature of the Van den Bergh reaction) is an end product of heme-protein catabolism from a series of enzymatic reactions by **heme-oxygenase and biliverdin reductase** and nonenzymatic reducing agents in the reticuloendothelial cells. It may also be partly caused by deposition of pigment from conjugated bilirubin, the end product from indirect, unconjugated bilirubin that has undergone conjugation in the liver cell microsome by the enzyme **uridine diphosphoglucuronic acid (UDP)–glucuronyl transferase** to form the polar, water-soluble glucuronide of bilirubin (**directreacting**).

Although bilirubin may have a physiologic role as an antioxidant, elevations of indirect, unconjugated bilirubin are potentially neurotoxic. Even though the conjugated form is not neurotoxic, direct hyperbilirubinemia indicates a potentially serious hepatic disorder or a systemic illness.

II. Bilirubin Metabolism: The TB level results from the balance of bilirubin production and excretion.

A. Bilirubin production. Bilirubin is derived from the breakdown of heme containing proteins in the reticuloendothelial system.

1. Red blood cell (RBC) hemoglobin is the major heme-containing protein.

Hemoglobin released from RBCs in the reticuloendothelial system or from ineffective erythropoiesis accounts for 80% to 90% of bilirubin production. One gram of hemoglobin produces 34 mg of bilirubin. Breakdown of other heme-containing proteins such as cytochromes and catalase contribute the remaining 10% to 20% of bilirubin.

2. Bilirubin metabolism. The microsomal enzyme heme oxygenase located in the liver, spleen, and nucleated cells oxidizes the heme ring (Iron protoporphyrin; which is the oxygen-carrying component of hemoglobin) from heme-containing proteins. This rate-limiting step produces **biliverdin**, free iron (which can be reutilized in hemoglobin synthesis) and carbon monoxide (CO) (which is excreted in the lungs) in equimolar amounts. bilirubin production can be indirectly assessed by measuring CO production.

The CO formed by heme degradation becomes bound to hemoglobin as carboxyhemoglobin (COHb) and is carried in the circulation to be excreted by the lung.

B. Bilirubin clearance and excretion

1. Transport. Bilirubin is nonpolar, insoluble in water, and is transported to liver cells bound to serum **albumin**. Bilirubin bound to albumin does not usually enter the central nervous system (CNS) and is thought to be nontoxic. Displacement of bilirubin from albumin by acidosis, drugs, such as ceftriaxone, or by free fatty acids (FFAs) at high molar ratios of FFA:albumin may increase bilirubin toxicity.

2. Hepatic uptake. Nonpolar, fat-soluble bilirubin (dissociated from albumin) crosses the hepatocyte plasma membrane and is bound mainly to cytoplasmic **ligandin** (Yprotein) for transport to the smooth endoplasmic reticulum.

3. Conjugation. In hepatocytes, the enzyme **uridine diphosphoglucuronate glucuronosyltransferase (UGT1A1)** catalyzes the conjugation of bilirubin with glucuronic acid, resulting in mostly bilirubin diglucuronides and some monoglucuronides that are more water-soluble than unconjugated bilirubin. Both forms of conjugated bilirubin are excreted into the bile canaliculi against a concentration gradient.

4. Excretion. Conjugated bilirubin is secreted into the bile and then excreted into the gastrointestinal (GI) tract where it is eliminated in the stool. Conjugated bilirubin is not reabsorbed from the bowel unless it is deconjugated by the intestinal enzyme **β glucuronidase**, present in the neonatal intestinal mucosa. Resorption of bilirubin from the GI tract and delivery back to the liver for re-conjugation is called the **enterohepatic circulation**. Intestinal bacteria, present in adults but to a limited extent in newborns, can prevent enterohepatic circulation of bilirubin by reducing conjugated bilirubin to **urobilin**, which is not a substrate for β -glucuronidase.

III. Nonpathological Hyperbilirubinemia; Physiologic jaundice:

Term neonate:

- ✓ Onset of jaundice on day 2-3 of life.
- ✓ Reaches a peak of 10-12 mg/dl by 2-3 days.
- ✓ Disappears by 4-5 days.

Preterm neonate:

- ✓ Onset of jaundice on day 3-4 of life.
- ✓ Reaches a peak of 15 mg/dl by 6-8 days.
- ✓ Disappears by 7-10 days.

This nonpathological jaundice [Physiologic Jaundice (Icterus Neonatorum)] is attributed to the following mechanisms:

A. Increased bilirubin production due to the following:

- ✓ Increased RBC volume per kilogram and decreased RBC survival (90 days vs. 120 days) in infants compared to adults.
- ✓ Increased ineffective erythropoiesis and increased turnover of nonhemoglobin heme proteins.

B. Defective uptake of bilirubin from plasma caused by decreased ligandin and binding of ligandin by other anions.

C. Decreased clearance due to decreased UGT1A1 activity.

D. Decreased hepatic excretion of bilirubin. Increased enterohepatic circulation caused by:

- ✓ High levels of intestinal β -glucuronidase,

- ✓ Preponderance of bilirubin monoglucuronide rather than diglucuronide,
- ✓ Decreased intestinal bacteria,
- ✓ Decreased gut motility with poor evacuation of bilirubin-laden meconium

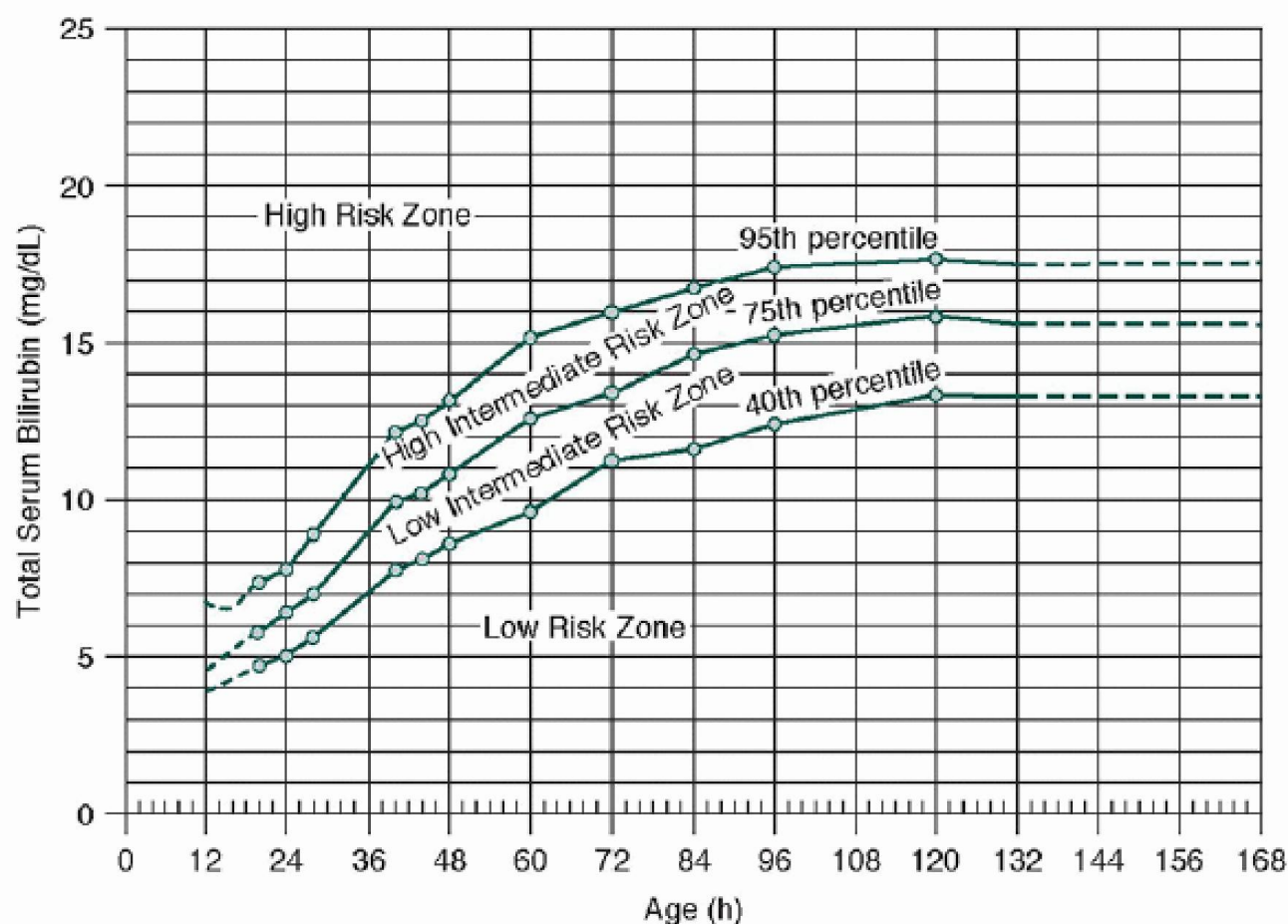
IV. Hyperbilirubinemia: is defined as a TB >95th percentile on the hour specific Bhutani nomogram.

A. The following situations suggest severe hyperbilirubinemia and require evaluation;

Pathologic jaundice:

- ✓ Onset of jaundice before 24 hours of age
- ✓ Rate of rise in total serum bilirubin (TSB) or transcutaneous bilirubin (TcB) level of >0.2 mg/dL/hour. Or >5 mg /dl/day.
- ✓ TSB > 95th percentile on age specific nomograms.
- ✓ Associated signs of illness such as vomiting, lethargy, poor feeding, excessive weight loss, apnea, tachypnea, or temperature instability; or Signs of bilirubin encephalopathy / Kernicterus
- ✓ Jaundice persisting for more than >2 weeks in term and 3 weeks in preterm.
- ✓ Direct serum bilirubin concentration >2 mg/dl or > 15% of total bilirubin.

Hour-specific bilirubin nomogram. Predictive ability of a predischarge hourspecific serum bilirubin for subsequent significant hyperbilirubinemia in healthy term and near-term newborns (From: Eichenwald. E.C., Hansen. AR., Martin. CR., Stark. AR., Cloherty and Starks Manual of Neonatal Care; 8th edition, 2016.)



(From: St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)

Designated Risk Status	Evaluation	Follow-up
High	For hemolysis and risk factors	In 6-12 hours; decide on intervention strategy
High-intermediate	For risk factors	May repeat TSB in 24 hours
Low-intermediate	For risk factors	May repeat in 48 hours
Low	Assess well-being	May not need TSB follow-up

There are two age-specific nomograms- one each for phototherapy and exchange transfusion. The nomograms have lines for three different risk categories of neonates.

- ✓ **Lower curve:** For lower risk babies (38 wk or more AND no risk factors),
- ✓ **Middle curve:** For medium risk babies (38 wk or more with risk factors, or 35 wk to 37 wk and without any risk factors)
- ✓ **Upper curve:** For higher risk (35 wk to 37 wk and WITH risk factors).

Risk factors include presence of isoimmune hemolytic anemia, G6PD deficiency, asphyxia, temperature instability, hypothermia, sepsis, significant lethargy, acidosis and hypoalbuminemia.

B. Causes of hyperbilirubinemia

1. Increased bilirubin production. Hemolytic disease is the most common cause of hyperbilirubinemia. This includes:

- ✓ Blood group incompatibility: e.g. Rh, ABO and minor blood group incompatibility
- ✓ RBC enzyme abnormalities: e.g. glucose-6-phosphatedehydrogenase or pyruvate kinase deficiencies
- ✓ RBC membrane defects: e.g. hereditary spherocytosis
- ✓ Sepsis
- ✓ Extravascular blood: e.g. sequestered blood due to bruising or cephalohematoma
- ✓ polycythemia.

2. Decreased bilirubin clearance

- ✓ Mutations in the gene that encodes UGT1A1 decrease bilirubin conjugation.
- ✓ Crigler-Najjar syndrome due to either absent UGT activity (type I) or reduced UGT activity (type II) results in severe hyperbilirubinemia.
- ✓ Gilbert syndrome results from a mutation in the promoter region of the UGT1A1 gene, reducing production of UGT, and is the most common inherited disorder of bilirubin glucuronidation.
- ✓ Decreased clearance may occur due to hormonal deficiency: in infants of diabetic mothers and with congenital hypothyroidism, hypopituitarism.

3. Increased enterohepatic circulation. Pathologic conditions leading to increased enterohepatic circulation include decreased enteral intake, including breast feeding failure; breast milk jaundice; or impaired intestinal motility due to intestinal atresias, meconium ileus, Hirschsprung disease, Pyloric stenosis.

a. Breast feeding failure jaundice. Breastfeeding failure jaundice typically occurs with lactation failure during the first postnatal week that leads to insufficient intake, with weight loss and sometimes hyponatremia. Hyperbilirubinemia is attributed mainly to the decreased intake of milk that leads to slower bilirubin elimination and increased enterohepatic circulation.

b. “Breast milk jaundice” (condition that may be due to genetic predisposition). Typically, it begins after the first 3 to 5 postnatal days, peaks within 2 weeks of age, and if breastfeeding is continued, gradually returns to normal levels over 3 to 12 weeks. If breastfeeding is stopped, the bilirubin level may fall rapidly in 48 hours. If nursing is then resumed, the bilirubin may rise by 2 to 4 mg/dL but usually will not reach the previous high level. Affected infants have good weight gain, normal liver function test (LFT) results, and no evidence of hemolysis. The mechanism of breast milk jaundice is thought to be due to a factor in human milk, possibly β -glucuronidase, that deconjugates intestinal bilirubin and promotes its absorption.

Risk Factors for Development of Severe Hyperbilirubinemia in Infants ≥ 35 Wk of Gestation (in Approximate Order of Importance) (From: American Academy of Pediatrics Subcommittee on Hyperbilirubinemia: Management of hyperbilirubinemia in the newborn infant 35 or more weeks of gestation, Pediatrics 114:297–316, 2004.)

Major risk factors

- ✓ Predischarge TSB or TcB level in the high-risk zone on hour specific nomogram
- ✓ Jaundice observed in the 1st 24 hr
- ✓ Deficiency of G6PD
- ✓ Incompatibility – Rh or ABO isoimmunization
- ✓ Gestational age Under 37 weeks
- ✓ Previous sibling received phototherapy
- ✓ Cephalohematoma or significant bruising
- ✓ Exclusive breastfeeding, particularly if nursing is not going well
- ✓ weight loss is excessive

Minor risk factors

- ✓ Predischarge TSB or TcB level in the high intermediate-risk zone
- ✓ Gestational age 37-38 wk
- ✓ Jaundice observed before discharge
- ✓ Previous sibling with jaundice
- ✓ Macrosomic infant of a diabetic mother
- ✓ Maternal age ≥ 25 yr
- ✓ Male gender

VI. Evaluation of infant with hyperbilirubinemia

A. History

Family history

- ✓ A family history of jaundice, anemia, splenectomy, or early gallbladder disease suggests hereditary hemolytic anemia (e.g., spherocytosis).
- ✓ A family history of liver disease may suggest galactosemia, $\alpha 1$ -antitrypsin deficiency, Gilbert disease, Crigler-Najjar syndrome types I and II.
- ✓ A sibling with jaundice or anemia may suggest blood group incompatibility or breast milk jaundice.

Pregnancy history

- ✓ Illness during pregnancy may suggest congenital infection.
- ✓ Infants of diabetic mothers are more likely to develop hyperbilirubinemia.
- ✓ Maternal drugs may interfere with bilirubin binding to albumin, making bilirubin toxic at relatively low levels (sulfonamides) or may trigger hemolysis in a G6PD-deficient infant (sulfonamides, nitrofurantoin, antimalarials).

Labor and delivery history

- ✓ Birth trauma may be associated with extravascular bleeding and hemolysis.
- ✓ Infants with hypoxic-ischemic insult may have elevated bilirubin levels; causes include inability of the liver to process bilirubin and intracranial hemorrhage.
- ✓ Delayed cord clamping may be associated with neonatal polycythemia.

Infant history

- ✓ Delayed or infrequent stooling may be caused by poor caloric intake or intestinal obstruction and lead to increased enterohepatic circulation of bilirubin.
- ✓ Poor caloric intake may decrease bilirubin uptake by the liver.

- ✓ Vomiting can be due to sepsis, pyloric stenosis, or galactosemia.

B. Physical examination.

Visual assessment of jaundice – in natural light without a yellow background. All newborns should be assessed every 12 hours in the first 3-5 days.

Blanching the skin with finger pressure makes it easier to observe jaundice.

Dermal pressure may reveal the anatomic progression of jaundice (face, ≈ 5 mg/dL; mid-abdomen, ≈ 15 mg/dL; soles, ≈ 20 mg/dL).

Jaundice: cephalocaudal progression so can roughly approximate serum bilirubin concentration based on degree of jaundice progression. (From: Polin RA., Lorenz JM., Pocket Clinicians Neonatology, CAMBRIDGE UNIVERSITY PRESS, 2008.)

- ✓ Head: 5.9 ± 0.3 mg/dL
- ✓ Chest: 8.9 ± 0.7 mg/dL
- ✓ Knees: 11.8 ± 0.8 mg/dL
- ✓ Ankles: >15 mg/dL

However, visual inspection is not a reliable indicator of serum TB level or the detection of rapidly rising levels, especially in infants with darkskin.

Jaundiced infants should have a bilirubin measurement and be examined for the following contributing factors:

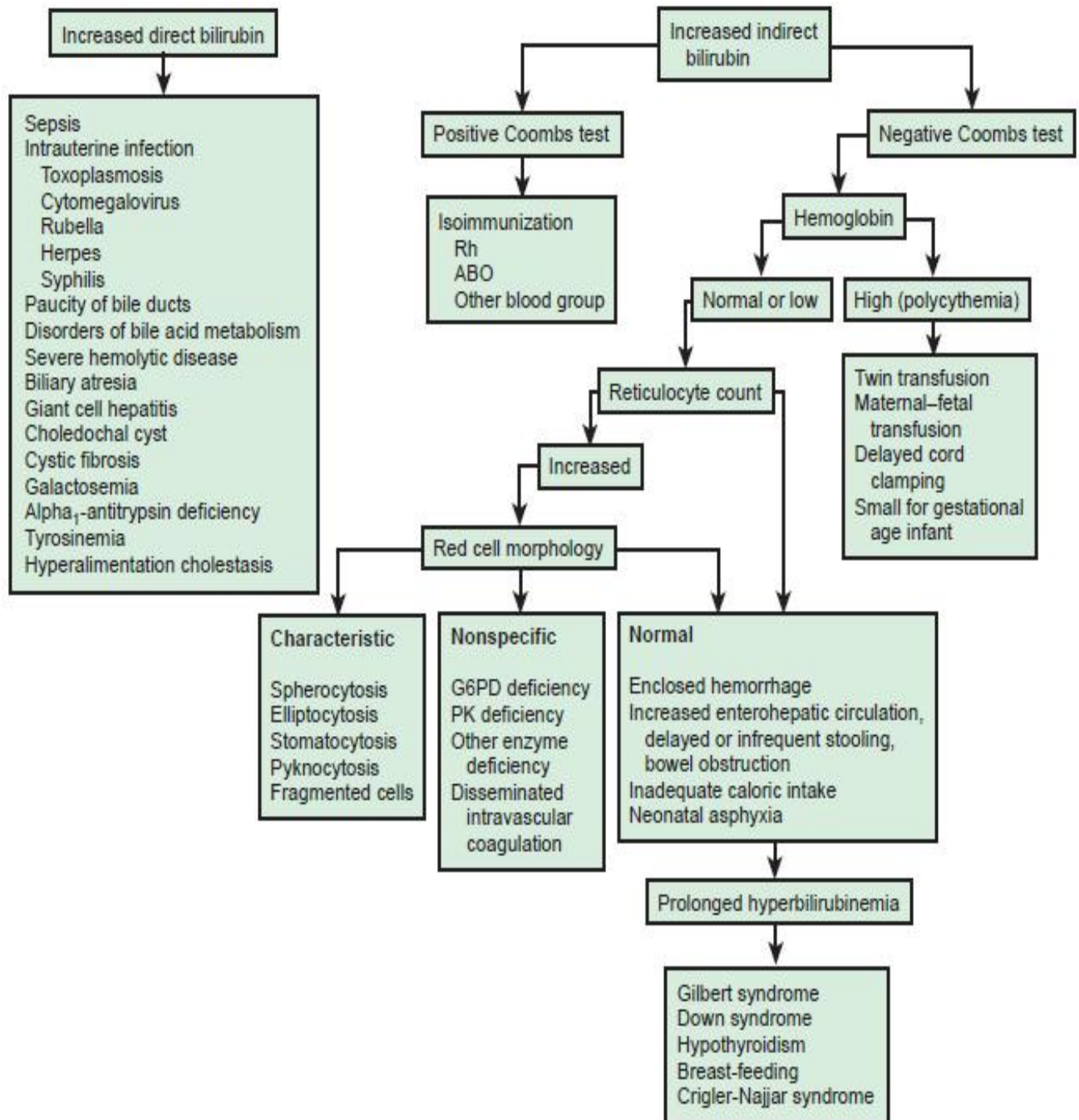
- ✓ Age in hours, weight, gestational age – SGA/AGA/LGA.
- ✓ Small for gestational age (SGA) may be associated with polycythemia and intrauterine infections.
- ✓ Plethora: polycythemia.
- ✓ Microcephaly may be associated with congenital infections.
- ✓ Extravascular blood bruising, cephalohematoma, or other enclosed hemorrhage
- ✓ Pallor associated with hemolytic anemia or extravascular blood loss
- ✓ Petechiae may suggest congenital infection, sepsis, or erythroblastosis.
- ✓ Hepatosplenomegaly may be associated with hemolytic anemia, congenital infection, or liver disease.
- ✓ Signs of sepsis.
- ✓ Chorioretinitis associated with congenital infection
- ✓ Evidence of hypothyroidism
- ✓ CNS examination: poor Moro's reflex, poor feeding, lethargy, tone abnormality, altered cry (signs of acute bilirubin encephalopathy).

C. Additional laboratory tests should be performed when serum TB is ≥ 95 th percentile for age in hours or at or near the threshold for initiation of phototherapy.

- ✓ The mother's blood type, Rh, and antibody screen.
- ✓ The infant's blood type, Rh, and direct Coombs test to assess for isoimmune hemolytic disease.
- ✓ Peripheral smear for RBC morphology and reticulocyte count to.
- ✓ Hematocrit or hemoglobin.
- ✓ Identification of specific antibody on the infant's RBCs (if result of direct Coombs test is positive)
- ✓ Direct or conjugated bilirubin
- ✓ With prolonged jaundice, tests for liver disease, congenital infections, septic workup, metabolic defects, or hypothyroidism are indicated.
- ✓ Serum albumin

- ✓ Abdominal Ultrasound
- ✓ G6PD measurement

Schematic approach to the diagnosis of neonatal jaundice (From Oski FA: Differential diagnosis of jaundice. In Taeusch HW, Ballard RA, Avery MA, editors: Schaffer and Avery's diseases of the newborn, ed 6, Philadelphia, 1991, Saunders.)



VII. Management of unconjugated hyperbilirubinemia.

Treatment modalities to reduce serum bilirubin concentration:

- ✓ **Hydration:** correction of dehydration
- ✓ **Phototherapy:** excretion of photoproducts in the bile and subsequent removal in the stool.
- ✓ **Exchange transfusion**
- ✓ **Drugs to increase conjugation:** by using enzyme inducers. e.g. phenobarbitone
- ✓ **Inhibition of reabsorption** (binding in the gut)
- ✓ **Inhibition of bilirubin production:** e.g. metalloporphyrin

Phototherapy is the initial intervention used to treat and prevent severe hyperbilirubinemia in asymptomatic infants and should be provided to infants with signs of acute bilirubin encephalopathy (ABE) while preparations are made for exchange transfusion. TB typically begins to decline within a few hours of treatment initiation.

In Phototherapy the rate of decline is increased by (efficacy can be increased by):

- ✓ Shorten the distance to as close as possible and maintain euthermia.
- ✓ Increased irradiance,
- ✓ More exposed surface area,
- ✓ Change position 2 hourly.
- ✓ Use double unit phototherapy, cover with aluminum
- ✓ Use the phototherapy unit with highest flux.
- ✓ Cover the phototherapy unit with white sheet to increase reflection
- ✓ Use special blue lights: eg-Philips TL 52 /20 W.
- ✓ The higher the initial TB, the more rapid the rate of decline (as much as 10 mg/dL [171 micromol/L] within a few hours).
- ✓ Phototherapy is less effective in infants whose hyperbilirubinemia is due to cholestasis or hemolysis with a positive direct antiglobulin test (DAT; Coombs test) than in infants with other causes.

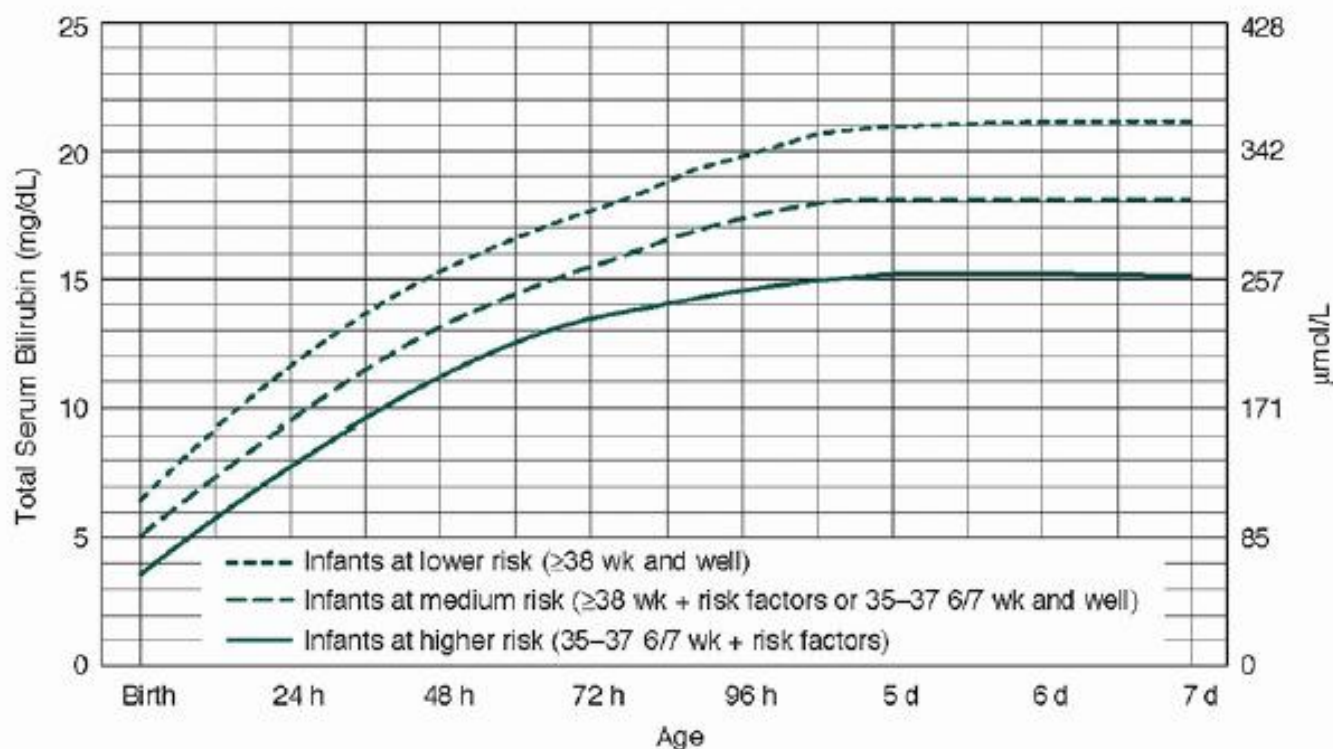
Mechanisms of bilirubin reduction by phototherapy

1. The main mechanism is **structural isomerization** by light that irreversibly converts bilirubin to lumirubin, a more soluble substance that can be excreted into bile and urine without conjugation; and is the most important mechanism of bilirubin elimination

2. Photoisomerization rapidly converts about 15% of the 4Z, 15Z bilirubin isomer to the less toxic 4Z, 15E form. Although the less toxic isomer can be excreted into bile without conjugation, the process is reversible and clearance is slow. Standard laboratory tests do not distinguish between the isomers, so TB levels may not change although may be less toxic.

3. Photo-oxidation is a slow process that converts bilirubin to small polar products that are excreted in the urine and is the least important mechanism of bilirubin elimination.

Guidelines for phototherapy in hospitalized infants of 35 or more weeks' gestation. TSB, total serum bilirubin. Management of hyperbilirubinemia in the newborn infant 35 or more weeks of gestation (from the American Academy of Pediatrics Subcommittee on Hyperbilirubinemia. Management of hyperbilirubinemia in the newborn infant 35 or more weeks of gestation. *Pediatrics* 2004; 114:297-316.)



- Use total bilirubin. Do not subtract direct reacting or conjugated bilirubin.
- Risk factors = isoimmune hemolytic disease, G6PD deficiency, asphyxia, significant lethargy, temperature instability, sepsis, acidosis, or albumin <3.0 g/dL (if measured).
- For well infants 35–37 6/7 wk, can adjust TSB levels for intervention around the medium risk line. It is an option to intervene at lower TSB levels for infants closer to 35 wk and at higher TSB levels for those closer to 37 6/7 wk.
- It is an option to provide conventional phototherapy in hospital or at home at TSB levels 2–3 mg/dL (35–50 μmol/L) below those shown, but home phototherapy should not be used in any infant with risk factors.

Guidelines for phototherapy in preterm LBW babies:

There are no evidence-based cut offs for preterm babies. The following table gives weight-based day specific guidelines. (From: St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)

Birth Wt (g)	Day 1	2	3	4	5	6	7
<1000		3	3	5	5	7	7
1000-1249	5	5	5	7	8	10	12
1250-1499	8	8	8	10	12	12	12
1500-1749	10	10	10	12	12	13	13
1750-1999	10	10	12	13	13	13	13
2000-2499	10	12	12	15	15	15	15

(From: 1. Maisels MJ, Watchko JF, Bhutani VK, et al. An approach to the management of hyperbilirubinemia in the preterm infant less than 35 weeks of gestation. *J Perinatol* 2012; 32:660-664.)

2. St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)

	Phototherapy	Exchange transfusion
Gestational age (week)	Initiate phototherapy total serum bilirubin (mg/ dl)	Total serum bilirubin (mg /dl)
<28 ^{0/7}	5–6	11–14
28 ^{0/7} –29 ^{6/7}	6–8	12–14
30 ^{0/7} –31 ^{6/7}	8–10	13–16
32 ^{0/7} –33 ^{6/7}	10–12	15–18
34 ^{0/7} –34 ^{6/7}	12–14	17–19

Risk Factors for Hyperbilirubinemia Neurotoxicity: Isoimmune hemolytic disease, G6PD deficiency, Asphyxia, Sepsis, Acidosis, Albumin <3.0 mg/dL, G6PD deficiency, Hyperthermia, Calori deprivation, Prolonged hyperbilirubinemia, Low birth weight, Prematurity.

Characteristics of devices. Although multiple devices are available for phototherapy, they are characterized by the following:

- ✓ Light emission in the blue-green spectrum (460 to 490 nm), which includes the region (460 nm) where bilirubin most strongly absorbs light
- ✓ Irradiance of at least 30 $\mu\text{W}/\text{cm}^2/\text{nm}$
- ✓ Illumination of maximal body surface area
- ✓ Shown to decrease TB during first 4 to 6 hours of exposure

Light sources

- ✓ **Blue light-emitting diodes (LEDs)** provide optimal high-intensity light in the absorption spectrum of bilirubin and are available as either overhead or underneath (mattress or fiberoptic pad) devices.
- ✓ **Fluorescent special blue light F20T12/BB and TL52 tubes** lower TB most effectively because they deliver light in the blue-green spectrum, providing maximal absorption and good skin penetration.
- ✓ **Halogen white lights** are placed at the recommended distance from the infant because they are hot and can cause thermal injury.
- ✓ **Fiberoptic blankets or pads** can be placed directly under the infant, generate little heat, and provide higher irradiance than fluorescent lights. Due to their small size, they rarely cover enough surface area to be effective when used alone in term infants and thus are typically used together with overhead lights.

If possible, in stable infants, Home phototherapy is effective and cheaper than hospital phototherapy.

Sunlight exposure. Although sunlight exposure effectively lowers TB level, safety concerns, including exposure to ultraviolet light, potential sunburn, and thermal effects, preclude use of sunlight as a reliable therapeutic tool. **Filtered sunlight:** has been

suggested as a clinical alternative in low-resource settings when phototherapy devices are not available.

Filtered sunlight selectively allows the transmission of blue light and removes harmful UV and infrared [IR] light rays, thereby reducing exposure to UV radiation and the risk of sunburn.

Practical aspects during phototherapy:

- ✓ Monitor temperature every 2 hourly.
- ✓ Monitor weight daily
- ✓ Record intake and output chart.
- ✓ Give frequent breastfeeding. Ensure baby is passing stools daily.
- ✓ Shield eyes from light source with appropriately sized eye pads. Due to risk of retinal degeneration seen in animals after 24 hours of fluorescent phototherapy.
- ✓ Give continuous phototherapy (except for feeding)
- ✓ Monitor side effects (dehydration, hypo/hyperthermia increased stool frequency, hypocalcemia, tanning).
- ✓ Keep the light source 45cm above the Newborn.
- ✓ Measure irradiance with use of flux meter monthly (intensive phototherapy - $> 30 \mu\text{w}/\text{cm}^2/\text{nm}$) & preferably use a chronometer to measure time of use of phototherapy lights.
- ✓ If flux meter is not available, change tubes of phototherapy every 3 months or after 1000 hours of use.
- ✓ Do not use TL-01, TL-05 blue tube-lights (commonly used as fly catchers) as they produce harmful ultraviolet rays. Do not use blue Perspex sheet as an alternative to blue lights.
- ✓ Watch for danger signs - Vomiting, lethargy, poor feeding, fever, high pitched cry, dark urine, light stools.
- ✓ Stop phototherapy when bilirubin is $<13 \text{ mg\%}$ in term / or two consecutive values are lower than age specific cut offs and $<10 \text{ mg\%}$ in preterm infants.
- ✓ Assess clinically for rebound after discontinuing phototherapy and do TB.
- ✓ Adequate hydration and urine output should be maintained to promote urinary excretion of bilirubin.

Monitoring

1. TB level is measured to monitor the response to therapy. The frequency of measurement depends on the initial TB value and the baby's age at initiation. When phototherapy is started during the birth hospitalization for a rising TB, TB is measured 4 to 6 hours after initiation and then repeated in 8 to 12 hours if TB has declined. For babies readmitted after birth hospitalization with a TB value that exceeds the 95th percentile for age in hours, TB is measured 2 to 3 hours after phototherapy is begun to ensure that TB is decreasing. TB is measured 18 to 24 hours after discontinuation of phototherapy.

Adverse effects. Phototherapy is generally considered safe. Temperature is monitored to avoid temperature instability. Monitoring of urine output and weight allows

early detection of increased insensible water loss that may lead to dehydration. Occurrence of loose stools or an erythematous rash, if present, is typically transient.

- ✓ **“Bronze baby” syndrome**, a dark bronze (dark, grayish brown skin) discoloration of the skin in infants undergoing phototherapy. thought to be related to impaired excretion of photoproducts of bile pigment, may occur with phototherapy in infants with direct hyperbilirubinemia (cholestatic jaundice) and usually resolves gradually within a few weeks after phototherapy is discontinued. The etiology is unknown, may result from photo-induced modification of porphyrins, which are often present during cholestatic jaundice and may last for many months. Despite the bronze baby syndrome, phototherapy can continue if needed, and whether the bronze pigments cause neurotoxicity is uncertain.
- ✓ **Increased Insensible water loss** (Give extra 20ml/kg fluid)
- ✓ **Retinal damage**
- ✓ **Erythematous skin rash**

Pharmacologic therapy.

1. Intravenous immunoglobulin (IVIG) is an adjunctive treatment for hyperbilirubinemia and has been used in infants with hemolytic disease caused by Rh or ABO incompatibility when TB continues to rise in infants receiving intensive phototherapy. The mechanism is unknown, but IVIG may act by occupying the Fc receptors on macrophages, decreasing removal of antibody-coated red cells from the circulation. We administer 0.5 to 1 g/kg IVIG over 2 hours and repeat the dose in 12 hours if needed.

2. Metalloporphyrins: A potentially important alternative therapy is the use of metalloporphyrins for hyperbilirubinemia. The metalloporphyrin Snmesoporphyrin (SnMP) offers promise as a drug candidate. The proposed mechanism of action is competitive enzymatic inhibition of the rate-limiting conversion of heme-protein to biliverdin (an intermediate metabolite in the production of unconjugated bilirubin) by hemeoxygenase.

A single intramuscular dose on the 1st day of life may reduce the need for subsequent phototherapy.

Other agents not for routine use:

3. Phenobarbitone:

Phenobarbital — Phenobarbital 5mg/kg/d for Crigler Najar syndrome(II) and Gilbert syndrome, increases the conjugation and excretion of bilirubin and decreases postnatal TB levels when given to pregnant women or infants.

4. Ursodeoxycholic acid — Ursodeoxycholic acid (UDCA) increases bile flow and helps to lower TB levels. It is useful in the treatment of cholestatic jaundice.

5. Clofibrate — Clofibrate is a peroxisome proliferator-activated receptor alpha agonist. It can be used in term neonates with ABO incompatibility.

Exchange transfusion:

Indications:

- Severe hyperbilirubinemia based on nomograms.
- If the TSB is above phototherapy level, exchange transfusion is indicated if baby has

received intensive phototherapy.

- If the value is above phototherapy cut off and baby has not received phototherapy previously, **start intensive phototherapy**. Arrange everything including blood for exchange, but repeat bilirubin in 2 hours and do exchange if there is no significant reduction in bilirubin (at least 2 mg./ dl). **By 6 hours, TSB should be below exchange cut off level.**

- Immediate exchange transfusion is indicated if baby shows features of BIND (hypertonia, arching, retrocollis, opisthotonus, fever, high pitched cry) or TSB is > 5 mg/dl than exchange level

- Use total bilirubin. Do not minus direct fraction unless > 50%.

- Measure serum albumin

As a rough guide, phototherapy should be started at a bilirubin level of 1% of the birth weight (in gm) in healthy low birth weight infants. An exchange transfusion should be considered at a value of 5 mg/dl higher than that for phototherapy (1% of birth weight + 5 mg/dl).

Other indications:

- Hyperbilirubinemia with BIND.

- Severe sepsis with sclerema or refractory metabolic problems.

- An exchange transfusion soon after birth is indicated in haemolytic jaundice if

- ✓ Cord bilirubin is > 4.5 mg/dl

- ✓ Cord Hb is <11 mg/dl, PCV<30

- ✓ The bilirubin is rising >1 mg/dl/hour despite phototherapy

- ✓ The hemoglobin level is between 11 and 13 mg/dl and bilirubin level is rising >0.5 mg/dl /hour despite phototherapy

- ✓ The bilirubin level is 20 mg/dl or it appears that it will reach 20 mg/dl

Exchange transfusion achieves:

(1) removal of antibody-coated red blood cells (a source of “potential” bilirubin),

(2) correction of anemia (if present), and

(3) removal of maternal antibody.

Single volume removes 63% of total patients blood volume. Double volume removes 86% and triple volume 95% of total patients' blood volume.

Immediately after a double volume exchange transfusion (about 160 to 180 mL/kg), TB values are typically half the value prior to the procedure. Thirty to 60 minutes, extravascular bilirubin rapidly equilibrates with the reduced vascular level, so that TB levels return to approximately **70-80% of previous level** i. e. two-thirds of pre exchange levels.

Choice of blood

- ✓ Fresh whole blood (preferably < 72 hrs, upto 5 days acceptable).

- ✓ O negative blood is the most preferable type for exchange transfusion.

- ✓ The blood should be warmed to 37°C with a blood warmer.

Selection of blood to be transfused to the newborn:

- ❖ If there is Rh hemolytic disease: give blood group compatible to the baby and Rh to the mother.

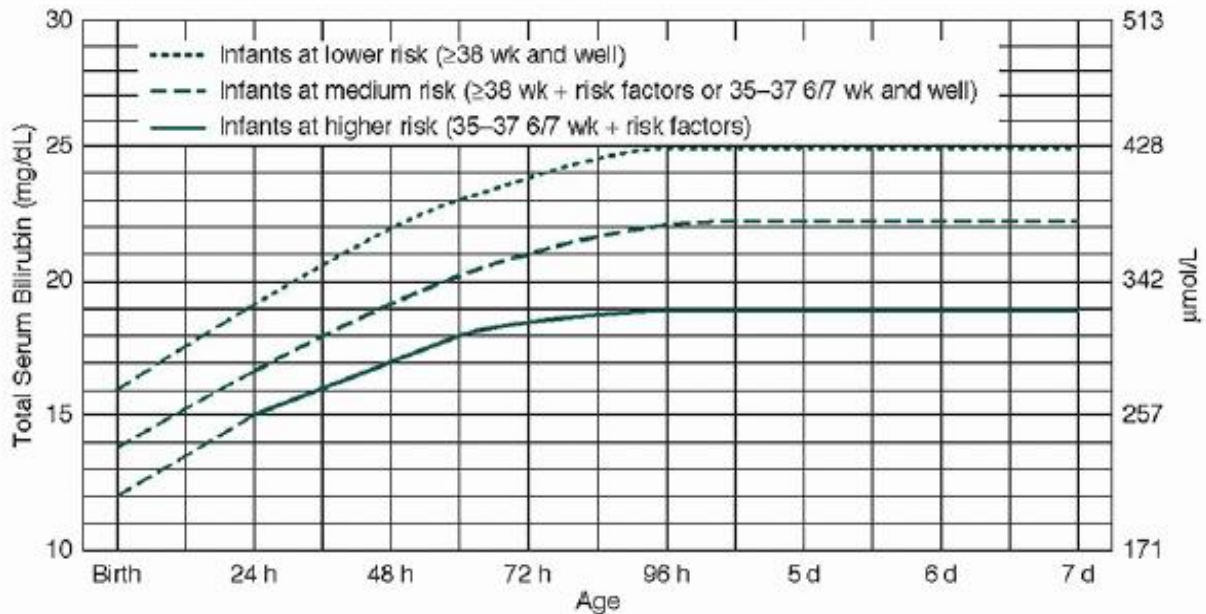
Or in Rh incompatibility, we use Type O negative packed red blood cells (RBCs) with low anti-A and anti-B titers and reconstituted in AB plasma, and:

- ✓ If the RBCs are made available before delivery of the sensitised infant the RBCs must be O Rh negative and cross matched against the mother.
 - ✓ If the RBCs are sourced after delivery the RBCs must be cross matched against the infant.
- ❖ If there is ABO hemolytic disease: give blood group of the mother and Rh compatible to the newborn.
- Or in ABO incompatibility, we use Type O negative packed cells, Rh compatible with mother and infant, reconstituted in AB plasma.
- ❖ In other situations – fresh ABO and Rh compatible blood should be used.
 - ❖ In minor blood group incompatibility (ie, anti-Kell, anti-E, anti-Duffy, anti-RhC), the donor blood should be cross matched to the mother's blood with the offending antigen excluded, and
 - ❖ In jaundice not related to isoimmune disorders, donor blood must always be cross-matched to infant's plasma and blood type.

Volume

Usually involves double the volume of the infant's blood (160ml/kg) plus additional volume to account for tubing losses (~30 mL) and is known as double volume exchange transfusion (DVET). This replaces 87% of the infant's blood volume with new blood. In preterms, the blood volume should be calculated from the nomogram.

Guidelines for exchange transfusion in hospitalized infants of 35 or more weeks' gestation. TSB, total serum bilirubin; G6PD, glucose-6-phosphate dehydrogenase (from the American Academy of Pediatrics Subcommittee on Hyperbilirubinemia. Management of hyperbilirubinemia in the newborn infant 35 or more weeks of gestation. *Pediatrics* 2004; 114:297-316.)



- The dashed lines for the first 24 h indicate uncertainty due to a wide range of clinical circumstances and a range of responses to phototherapy.
- Immediate exchange transfusion is recommended if infant shows signs of acute bilirubin encephalopathy (hypertonia, arching, retrocollis, opisthotonos, fever, high pitched cry) or if TSB is ≥ 25 mg/dL ($85 \mu\text{mol/L}$) above these lines.
- Risk factors—isoimmune hemolytic disease, G6PD deficiency, asphyxia, significant lethargy, temperature instability, sepsis, acidosis.
- Use total bilirubin. Do not subtract direct reacting or conjugated bilirubin.
- If infant is well and 35–37 6/7 wk (medium risk) can individualize TSB levels for exchange based on actual gestational age.

Bilirubin-albumin ratio

The ratio of bilirubin (mg/dL) to albumin (g/dL) correlates with unconjugated bilirubin levels in newborns and therefore can be used as a surrogate for unconjugated bilirubin or residual binding capacity of albumin.

Sick infants have an impaired albumin binding of bilirubin and can have ratios of 0.7, and it is recommended that the ratio should not exceed 0.5.

In addition, albumin levels and the ability of albumin to bind bilirubin vary significantly in sick infants.

It has also been shown that binding increases with increasing gestational and postnatal age.

Therefore, it is recommended that the bilirubin-to-albumin ratio should be used in concert with measurements of TSB in determining exchange transfusion

(From: Subcommittee on Hyperbilirubinemia, Management of Hyperbilirubinemia in the Newborn Infant 35 or More, Weeks of Gestation American academy of pediatrics, Clinical practice guideline. *Pediatrics* 2004;114;297 DOI: 10.1542/peds.114.1.297.)

Risk Category	B/A Ratio at Which Exchange Transfusion Should be Considered	
	TSB mg/dL/ Alb, g/dL	TSB μ mol/L/ Alb, μ mol/L
Infants $\geq$ 38 0/7 wk	8.0	0.94
Infants 35 0/7–36 6/7 wk and well or $\geq$ 38 0/7 wk if higher risk or isoimmune hemolytic disease or G6PD deficiency	7.2	0.84
Infants 35 0/7–37 6/7 wk if higher risk or isoimmune hemolytic disease or G6PD deficiency	6.8	0.80

Guidelines for Exchange Transfusion in Preterm LBW babies without risk

factors for Kernicterus or hemolysis: (From: St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)

Birth Wt (g)	Day 1	2	3	4	5	6	7
<1000	10	10	10	12	12	12	12
1000-1249	12	12	12	12	15	15	15
1250-1499	15	15	15	15	16	16	16
1500-1749	15	17	17	17	18	18	18
1750-1999	17	17	18	18	19	19	19
2000-2499	18	18	19	19	19	20	20

- Check potassium if blood is >7days old and particularly if metabolic abnormalities are noted following ET,
- Ensure the stomach is empty. Aspirate stomach and keep OG on free drainage. Do not feed during the procedure.

• Equipment: Cap, mask, Sterile gown, Surgical gloves, extension tubing Set, Blood & Infusion Warmer, vacutainers for biochemistry & hematology, Exchange Transfusion Record Chart, Cardiorespiratory monitor, Consent form for parents, antiseptic solution for Line insertion

- Push-pull technique - Umbilical exchange through UVC, the tip should be in the IVC to get a good backflow few cm. Do not perform Exchange Transfusion through a UVC if the tip is in the portal circulation. This may cause NEC by reducing bowel blood flow. Avoid umbilical exchange in SGA and preterm babies who are more prone to develop NEC. **A large central vein like femoral vein may also be used if vascular access difficult esp in an older newborn**

4. Peripheral exchange – Withdraw from artery and infuse through vein.

Isovolumetric exchange (simultaneous pulling blood from umbilical artery and pushing blood through the umbilical vein). Continuous infusion of blood into the UVC balanced by controlled removal of blood from the UAC.

Aliquots <3-5% of blood volume, the sicker and smaller baby the smaller is the aliquot 5 ml for infants under 1500 gm.

10 ml for infants 1500 to 2500 gm.

15 ml for infants 2500 to 3500 gm.

Each aliquot should take 3-5 minutes.

. Individual aliquots should be approximately 10% or less of the infant's blood volume, with a maximum volume of 20 mL for a term baby who weighs more than 3 kg and smaller volumes in babies with physiologic instability.

- If there is a risk of thrombocytopenia in a sick child esp if the exchange is done with PRBC and FFP reconstituted, consider platelet transfusion at mid point and at end of ET.

- Routine calcium injections during exchange transfusion is not indicated. But if

unexplained tachycardia or arrhythmia give slow IV calcium.

- After 4-6 hours – send a repeat serum bilirubin.

Intensive phototherapy should be resumed after the transfusion, and TB should be monitored at 2, 4, and 6 hours after the transfusion and then at least every 12 to 24 hours until TB declines sufficiently to discontinue phototherapy. Increasing TB or recurrent neurologic signs are followed to assess need for repeat exchange transfusion.

- **Monitoring during transfusion:** every 10 min during the procedure, check and document HR (vital monitor), respiratory rate, temperature, color, SaO₂ probe. Monitor vitals and abdominal girth. hrly for 2 hours. If constant, start feeds. Check blood sugar at end of procedure, 30 min, 1 hour and 2 hours.

Complications related to procedure and blood

- Hypocalcemia and hypomagnesemia.
- Hypoglycemia.
- Acid base imbalance.
- Hyperkalemia.
- Infection.
- Bleeding.
- Thrombocytopenia
- Thrombosis, embolisation.
- Perforation of vessel, vasospasm.
- Volume overload.
- Cardiac arrhythmias and arrest.
- Hemolysis (hemoglobinuria, hyperkalemia)
- GVHD
- Hypo/hyperthermia

VIII. BILIRUBIN TOXICITY

A. Brain injury by bilirubin. Unconjugated bilirubin that is not bound to albumin is a potential toxin that can enter the brain and cause apoptosis and/or necrosis. FFAs and certain drugs (e.g., ceftriaxone) may displace bilirubin from albumin and promote entry into the brain. If the blood-brain barrier is disrupted by factors including hyperosmolarity, asphyxia, and hypercarbia, bilirubin can also enter the brain bound to albumin. Acidosis affects bilirubin solubility and promotes its deposition into brain tissue. The blood-brain barrier may be more vulnerable in preterm than in term infants.

B. Bilirubin deposited in the brain can result in bilirubin-induced neurologic dysfunction (BIND). Severe hyperbilirubinemia (TB >25 mg/dL) is associated with increased risk of BIND in term or late preterm infants; the level of risk in preterm infants is uncertain but presumed to be lower. The brain regions typically affected by bilirubin toxicity include the basal ganglia, cerebellum, white matter, and the brainstem nuclei for oculomotor and auditory function.

C. Neurologic manifestations of bilirubin toxicity reflect the areas of the brain that are most often affected and can be reversible or permanent.

1. ABE is the clinical manifestation of bilirubin toxicity seen in the neonatal period. The clinical presentation consists of three phases:

- ✓ **Early phase.** Signs are subtle and may include lethargy, hypotonia, high-pitched cry, and poor suck.
- ✓ **Intermediate phase** progresses in the absence of intervention for hyperbilirubinemia and is characterized by hypertonia of extensor muscles (rigidity, opisthotonus, and retrocollis), oculogyric crisis, irritability, fever, and seizures. Some infants die in this phase. All infants who survive this phase are likely to develop chronic bilirubin encephalopathy (clinical diagnosis of kernicterus).
- ✓ **Advanced phase.** Signs include pronounced opisthotonus and retrocollis, cry that can be weak or shrill, apnea, seizures, and coma. Affected infants die from intractable seizures or respiratory failure.

2. Kernicterus refers to the chronic and permanent sequelae of bilirubin toxicity that develop during the first year of age. Most infants who develop kernicterus have had signs of ABE in the neonatal period although some have a history of high TB level with few or no signs of ABE. The signs of kernicterus are as follows:

- ✓ Choreoathetoid cerebral palsy with neuromotor impairments
- ✓ Sensorineural hearing loss (auditory neuropathy), characterized by abnormal brainstem auditory evoked response with normal otoacoustic emission testing
- ✓ Limitation of upward gaze
- ✓ Dental enamel dysplasia

Incidence and prognosis:

By pathologic criteria, kernicterus develops in 30% of infants (all gestational ages) with untreated hemolytic disease and bilirubin levels >25-30 mg/dL. Overt neurologic signs have a grave prognosis; more than 75% of infants die, and 80% of affected survivors have bilateral choreoathetosis with involuntary muscle spasms. Mental retardation, deafness, and spastic quadriplegia are common.

IX. Cholestasis or conjugated hyperbilirubinemia is due to failure to excrete bile.

This may be caused by defects in intrahepatic bile production, defects in transmembrane transport of bile, or mechanical obstruction to flow. Conjugated hyperbilirubinemia is defined by a direct or conjugated bilirubin level >1 mL/dL or >15% of the TB level. It may be associated with hepatomegaly, splenomegaly, pale stools, and dark urine.

An infant with jaundice at 2 weeks of age should be evaluated for cholestasis by measuring total and direct bilirubin level. Rapid diagnosis is important so that therapy for treatable disorders can be started promptly.

A. Many disorders cause neonatal cholestasis;

1. Obstructive bile duct disorders. Biliary atresia is a frequent cause and must be identified promptly so that intervention (hepatoportoenterostomy) can be performed before 2 months of age. This condition may be associated with situs inversus, polysplenia or asplenia, and cardiac anomalies. Another cause is Alagille syndrome, which is characterized by unusual facial

appearance, ocular abnormality (posterior embryotoxon), cardiac abnormalities (pulmonic stenosis), and vertebral anomalies (butterfly vertebrae). Choledochal duct cysts are an uncommon but surgically treatable cause of cholestasis.

2. Infectious causes include sepsis and urinary tract infections as well as infections caused by numerous viral, bacterial, and other organisms.

3. Metabolic disorders include α 1-antitrypsin deficiency, cystic fibrosis, galactosemia, tyrosinemia, galactosemia, storage diseases (Gaucher, Niemann-Pick), Zellwegersyndrome, mitochondrial disorders, and congenital disorders of glycosylation.

4. Immunologic disorders include gestational alloimmune liver disease (formerly neonatal hemochromatosis) and neonatal lupus erythematosus.

5. Endocrine disorders include hypothyroidism and panhypopituitarism.

6. Toxic disorders. A frequent cause of cholestasis in the neonatal intensive care unit (NICU) occurs in infants unable to take enteral feeding who have prolonged courses of total parenteral nutrition (PN) including lipid. This condition typically resolves with introduction of enteral feedings.

7. Isoimmune hemolysis. Conjugated hyperbilirubinemia occurs in a small proportion of infants with excessive hemolysis such as ABO/Rh incompatibility and may persist for 2 weeks.

B. Diagnosis

1. History and findings on physical examination may support a specific diagnosis. Acholic stools suggest obstruction.

2. Laboratory studies to evaluate liver function should include total and direct or conjugated bilirubin, serum alanine aminotransferase (ALT), aspartate aminotransferase (AST), gamma-glutamyl transpeptidase (GGT), alkaline phosphatase, and coagulation studies. Specific laboratory studies should be performed based on findings from the history and physical examination. These include tests for infections and metabolic, genetic, or endocrine disorders.

3. Abdominal ultrasonography may suggest biliary atresia by failure to visualize the gall bladder or presence of the triangular cord sign. Choledochal duct cyst, gallstones, or vascular malformations may be identified.

4. Hepatobiliary scintigraphy with technetium-labeled iminodiacetic acid analogs may distinguish biliary atresia from other causes of cholestasis such as neonatal hepatitis.

Bilirubin concentration measured in a duodenal aspirate and compared to serum concentration is an alternative to scintigraphy to assess bile excretion.

5. Percutaneous liver biopsy is recommended to evaluate cholestatic jaundice

6. If studies support a diagnosis of biliary atresia, intraoperative cholangiography is performed. If biliary obstruction is demonstrated, a hepatoportoenterostomy (Kasai procedure) is performed.

Hypoglycemia and Hyperglycemia in Neonates

Definition:

Clinically significant neonatal hypoglycemia requiring intervention cannot be defined by a precise numerical blood glucose concentration because of the following:

- ✓ Low blood glucose concentrations normally occur in the first hours after birth.
- ✓ Lack of outcome data: because the blood glucose concentration and duration of hypoglycemia associated with poor neurodevelopmental outcome has not been established.

Hypoglycemia is common metabolic problem in NICUs. This is because of abrupt cease in glucose supply following clamping of the umbilical cord at birth. Some neonates are symptomatic whereas most are asymptomatic despite very low blood glucose levels. This variability is due to number of factors including:

- ✓ Gestational age
- ✓ Birth weight
- ✓ Post natal age
- ✓ Feeding status
- ✓ Presence or absence of associated illnesses

Hypoglycemia is defined as a glucose concentration low enough to cause signs or symptoms of impaired brain function (neuroglycopenia).

In newborns, a plasma glucose value of ≤ 50 mg/dL is an appropriate threshold to distinguish infants who warrant further diagnostic testing.

For intervention or further evaluation, hypoglycemia could be defined as blood glucose level less than 40mg/dl.

Hypoglycemia may be Transient or Persistent.

Transient hypoglycemia implies low glucose values that last only a short time if not corrected and that it is confined to the newborn period.

Persistent and recurrent hypoglycemia implies a form that requires prolonged management (glucose infusions for several days at high rates of infusion) and perhaps pharmacologic intervention. Several of these hypoglycemia syndromes may continue throughout infancy and childhood.

Causes of the Two Types of Neonatal Hypoglycemia:

I. Transient hypoglycemia (Most cases)

A. Associated with changes in maternal metabolism

1. Intrapartum administration of glucose
2. Drug treatment
 - a. Terbutaline, ritodrine, propranolol
 - b. Oral hypoglycemic agents
3. Diabetes in pregnancy: infant of diabetic mother

B. Associated with neonatal problems

1. Idiopathic condition or failure to adapt
2. Intrauterine growth retardation
3. Birth asphyxia
4. Infection
5. Hypothermia
6. Hyperviscosity
7. Erythroblastosis fetalis
8. Other
 - a. Iatrogenic causes
 - b. Congenital cardiac malformations

II. Persistent or recurrent hypoglycemia

A. Hyperinsulinism

1. Beta cell hyperplasia, nesidioblastosis-adenoma spectrum, sulfonylurea receptor defect
2. Beckwith-Wiedemann syndrome (macrosomia, mild microcephaly, omphalocele, macroglossia, hypoglycemia, and visceromegaly)

B. Endocrine disorders

1. Pituitary insufficiency
2. Cortisol deficiency
3. Congenital glucagon deficiency
4. Epinephrine deficiency

C. Inborn errors of metabolism

1. Carbohydrate metabolism
 - a. Galactosemia
 - b. Hepatic glycogen storage diseases
 - c. Fructose intolerance

2. Amino acid metabolism

- a. Maple syrup urine disease
- b. Propionic acidemia
- c. Methylmalonic acidemia
- d. Hereditary tyrosinemia
- e. 3-hydroxy, 3-methyl glutaric acidemia
- f. Ethylmalonic-adipic aciduria
- g. Glutaric acidemia type II

3. Fatty acid metabolism

- a. Defects in carnitine metabolism
- b. Acyl-coenzyme dehydrogenase defects

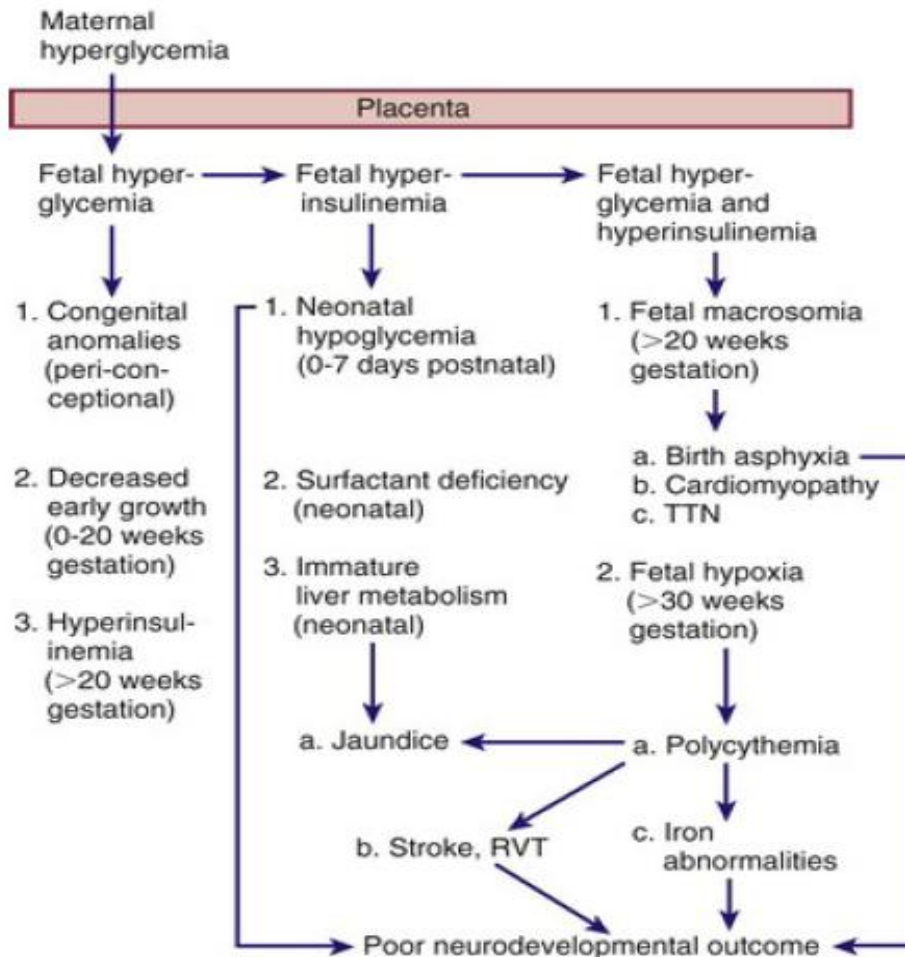
D. Neurohypoglycemia (hypoglycorrachia) due to defective glucose transport

Clinical Symptoms and Signs of Hypoglycemia:

- ✓ Weak or high-pitched cry, Irritability
- ✓ Apnea, cyanotic spells, Grunting, tachypnea, Tachycardia, Hypothermia
- ✓ Feeding difficulty
- ✓ Lethargy, stupor, apathy
- ✓ Seizures, Jitteriness, tremors,
- ✓ Hypotonia, limpness
- ✓ Sweating
- ✓ Some infants may be asymptomatic

Morbidity in Infants of Diabetic Mothers:

- ✓ Congenital anomalies, Small left colon, Heart failure and septal hypertrophy of heart
- ✓ Surfactant deficiency, respiratory distress syndrome, transient tachypnea of the newborn, persistent pulmonary hypertension
- ✓ Hyperbilirubinemia
- ✓ Hypoglycemia, hypocalcemia, hypomagnesemia
- ✓ Macrosomia, nerve injury related to birth trauma, Visceromegaly
- ✓ Renal vein thrombosis
- ✓ Unexplained intrauterine demise
- ✓ Polycythemia
- ✓ Predisposition to later-life obesity, insulin resistance, and diabetes



The fetal and neonatal events attributable to fetal hyperglycemia (*column 1*), fetal hyperinsulinemia (*column 2*), or both in synergy (*column 3*). Time of risk is denoted in parentheses. RVT, Renal vein thrombosis; TTN, transient tachypnea of the newborn. (From: Nold JL, Georgieff MK: Infants of diabetic mothers, *Pediatr Clin North Am* 51:619–637, 2004.)

Monitor:

- All infants admitted in NICU.
- In postnatal wards -
 - ✓ low birth weight infants < 2500 g,
 - ✓ SGA infants, fatally malnourished infants (low CAN score)
 - ✓ LGA infants, > 4000 g and infants of diabetic mother
 - ✓ Infants who are not feeding well with > 10 % weight loss
- Other high-risk infants:
 - ✓ Infant of a diabetic mother
 - ✓ Babies with perinatal asphyxia
 - ✓ Polycythemia
 - ✓ Hypothermia

- ✓ Sepsis
- ✓ Following exchange transfusion
- ✓ Infants born to mothers on tocolytics – β sympathomimetic agents - terbutaline, isoxsuprine, chlorpropamide, propranolol

Time of monitoring:

- ✓ In Post-natal ward (refer algorithm)
- ✓ In NICU (refer algorithm)
- ✓ Infant of diabetic mother (cord blood sugar for hyperglycemia must be sent).
Perform Hg stix at 1, 2, 3, 6, 12, 24, 36 and 48 hours.

Investigations:

- ✓ GRBS using glucometer: Calibrate glucometer. Give a bold heel prick with lancet. Use the glucometer as per manufacturer's instructions. Reagent strips measure whole blood glucose, which is 15% lower than plasma levels.
- ✓ Blood Sugar estimation: When hypoglycemia is detected on the GRBS confirm with random blood sugar estimation. Sample must be processed immediately as delay causes glucose oxidation giving falsely low values.
- ✓ Other investigations as indicated. SGA/Polycythemia - perform PCV. At risk for sepsis do septic screen / CSF.
- ✓ In case of persistent hypoglycemia or need for large glucose infusion over a week, or if persistent GIR > mg/kg/min, re-evaluate for other causes.
 - Persistent bacterial or fungal infections.
 - Inborn errors of metabolism.
 - Defects in carbohydrate metabolism - Glycogen storage diseases, Galactosemia.
 - Defects in amino acid metabolism.
 - Endocrinal disorders - Persistent hyperinsulinemic hypoglycemia of infancy (PHHI-islet cell hyperplasia), adrenal insufficiency, hypopituitarism, hypothalamic deficiency, glucagon deficiency, epinephrine deficiency,
- ✓ Following investigations must be done in case of persistent hypoglycemia.
 - Serum insulin with simultaneous blood glucose estimation: Insulin > 2 uIU/ml at time of hypoglycemia is suggestive of hyperinsulinemia. Insulin: glucose ratio - > 0.4 is suggestive of hyperinsulinemia.
 - Endocrine Profile: Cortisol, TSH, free T4. For other hormonal evaluation like Growth hormone, ACTH, Glucagon,
 - Metabolic Profile: Plasma amino acid, urinary ketones, urine for reducing substances. Urine amino acid, urine organic acids, ABG.
 - 18 F L Dopa PET Scan: to differentiate between focal and diffuse PHHI.

Management of neonatal hypoglycemia

The overall management of neonatal hypoglycemia should include:

- ✓ Anticipation and prevention in those who are at high risk.
- ✓ Correction of hypoglycemia in those who are symptomatic and
- ✓ Investigation and treatment of the cause of hypoglycemia, when it is possible to identify the cause.

a) Treatment of asymptomatic hypoglycemia

Feeding: is the initial treatment in an asymptomatic term infant,

- ✓ Immediately offer breast-feeding.
- ✓ Check blood glucose 30 minutes after feeding to insure normal glucose level before the next feeding.
- ✓ If repeated blood glucose is $> 40\text{mg/dl}$ continue to offer feedings at 2-3 hours interval.

Indications of IV infusions in asymptomatic hypoglycemia (use same infusion as symptomatic hypoglycemia)

- ✓ Blood glucose $< 25\text{mg/dl}$.
- ✓ Blood glucose remains $< 40\text{mg/dl}$ after one attempt of feeding
- ✓ If infant becomes symptomatic
- ✓ If oral feeding is contraindicated

b) Treatment of symptomatic hypoglycemia

Immediate treatment:

- ✓ Secure IV line
- ✓ Give 2ml/kg of 10% glucose IV bolus over one minute if signs other than seizure
- ✓ Give 4 ml/kg of 10% glucose as a bolus over one minute if seizure is present.
- ✓ The small bolus minimizes hyperglycemia that can provoke insulin secretion and possibly prolong hypoglycemia.

10% dextrose for IV bolus can be prepared using 40% dextrose, which is available in our country by taking one part of 40% dextrose and three parts of distilled water.

e.g: for a symptomatic infant weighing 4kgs ,the total volume of 10% dextrose will be $4 \times 2\text{ml} = 8\text{ml}$. To prepare this 8 ml take one part(2ml)from 40% dextrose and three parts (6ml) from distilled water.

Continuous therapy:

- ✓ Put on 10% glucose infusion at glucose infusion rate (GIR) of 6mg/kg/minutes (~ 90ml/kg/day) as maintenance.
- ✓ Recheck blood glucose after 30 minutes and if it remains above 40mg/dl frequency of checking can be decreased to one hourly then every six hourly.
- ✓ If blood glucose remains <40mg/dl, increase the GIR by 2mg/kg/minutes every 30 minutes until repeat values are above 40mg/dl.
- ✓ Once the blood glucose values stabilize above 40mg/dl for 24 hours, the GIR can be tapered off at 2 ml/kg/min every six hourly with proportional increment of oral feeds. Breastfeeding should be continued.
- ✓ If the neonate requires GIR > 12.5mg/kg/minutes, persistent hypoglycemia should be considered.
- ✓ **Glucose infusion rate (GIR) can be calculated using the following formula:**
Concentration X volume ÷ 144

$$\text{GIR in mg/kg/min} = \frac{\text{dextrose \% []} \times \frac{\text{ml}}{\text{kg}}/\text{day}}{144}$$

For e.g. If a baby is receiving 100 cc/kg/day of D10, the glucose delivery rate will be
GDR = 10 X 100 ÷ 144 = 6 – 7 mg/kg/min. (~ 0.07 ml/kg/min) i.e.: 100 mg/kg of glucose is equivalent to 1 mL/kg of dextrose 10% in water (D10W).

Practical points:

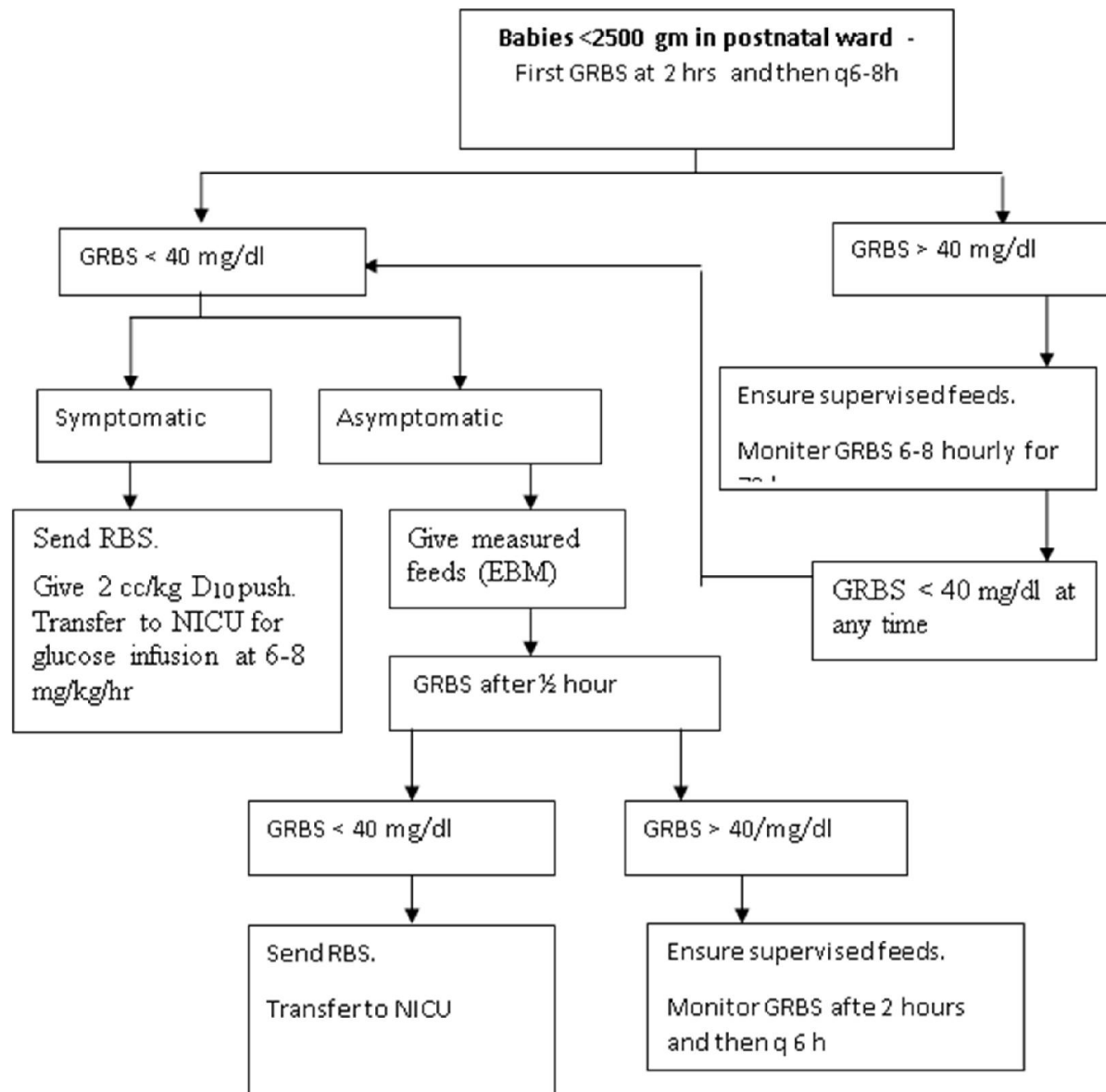
- ✓ Maximum glucose concentration that can be given through the peripheral vein is 12.5%. Glucose concentration of more than 12.5% has to be given through central venous line.
- ✓ In addition to glucose infusion and monitoring, reduce energy needs by correcting acidosis, maintaining a thermo neutral environment and treatment of other underlying conditions like sepsis.
- ✓ Do not stop an IV infusion of glucose abruptly, severe rebound hypoglycemia may occur.
- ✓ If the patient needs repeated boluses, this may be an indication for increasing the rate of continuous glucose infusion, and for considering other causes.

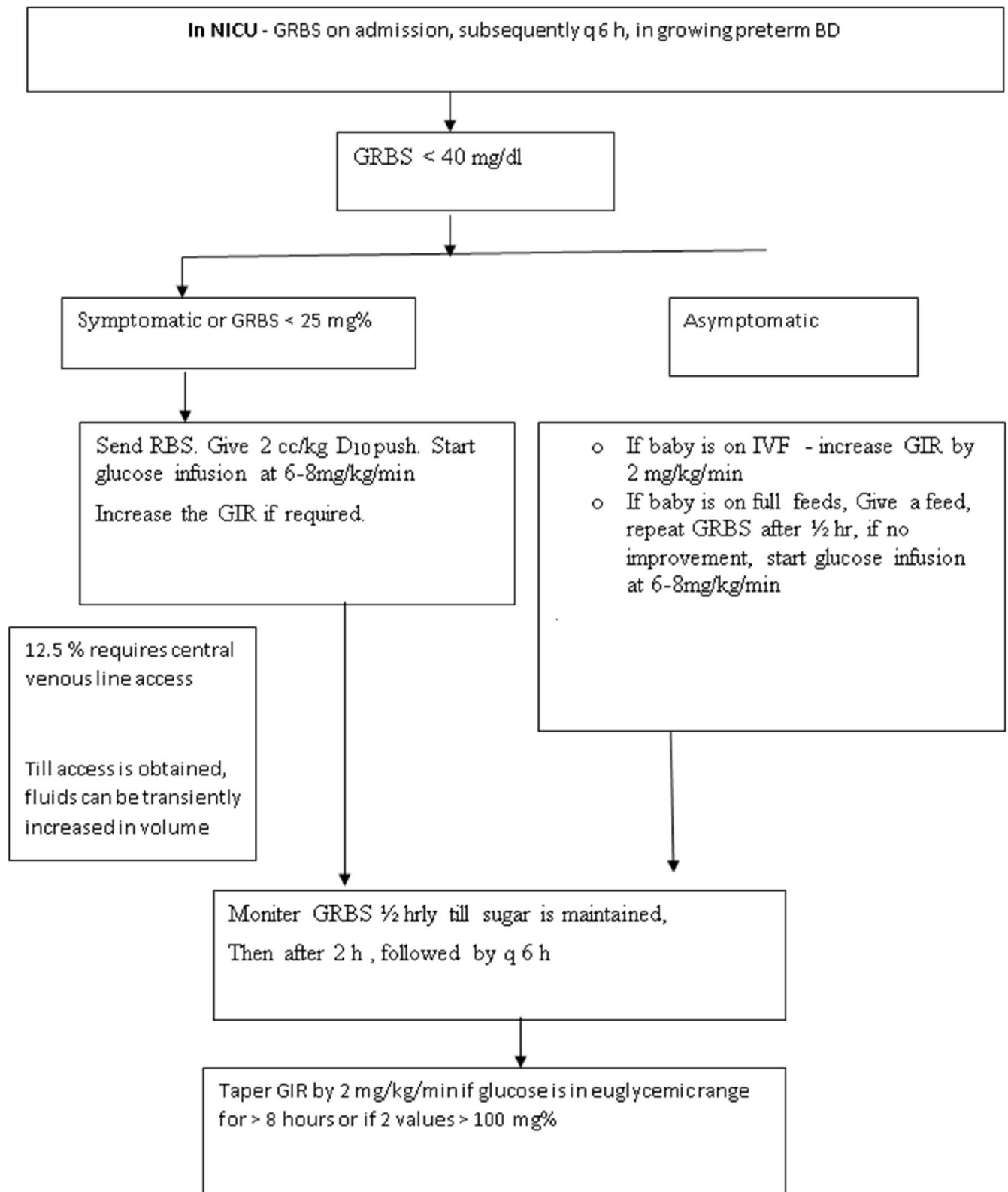
Persistent hypoglycemia:

If the infant requires more than 12 - 15 mg/kg/min. of glucose infusion, the appropriate investigation must be sent.

1. **Consider IV hydrocortisone.** Historically, providers have administered **hydrocortisone**, 10 mg/kg/day intravenously in two divided doses, if it is difficult to maintain glucose values in the normal range despite 12 to 15 mg of glucose per kilogram per minute. Hydrocortisone reduces peripheral glucose utilization, increases gluconeogenesis, and increases the effects of glucagon.
2. **Diazoxide** (8 to 15 mg/kg/day in divided doses every 8 to 12 hours) may be given orally for infants who are persistently hyperinsulinemic. This drug inhibits insulin release by acting as a specific ATP-sensitive potassium channel agonist in normal pancreatic beta cells and decreases insulin release. It can take up to 5 days for a positive effect to be seen. Side effects include fluid retention, and coadministration with a diuretic such as hydrochlorothiazide may be considered.
3. **Octreotide** (5 to 20 µg/kg/day subcutaneously or intravenously divided every 6 to 8 hours). A long-acting somatostatin analog that inhibits insulin secretion. It can be used when diazoxide does not successfully control the glucose levels. Tachyphylaxis can develop.
4. **Glucagon** (0.2 mg/kg intramuscularly, subcutaneous (SC), or IV, maximum 1.0 mg) is rarely used. It may be given to hypoglycemic infants with good glycogen stores, but it is only a temporizing measure to mobilize glucose for 2 to 3 hours in an emergency until IV glucose can be given. The glucose level will often fall after the effects of glucagon have worn off, and it remains important to obtain IV access to adequately treat these infants. For IDMs, the dose is 0.3 mg/kg (maximum dose is 1.0 mg).
5. **Surgical management:** If medical treatment does not control the blood glucose level, consider a 18F-fluoro-L-DOPA positron emission P.322 tomography (PET) scan to identify focal lesions in the pancreas and consider surgical treatment by **subtotal pancreatectomy**.

Approach to Neonatal Hypoglycemia (From: St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)





Persistent hypoglycemia

Investigate if > 3 days



LGA

- Consider hyperinsulinemia
- PHHI
- Beckwith Weidman syndrome
- Send HbA_{1c} in mother

Central line access for infusions > 12.5 %
concentration



At the time of hypoglycemia:

- Send insulin, RBS, Cortisol.
- If GIR is > 12 mg/kg/min, hydrocortisone may be considered at 10 mg/kg after collecting cortisol
- **Work up for Inborn Errors of Metabolism**

II. Hyperglycemia is usually defined as a whole blood glucose level higher than 125 mg/dL or plasmaglucoase values higher than 145 mg/dL. It is characterized by: Polyuria, dehydration, and ketoacidosis.

Etiology:

- ✓ Iatrogenic. Exogenous parenteral glucose administration.
- ✓ Drugs: glucocorticoids, caffeine, theophylline, phenytoin, and diazoxide.
- ✓ Extremely low birth weight infants, possibly due to variable insulin response.
- ✓ Lipid infusion. Free fatty acids are associated with increased glucose levels.
- ✓ Sepsis, possibly due to depressed insulin release, cytokines, or endotoxin, resulting in decreased glucoseutilization. Stress hormones such as cortisol and catecholamines are elevated in sepsis.
- ✓ “Stressed” preterm infants requiring painful procedures.”
- ✓ Hypoxia, possibly due to increased glucose production in the absence of a change in peripheral utilization.
- ✓ Surgical procedures: due to the secretion of epinephrine, glucocorticoids, and glucagon as well as excess administration of glucose-containing IV fluids.
- ✓ Diabetes due to pancreatic lesions.
- ✓ Transient hyperglycemia associated with ingestion of hyperosmolar formula.
- ✓ Neonatal diabetes mellitus: requires insulin treatment in the first months of life. They characteristically are SGA term infants, without gender predilection, and a third have a family history of diabetes mellitus. They present with marked glycosuria, hyperglycemia (240 to 2,300 mg/dL), polyuria, severe dehydration, acidosis, mild or absent ketonuria, reducedsubcutaneous fat, and failure to thrive.

Diagnosis of hyperglycemia will be made based on:

- ✓ Perinatal risk factors (asphyxia, sepsis...)
- ✓ Clinical manifestations (weight loss, signs of dehydration, polyuria).
- ✓ Laboratory (blood glucose >125 mg/dl., ketone & glucose on urine analysis)

Management of neonatal hyperglycemia:

- ✓ Prevent of hyperglycemia by carefully adjusting GIR and frequent monitoring of blood glucose should be the primary goal.
- ✓ If the neonate has signs of dehydration or in shock treat accordingly
- ✓ Decrease glucose infusion by 2mg/kg/min (30ml/kg/day) every 6 hrs and stop gradually.
- ✓ Look for and treat underlying causes.
- ✓ Exogenous insulin therapy has been used when glucose values exceed 250 mg/dL despite efforts to lowerthe amount of glucose delivered.
- ✓ Insulin infusion: The standard dilution is 15 units regular human insulin (0.15 mL) added to 29.85 mL normal saline for aconcentration of 0.5 units/mL.

i. Bolus insulin infusion

- Dose 0.05 to 0.1 units/kg every 4 to 6 hours as needed (PRN)
- Infuse over 15 minutes via syringe pump
- Monitor glucose every 30 minutes to 1 hour
- If glucose remains >200 mg/dL after three doses, consider continuous infusion of insulin.

ii. Continuous insulin infusion

- Rate of infusion is 0.05 to 0.1 units/kg/hour (usual starting dose is 0.05 units/kg/hour).
- Check glucose levels every 30 minutes until stable to adjust the infusion rate.
- If glucose remains >180 mg/dL, titrate in increments of 0.01 unit/kg/hour.
- If hypoglycemia occurs, discontinue insulin infusion and administer IV bolus of D10W at 2 mL/kg × 1 dose.

Meconium Aspiration Syndrome (MAS)

I. BACKGROUND

MAS: is a syndrome that includes:

- ✓ Chemical pneumonitis,
- ✓ Mechanically obstructs the small airways, causes atelectasis, and
- ✓ A “ball-valve” effect with resultant air trapping and possible air leak.
- ✓ Persistent pulmonary hypertension of the newborn (PPHN)
- ✓ Aspirated meconium also inhibits surfactant function.

A. Cause. Acute or chronic hypoxia and/or infection can result in the passage of meconium *in utero*. In this setting, gasping by the fetus or newly born infant can cause aspiration of amniotic fluid contaminated by meconium. Meconium aspiration before or during birth can obstruct airways, interfere with gas exchange, and cause severe respiratory distress.

B. Incidence. Meconium-stained amniotic fluid (MSAF) complicates approximately 10% to 15% of deliveries. The incidence of MSAF in preterm infants is very low. Most babies with MSAF are 37 weeks or older, and many meconium-stained infants are postmature and small for gestational age. Approximately 3% to 4% of neonates born through MSAF develop meconium aspiration syndrome (MAS), and approximately 30% to 50% of these infants require continuous positive airway pressure (CPAP) or mechanical ventilation and 3-5% die.

II. Pathophysiology. Meconium is a sterile, thick, black-green odorless material that results from the accumulation of debris in the fetal intestine starting in the third month of gestation.

The components of meconium include water (72% to 80%), desquamated cells from the intestine and skin, gastrointestinal mucin, lanugo hair, fatty material from the vernix caseosa, amniotic fluid, intestinal secretions, blood group-specific glycoproteins, bile, and enzymes including phospholipase A2.

A. Passage of meconium *in utero*. MSAF occurs more commonly in term or postterm pregnancies and rarely prior to 34 weeks' gestation.

MSAF may result from:

- ✓ A postterm fetus with rising motilin levels and normal gastrointestinal function,
- ✓ Vagal stimulation produced by cord or head compression, or
- ✓ *In utero* fetal stress.

Amniotic fluid that is:

- ✓ Thinly stained is described as *watery*.
- ✓ Moderately stained fluid is opaque without particles, and
- ✓ Fluid with thick meconium with particles is sometimes called *pea soup*.

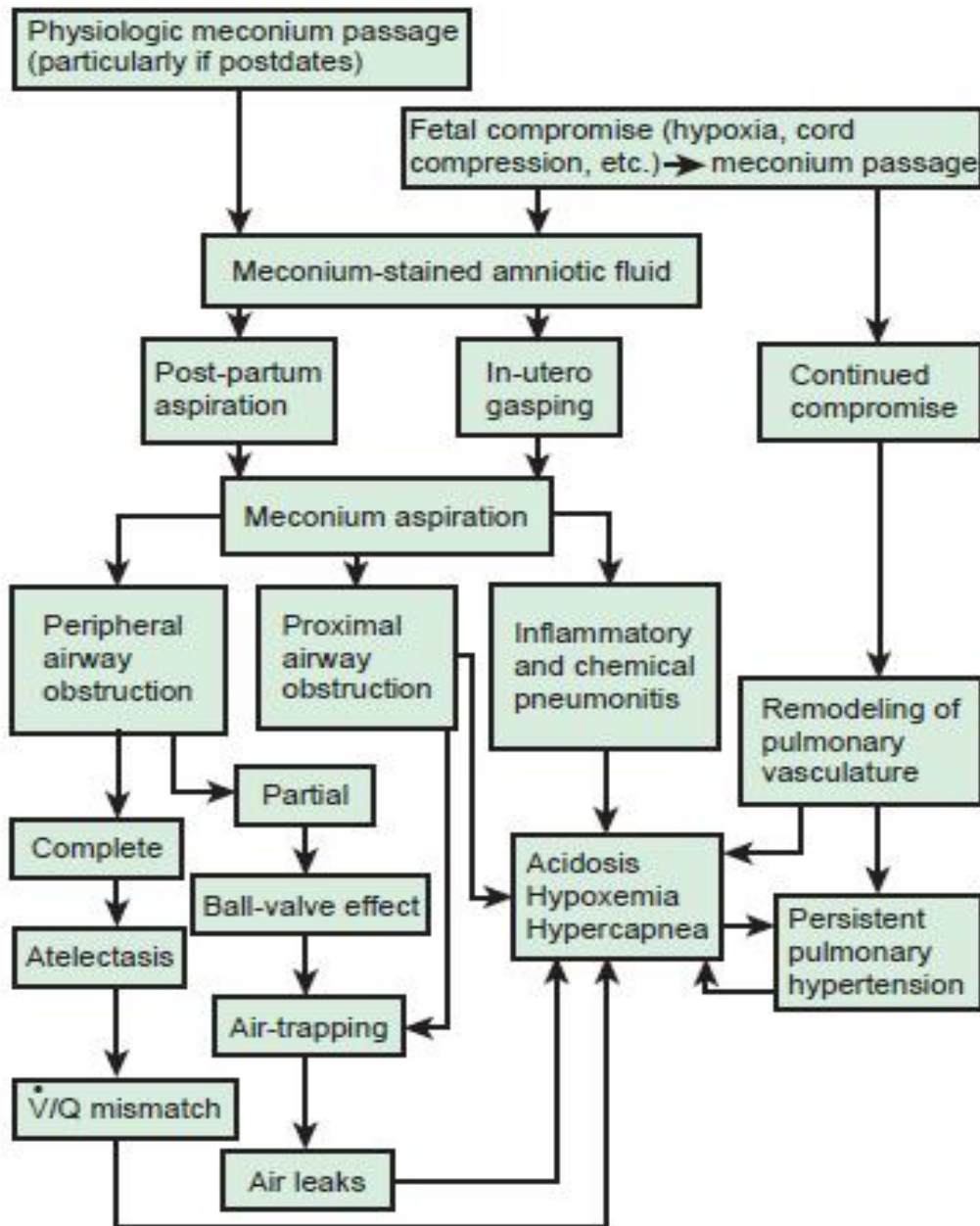
B. Aspiration of meconium. In the presence of fetal stress, gasping by the fetus can result in aspiration of meconium before, during, or immediately following delivery. Severe MAS appears to be caused by pathologic intrauterine processes, primarily chronic hypoxia, acidosis, and infection.

C. **Severity.** MAS is considered:

- ✓ **Mild** in infants requiring <40% oxygen for <48 hours,
- ✓ **Moderate** in infants requiring >40% oxygen for >48 hours without air leak,
- ✓ **Severe** in infants who require assisted ventilation for >48 hours and is often associated with PPHN.

Pathophysiology of meconium passage and the meconium aspiration syndrome

(From: Nelson Textbook of Pediatrics.)



Downe-Vidyasagar Score for grading respiratory distress in a term baby (From: AIIMS Protocols in Neonatology, Section 4, Respiratory system, 2019.)

SCORE	Respiratory Rate	Retractions	Cyanosis	Grunt		Aus
0	<60	Nil	Absent	Absent		Good
1	60 – 80	Mild	Present on room air	Audible on stethoscope		Breathless or difficult
2	>80 or apnea	Moderate to Severe	Present on FiO2 40%	Audible without the stethoscope		Breathless and audible

Grades of severity according to the total score: 1 – 3 - mild; 4 – 6 - moderate, >6 severe
Monitor closely for deterioration in clinical status or drop in SaO₂ -suggestive of air leak Syndrome, lung collapse, PPHN.

Risk factors

- ✓ Prolonged labor
- ✓ Post maturity
- ✓ Maternal illness, diabetes, hypertension
- ✓ Umbilical cord complications
- ✓ Intrauterine growth retardation

Clinical Features

- ✓ Meconium staining, nails, skin, umbilical cord
- ✓ Increased AP diameter of the chest
- ✓ Tachypnea, retractions, grunting, and cyanosis
- ✓ Pneumothorax, or pneumomediastinum, or both.

Meconium aspiration syndrome is defined as at least 2 of the following 3 Criteria:

- ✓ Meconium stained amniotic fluid
- ✓ Respiratory distress
- ✓ CXR findings of MAS

MAS infants may be hypoxic but appear well. Always check the saturations.
The condition usually improves within 72 hr, but when its course requires assisted ventilation, it may be severe with a high risk for mortality.

Investigations if symptomatic

- ❖ Cord ABG
- ❖ ABG on admission (if cord ABG not collected or clinically indicated)
- ❖ CXR if mod -severe RD or mild RD persisting > 2 hours. The classic CXR findings are:
 - Diffuse, asymmetric patchy infiltrates,
 - Areas of consolidation, often worse on the right,
 - Coarse streaking of both lung fields
 - Hyperinflation signs
 - Air leak signs.
- ❖ Asphyxia workup
- ❖ Echocardiography

III. Prevention of MAS:

A. Prevention of passage of meconium *in utero*. Mothers at risk for uteroplacental insufficiency and thus MSAF include those with preeclampsia or increased blood pressure, chronic respiratory or cardiovascular disease, poor intrauterine fetal growth, postterm pregnancy, and heavy smokers. These women should be carefully monitored during pregnancy.

B. Timing and mode of delivery. In pregnancies that continue past the due date, induction as early as 41 weeks may help prevent MAS by avoiding passage of meconium. Delivery mode does not appear to significantly impact the risk of aspiration.

IV. Management of neonate with meconium aspiration syndrome

Assess the gestational age of the baby. A preterm < 34 weeks who passes meconium in utero has either infection or is severely hypoxic. Look for the signs of IUGR. Note whether umbilical cord, nails and skin are stained with meconium.

- Staining of cord – 1-3 hours
- Staining of nails – 4-6 hours
- Staining of skin – 12-24 hours

Infants born with MSAF should be observed closely for respiratory distress suggestive of MAS.

Monitoring of oxygen saturation during this period aids assessment of the severity of the infant's condition and avoids hypoxemia.

1. Routine suctioning or gastric lavage with normal saline has not been found to be beneficial.

- ✓ Oropharyngeal and nasopharyngeal suctioning on the perineum and routine tracheal intubation and aspiration of meconium in vigorous infants are not effective in preventing MAS. Although intubation and endotracheal suctioning for meconium in nonvigorous infants was previously recommended, evidence to support this intervention in improving respiratory and neurodevelopmental outcomes is insufficient, and this procedure is no longer recommended.
- ✓ Current recommendation is interventions to support ventilation and oxygenation as needed, which may include intubation and suction if the airway is obstructed.
- ✓ Endotracheal intubation and suctioning is indicated for depressed infants (those with hypotonia, bradycardia, or apnea). Those who fail to breathe spontaneously should not be stimulated until the airway is cleared to prevent aspiration during the first breath.
- ✓ If an infant does not improve with intubation and positive pressure ventilation, the trachea may be obstructed by thick secretions, including meconium. The trachea should be suctioned using a suction catheter inserted through the endotracheal tube or directly suctioned through the tube using a meconium aspirator attached

to a suction source. The level of suction should be a pressure of 80 to 100 mm Hg.

2. Prevent hypothermia, hypoxia, hypercarbia, hypoglycemia and acidosis. This could result in PPHN.
3. Circulatory support with normal saline or packed red blood cells (maintain a hemoglobin concentration above 15 g (hematocrit above 40%). Administer IV fluids as required if ventilated or severe asphyxia. For infants with mild RD, give gavage feeds.
4. Infants may also require specific therapy for hypotension and poor cardiac output, including cardiotonic medications such as dopamine.
5. Renal function should be continuously monitored.
6. We avoid chest physiotherapy because of the potential adverse effect of exacerbating PPHN.
7. Airway and oral suctioning may be required to facilitate airway clearance, but potential benefits must be balanced against the risk of hypoxic episodes and subsequent worsening of PPHN.
8. **Oxygen therapy:** It is crucial to provide sufficient oxygen because repeated hypoxic insults may result in ongoing pulmonary vasoconstriction and contribute to the development of PPHN.
9. **Treat the complications** (pneumothorax, PPHN and pneumonia, respiratory failure).

10. Assisted ventilation in MAS

Indications for ventilation

- ✓ RDscore > 6
- ✓ ABG: $\text{paO}_2 < 50$ on 0.8 FiO_2 or $\text{paCO}_2 > 60$ or $\text{pH} < 7.25$

❖ **CPAP.** If FiO_2 requirements exceed 0.40, a trial of CPAP may be considered. **CPAP** can be tried especially in low volume atelectatic lungs. but the large infant may not tolerate it. Do not use CPAP > 6 cm H₂O, Do not give CPAP if air leak, hyperinflated lungs, or high PaCO₂

- ❖ **Mechanical ventilation.** Infants with severe disease may have substantial gas exchange abnormalities. Mechanical ventilation is indicated for excessive carbon dioxide retention ($\text{PaCO}_2 > 60$ mm Hg) or for persistent hypoxemia ($\text{PaO}_2 < 50$ mm Hg).

Mechanical Ventilation: Start with patient triggered ventilation. HFV as indicated.

- PIP - 18-20 cm H₂O
- Low to moderate PEEP – 3-4 cm H₂O
- $\text{Ti} = 0.3\text{--}0.5$ sec. Adequate $\text{Te} = 0.5\text{--}0.7$ sec.
- Relatively rapid rate (40-60/min). If PS available, use rates < 30 /min and PS to support. set Vt
- If gas trapping occurs, increase Te to 0.7-1.0 sec and decrease PEEP to 3-4 cm H₂O. The findings on the CXR can guide the ventilatory settings. If it shows low volume, higher PEEP can be used. If it shows hyperinflation avoid auto PEEP. Give adequate time for expiration.

- ❖ **Extracorporeal membrane oxygenation** for infants with refractory respiratory failure.

11. Medications

- ❖ **Antibiotics.** Differentiating between bacterial pneumonia and meconium aspiration by clinical course and chest x-ray findings may be difficult. Although few infants with MAS have documented infections, routine antibiotics in MAS are not useful. Administer antibiotics if risk for sepsis and symptomatic or if septic screen at 6 -12 hours is suggestive. MAS in preterm neonates may be a sign of infection. Blood cultures should be obtained to identify bacterial disease, if present, and to determine length of antibiotic course.
- ❖ **Surfactant.** Endogenous surfactant activity may be inhibited by meconium and is a secondary cause of surfactant deficiency. Several studies suggest that the enzymatic and sterol components of meconium disrupt the surfactant phospholipids and limit the ability for surfactant to lower surface tension.

Surfactant replacement in MAS improves oxygenation, reduces the need for ECMO.

Consider surfactant if the baby is requiring high ventilatory parameters – $\text{FiO}_2 > 0.6$ or $\text{PIP} > 20 \text{ cm H}_2\text{O}$, especially if the CXR shows low volume lungs with atelectasis. Give 4 ml/kg surfactant.

- ❖ **Corticosteroids.** We do not routinely recommend the use of corticosteroids in MAS. This approach has been proposed to reduce meconium-induced inflammation and to minimize prostaglandin-mediated pulmonary vasoconstriction.
- ❖ **Sedatives.** The use of sedation and muscle relaxation may be warranted in infants who require mechanical ventilation.
- ❖ **Inhaled nitric oxide (iNO):** In severely ill infants with MAS and PPHN, inhaled nitric oxide (iNO) reduces the need for ECMO support by approximately 40%. **Nitric Oxide** is an endothelium-derived signaling molecule that relaxes vascular smooth muscle and can be delivered to the lung by inhalation.
- ❖ **Antioxidants.** The use of antioxidants, in particular *N*-acetylcysteine, to mitigate reactive oxygen species (ROS)-induced lung injury in MAS has been recently investigated in animal models.

Targets in MAS

- ✓ Target SaO_2 92-98 %.
- ✓ Target ABG: (without PPHN)
 - PH 7.3-7.4,
 - PaO_2 60-80 mm Hg,
 - PaCO_2 35-45 mm Hg

G. Complications

1. Air leak. Pneumothorax or pneumomediastinum occurs in approximately 15% to 33% of patients with MAS. Air leaks occur more frequently with mechanical ventilation, especially in the setting of air trapping.

2. PPHN: Aspirated meconium leads to vasospasm, hypertrophy of the pulmonary arterial musculature, and pulmonary hypertension that lead to extrapulmonary right-to-left shunting through the ductus arteriosus or the foramen ovale, resulting in worsened ventilation-perfusion (V/Q) mismatch and severe arterial hypoxemia. Approximately one-third of infants with MAS develop persistent pulmonary hypertension of the newborn (PPHN), which contributes to the mortality associated with this syndrome.

Depending on the extent of hypoxemia, echocardiography should be performed to ascertain the degree to which the right-to-left shunting is contributing to the infant's overall hypoxemia and to exclude congenital heart disease as the etiology.

3. Pulmonary sequelae. Approximately 5% of survivors require supplemental oxygen at 1 month, and a substantial proportion may have abnormal pulmonary function, including increased functional residual capacity, airway reactivity with symptomatic cough, wheezing, and higher incidence of pneumonia and persistent hyperinflation for up to 5-10 yr.

Necrotizing Enterocolitis

Definition. NEC is an acute inflammatory injury of the distal small and often proximal large intestine. The colon appears to be the most commonly affected site. The disease is characterized by various degrees of mucosal or transmural necrosis of the intestine. It is associated with inflammation, invasion of enteric gas-forming organisms, and dissection of gas into the muscularis and portal venous system.

The triad of intestinal ischemia (injury), enteral nutrition (metabolic substrate), and bacterial translocation has classically been linked to NEC.

Epidemiology: An estimated 0.3 to 2.4 cases occur in every 1,000 live births. In most centers, NEC occurs between 1% and 5% of all NICU admissions and 5% to 10% of very low birth weight (VLBW) infants. Approximately 10% of infants with NEC are **term**. The postnatal age at onset is inversely related to birth weight and gestational age, with a mean age at onset of 12 days.

Commonly the onset of NEC is related with gestational age and is as follows:

- ✓ In <31 weeks onset is 23rd day
- ✓ >31 weeks – 11 days
- ✓ Full term – 3rd day

Pathogenesis:

The following Risk factors have been implicated in the pathogenesis of NEC:

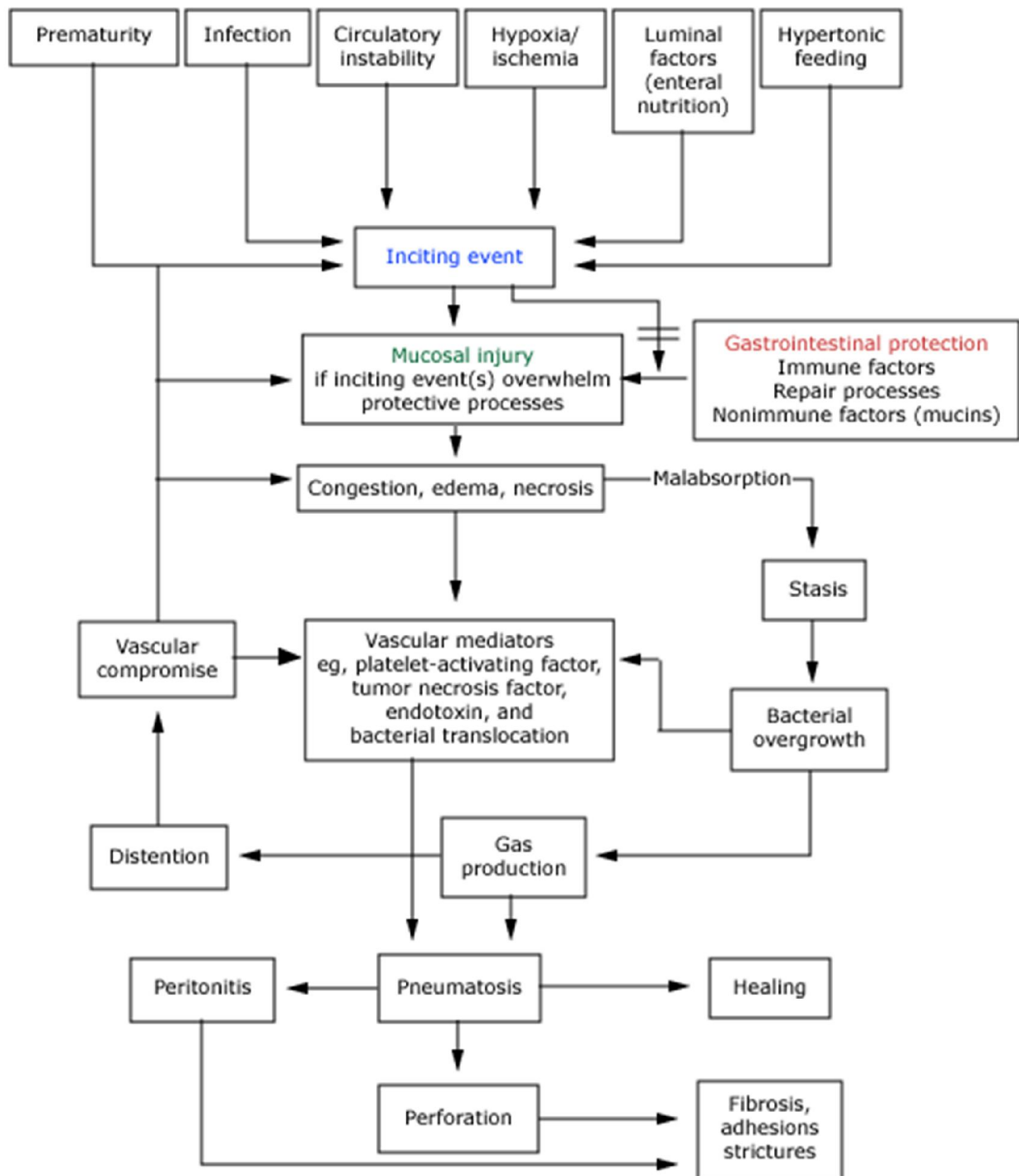
- ✓ **Prematurity:** Bacterial overgrowth and increased intestinal permeability are factors that contribute to greater bacterial translocation from the intestinal lumen into the intestinal tissue leading to an inflammatory response with activation of cytokines
- ✓ **Microbial bowel overgrowth:** because NEC does not occur in utero when the gut is sterile.
- ✓ **Enteral feeding:** serve as substrates for bacterial proliferation in the gut.
- ✓ **Impaired mucosal defense:** immunologic and GI immaturity results in altered host resistance.
- ✓ **Circulatory instability of the intestinal tract:** Ischemic insult to the GI tract has been proposed as a major contributor to NEC. Circulatory events that have been implicated in the development of NEC include perinatal asphyxia, recurrent apnea, hypoxia from severe respiratory distress syndrome, hypotension, congenital heart disease, patent ductus arteriosus, heart failure, umbilical arterial catheterization, anemia, polycythemia.
- ✓ **Medications** that cause intestinal mucosal injury or enhance microbial overgrowth: The administration of hyperosmolar medications and/or formulas can cause mucosal injury and may result in NEC. Oral medications such as theophylline, multivitamins, or phenobarbital contain hypertonic additives that might irritate the intestinal mucosa. Histamine type 2 receptor (H2) antagonists,

such as cimetidine, ranitidine, and famotidine, are associated with higher rates of NEC.

Anticipate NEC in the following conditions

- Prematurity < 32 weeks,
- Hypoxia – perinatal asphyxia, in utero hypoxia (SGA with reverse end diastolic flow), postnatal recurrent apnea and hypoxia, PDA (ductal steal)
- Rapid increase in feeds (>20-35 ml/kg/d), Formula feeds (6 times more common than if only breast milk fed)
- Sepsis
- Polycythemia
- Umbilical catheterization
- Inappropriate weight gain

Contributing mechanisms to the pathogenesis of NEC (from: UPTODATE 2018)



DIAGNOSIS;

- ❖ **Clinical characteristics:** The clinical features of NEC can be divided into systemic and abdominal signs.

Systemic signs. Respiratory distress, apnea and/or bradycardia, lethargy, temperature instability, irritability, poor feeding, hypotension (shock), decreased peripheral perfusion, acidosis, oliguria, bleeding diathesis.

Abdominal (enteric) signs. Abdominal distension or tenderness, feeding intolerance, delayed gastric emptying, gastric aspirates (feeding residuals), vomiting (of bile, blood, or both), ileus (decreased or absent bowel sounds), hematochezia (grossly bloody stools), abdominal wall erythema or induration, persistent localized abdominal mass, or ascites.

- ❖ **Laboratory Investigations:**

Imaging studies:

- ✓ The abdominal **radiograph** will often reveal an abnormal gas pattern consistent with ileus. Both anteroposterior (AP) and cross-table lateral or left lateral decubitus views should be included. These films may reveal bowel wall edema, a fixed-position loop on serial studies, the appearance of a mass, pneumatosis intestinalis (the radiologic hallmark used to confirm the diagnosis), gasless abdomen indicating ascites, portal or hepatic venous air, pneumobilia, or pneumoperitoneum with the appearance of gas under the diaphragm. pneumatosis intestinalis. Portal venous gas, perforation. Gas under diaphragm, rigler's sign, cupola sign, football sign, ligamentum teres sign are signs of PERFORATION. (pneumatosis intestinalis. Portal venous gas, perforation)
- ✓ Abdominal **ultrasound** can be a more sensitive method to detect intramural air or pneumatosis, portal venous gas and pneumoperitoneum
- ✓ Doppler studies can confirm bowel necrosis by absent blood flow.

Blood and serum studies:

- ✓ Thrombocytopenia, persistent metabolic acidosis, and severe refractory hyponatremia constitute the most common triad of signs and help to confirm the diagnosis; (CBC, ABG, Serum electrolyte)
- ✓ Septic work-up: Blood cultures, CRP, CBC...

Analysis of stool for blood has been used to detect infants with NEC based on changes in intestinal integrity.

- ❖ **Bell staging criteria** with the Walsh and Kleigman modification allow for uniformity of diagnosis across centers.

Differential diagnosis:

- ✓ **Pneumonia and sepsis:** are common and frequently associated with intestinal ileus.
- ✓ **Surgical abdominal catastrophes** include malrotation with obstruction (complete or intermittent), malrotation with midgut volvulus, intussusception, ulcer, gastric perforation, and mesenteric vessel thrombosis.
- ✓ **Infectious enterocolitis** is rare in this population but must be considered if diarrhea is present.

- ✓ **Severe allergic colitis** can present with abdominal distension and bloody stools. Usually, these infants are well appearing and have normal abdominal radiographs and laboratory studies.
- ✓ **Feeding intolerance.**

Spontaneous intestinal Perforation (From: St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)

Characteristics	Spontaneous Intestinal Perforation	NEC
Age of onset	0-14 days	2-6 weeks
Enteral feeds	No	Yes
Pneumatosis intestinalis	No	Yes
Histology of villous necrosis	No	Yes
Mortality	Lower	Higher

Management of NEC:

- ✓ **Start with ABC of life.**
- ✓ **Manage based on the Bell staging.**

Modified Bell staging criteria for necrotizing enterocolitis (NEC) (from: Neu J. Necrotizing enterocolitis: the search for a unifying pathogenic theory leading to prevention. *Pediatr Clin North Am* 1996; 43:409. Graphic 54166 Version 5.0.)

Stage	Classification of NEC	Systemic signs	Abdominal signs	Radiographic signs
IA	Suspected	Temperature instability, apnea, bradycardia, lethargy	Gastric retention, abdominal distention, emesis, heme-positive stool	Normal or mild intestinal dilation, mild ileus
IB	Suspected	Same as above	Grossly bloody stool	Same as above
IIA	Definite, mildly ill	Same as above	Same as above, plus absent bowel sounds with or without abdominal tenderness	Intestinal dilation, ileus, pneumatosis intestinalis
IIB	Definite, moderately ill	Same as above, plus mild metabolic acidosis and thrombocytopenia	Same as above, plus absent bowel sounds, definite tenderness, with or without abdominal cellulitis or right lower quadrant mass	Same as IIA, plus ascites
IIIA	Advanced, severely ill, intact bowel	Same as IIB, plus hypotension, bradycardia, severe apnea, combined respiratory and metabolic acidosis, DIC, and neutropenia	Same as above, plus signs of peritonitis, marked tenderness, and abdominal distention	Same as IIA, plus ascites
IIIB	Advanced, severely ill, perforated bowel	Same as IIIA	Same as IIIA	Same as above, plus pneumoperitoneum

Management of Necrotizing Enterocolitis: (From: St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)

Bell Staging Criteria	Diagnosis	Management
Stage I (suspected)	Clinical signs Nondiagnostic radiograph	NPO with IV fluids Gastric decompression Ampicillin and gentamicin × 48 hours
Stage II (definite)	Clinical signs Pneumatosis intestinalis and/or portal venous gas on radiograph	NPO with parenteral nutrition Gastric decompression Ampicillin, gentamicin, and metronidazole × 7- days for stage IIA and 10-14 days for stage IIB Surgical consultation
Stage III (advanced)	Clinical signs Critically ill Pneumatosis intestinalis or pneumoperitoneum on radiograph	NPO with parenteral nutrition Nasogastric drainage, fluid resuscitation, inotropic support, ventilator therapy, paracentesis Gastric decompression Ampicillin, gentamicin, and metronidazole 14 days, Surgical consultation

NPO, nothing by mouth; IV, intravenous; CBC, complete blood count; KUB, kidney, urethra, bladder xray; AP, anteroposterior; PRN, as needed.

- ✓ **Surgical intervention:** At least 30% of NEC cases result in surgical resection of affected tissue.

The specific type of surgical treatment includes:

- peritoneal drainage,
- laparotomy with diverting ostomy alone,
- laparotomy with intestinal resection and primary anastomosis,
- “clip and drop,” or stoma creation,

In case of “NEC totalis” (bowel necrosis from duodenum to rectum), mortality is almost certain and resection is typically not attempted.

In ELBW infants (<1,000 g) and extremely unstable infants, peritoneal drainage under local anesthesia may be a management option as an alternative to exploratory laparotomy.

The incidence of recurrent NEC is 4% and appears to be independent of type of management.

Indications for surgery include:

- GI perforation is probably the only absolute indication for surgical intervention. Perforation occurs in 20% to 30% of patients, usually 12 to 48 hours after the onset of NEC. evidence of perforation on abdominal X-ray (pneumoperitoneum) or positive result of abdominal paracentesis.

- Failure of medical management, a single fixed bowel loop on radiographs, abdominal wall erythema, and a palpable mass are relative indications for exploratory laparotomy.
- ✓ Ideally, surgery should be performed after intestinal necrosis develops but before perforation and peritonitis occurs.

Prevention of NEC:

- ✓ **Exclusive feeding:** Use of human milk
- ✓ **Probiotics** in infants < 32 weeks, < 1200g: **Enterally fed probiotics** (live nonpathogenic microbial preparations that colonize the intestine) fed to preterm infants may help to normalize intestinal microflora colonization. (e.g., Lactobacillus.
- ✓ Hand hygiene
- ✓ KMC
- ✓ Antenatal Steroid for **Induction of GI maturation**
- ✓ **Nutritional supplements** (e.g., polyunsaturated fatty acids [PUFA],
- ✓ **immune modulators** such as immunoglobulins, **Lactoferrin...**

Complications:

Acute complications:

- ✓ Infectious complications – Sepsis, meningitis, peritonitis, and abscess formation
- ✓ Disseminated intravascular coagulation,
- ✓ Respiratory and cardiovascular complications – Hypotension, shock, and respiratory failure
- ✓ Metabolic complications – Hypoglycemia and metabolic acidosis

Late complications:

- ✓ intestinal narrowing (ie, stricture formation)
- ✓ short bowel syndrome.

Prognosis:

Medical management fails in approximately 20-40% of patients with pneumatosis intestinalis at diagnosis; of these, 10-30% die.

Mortality can approach 100% in case of NEC totalis. Overall, NEC is responsible for 12% of deaths in extreme premature infants <27 weeks of gestational age.

Neonatal Sepsis

Introduction.

Definitions of Systemic Inflammatory Respiratory Response Syndrome (SIRS) and sepsis in pediatric patients (From Nelson Textbook of Pediatrics 21st Ed.)

SIRS: the systemic inflammatory response to a variety of clinical insults, manifested by 2 or more of the following conditions 1 of which must be abnormal temperature or leukocyte count:

Temperature instability $<36^{\circ}\text{C}$ or $>38.5^{\circ}\text{C}$ (38°C in preterm)

Respiratory dysfunction:

- ✓ Tachypnea >2 SD above the mean for age
- ✓ Hypoxemia ($\text{PaO}_2 < 70$ mm Hg on room air)

Cardiac dysfunction:

- ✓ Tachycardia >2 SD above the mean for age
- ✓ Delayed capillary refill >3 sec
- ✓ Hypotension >2 SD below the mean for age

Perfusion abnormalities:

- ✓ Oliguria (urine output <0.5 mL/kg/hr)
- ✓ Lactic acidosis (elevated plasma lactate and/or arterial pH <7.25)
- ✓ Altered mental status

Leukocyte count increased or decreased for age (not secondary to chemotherapy-induced leukopenia) or $>10\%$ immature neutrophils (In preterm - 20% OR CRP > 10 mg/dl)

Sepsis: SIRS in the presence of or as a result of suspected or proven infection.

Severe sepsis: Sepsis plus 1 of the following: cardiovascular organ dysfunction OR ARDS

OR 2 or more other organ dysfunctions (See sepsis protocol annexure for definition of organ

dysfunction as defined by International pediatric sepsis consensus)

Definite sepsis- bacteremia (culture positive) with clinical features

Probable sepsis- clinical suspicion with positive screen

Suspect sepsis- clinical suspicion only

Neonatal sepsis is classified according to the infant's age at the onset of symptoms:

- ✓ **Early-onset sepsis** is defined as the onset of symptoms before 7 days of age, although some experts limit the definition to infections occurring within the first 72 hours of life.
- ✓ **Late-onset sepsis** is defined as the onset of symptoms at ≥ 7 days of age. Similarly, to early-onset sepsis, there is variability in its definition, ranging from an onset at >72 hours of life to ≥ 7 days of age.

Early-onset sepsis: Early-onset infection is usually due to vertical transmission by ascending contaminated amniotic fluid or during vaginal delivery from bacteria colonizing or infecting the mother's lower genital tract.

Late-onset sepsis: Late-onset infections can be acquired by the two following mechanisms:

- ✓ Maternal vertical transmission, resulting in initial neonatal colonization that evolves into later infection.
- ✓ Horizontal transmission from direct contact with care providers or environmental sources (ie, health care-associated infections). Disruption of the intact skin or mucosa, which can be due to invasive procedures (eg, intravascular catheter), increases the risk of late-onset infection.

Very-late-onset infections (onset after age 1 mo) may also occur, particularly in very-low-birth weight (VLBW) preterm infants or term infants requiring prolonged neonatal intensive care.

Health care-associated infection (HCAI), also referred to as "**nosocomial**" or "**hospital**" **acquired infection**, is an infection occurring in a patient during the process of care in a hospital or other health care facility which was not present or incubating at the time of admission. Any new bloodculture positive sepsis after 48 hours of hospital stay is considered nosocomial sepsis.

Risk factors for Early Onset Sepsis are as follows:

- ✓ Intra-partum per vaginal examinations ≥ 3 , (or all per vaginal examinations > 6)
- ✓ **Chorioamnionitis** was thought to result from infection of the amniotic fluid but is now better defined by the term intrauterine inflammation or infection at birth (Triple I). This is defined by fetal tachycardia, maternal leukocytosis ($>15,000$ cells in the absence of corticosteroids), purulent fluid from the cervical os, biochemical or microbiologic amniotic fluid changes consistent with infection, and fever ($\geq 39.0^{\circ}\text{C}/10.2^{\circ}\text{F}$).

Clinical chorioamnionitis: maternal fever of greater than 38°C (100.4°F) and at least two of the following criteria:

- Maternal leukocytosis (greater than 15 000 cells/ mm^3),
- Maternal tachycardia (greater than 100 beats/minute),
- Fetal tachycardia (greater than 160 beats/minute),
- Uterine tenderness,
- Foul odor of the amniotic fluid)

Classification of Triple I and Isolated Maternal Fever (From: Nelson Textbook of Pediatrics.)

TERMINOLOGY FEATURES	
Isolated maternal fever	Maternal oral temperature $\geq 39^{\circ}\text{C}$ is considered a “documented fever.” If the oral temperature is $\geq 38^{\circ}\text{C}$ but $\leq 39^{\circ}\text{C}$, repeat the measurement in 30 min. If the repeat value is $\geq 38^{\circ}\text{C}$, it is considered a “documented fever.”
Suspected Triple I	Fever without a clear source with <i>any</i> of the following: 1. Baseline fetal tachycardia (>160 beats/min for 10 min) 2. Maternal WBC $>15,000/\text{mm}^3$ 3. Purulent fluid from the cervical os
Confirmed Triple I	All the above (from suspected Triple I) with <i>any</i> of the following: 1. Amniocentesis-proven infection through positive Gram stain 2. Low glucose of amniotic fluid or positive amniotic fluid culture 3. Placental pathology consistent with infection

- ✓ Very low birth weight (<1500 grams)
- ✓ Prematurity (<37 weeks ‘gestation)
- ✓ Febrile illness in the mother with evidence of bacterial infection / UTI within 2 weeks prior to delivery.
- ✓ Perinatal asphyxia (Apgar score <4 at 1 minute)
- ✓ Prolonged Labor >24 hours.
- ✓ PPRM / Spontaneous preterm
- ✓ Prolonged rupture of membranes (ROM) (>18 hours).
- ✓ Congenital immune defects, Inadequate transplacental antibody to infecting strain, Asplenia, Decreased local and/or Systemic immune function.
- ✓ Malformations leading to high inocula of bacteria (e.g., obstructive uropathy).

Initial Signs and Symptoms of Infection in Newborn Infants:

General

- ✓ Fever, temperature instability
- ✓ “Not doing well”, Poor feeding
- ✓ Edema

Gastrointestinal System

- ✓ Vomiting, Diarrhea
- ✓ Hepatomegaly, Abdominal distention

Respiratory System

- ✓ Apnea, dyspnea
- ✓ Tachypnea, retractions, Flaring, grunting, Cyanosis

Renal System

Oliguria

Cardiovascular System

- ✓ Pallor; mottling; cold, clammy skin
- ✓ Tachycardia, Hypotension, Bradycardia

Central Nervous System

- ✓ Irritability, High-pitched cry, lethargy
- ✓ Tremors, seizures, Hyporeflexia, hypotonia, Abnormal Moro reflex
- ✓ Irregular respirations
- ✓ Full fontanel,

Hematologic System

- ✓ Jaundice
- ✓ Splenomegaly
- ✓ Pallor, Petechiae, purpura, Bleeding

IMCI and WHO Criteria for the Diagnosis of Sepsis in young infants (from WHO: *Pocket book of hospital care for children: guidelines for the management of common childhood illnesses*, ed 2, Geneva, 2013, WHO, pp 45–69.)

- ✓ Neurologic: convulsions, drowsy or unconscious, decreased activity, bulging fontanel
- ✓ Respiratory: respiratory rate >60 breaths/min, grunting, severe chest indrawing, central cyanosis
- ✓ Cardiac: poor perfusion, rapid and weak pulse
- ✓ Gastrointestinal: jaundice, poor feeding, abdominal distention
- ✓ Dermatologic: skin pustules, periumbilical erythema or purulence
- ✓ Musculoskeletal: edema or erythema overlying bones or joints
- ✓ Other: temperature >37.7°C (99.9°F; or feels hot) or <35.5°C (95.9°F; or feels cold)

Evaluation of a Newborn for Infection or Sepsis (From; Nelson Textbook of Pediatrics 21st Ed.)

History (Specific Risk Factors)

- ✓ Maternal infection during gestation or at parturition (type and duration of antimicrobial therapy):
 - Urinary tract infection
 - Chorioamnionitis
 - Maternal colonization with group B streptococci, *Neisseria gonorrhoeae*, herpes simplex
 - Low gestational age/birthweight, Multiple birth
 - Duration of membrane rupture
 - Complicated delivery
 - Fetal tachycardia (distress)
 - Age at onset (in utero, birth, early postnatal, late)
 - Location at onset (hospital, community)
- ✓ Medical intervention:
 - Vascular access
 - Endotracheal intubation
 - Parenteral nutrition
- ✓ Surgery

Evidence of other Diseases

- ✓ Congenital malformations (heart disease, neural tube defect)
- ✓ Respiratory tract disease (respiratory distress syndrome, aspiration)
- ✓ Necrotizing enterocolitis
- ✓ Metabolic disease (e.g., galactosemia)

Evidence of focal or systemic Disease

- ✓ General appearance, neurologic status
- ✓ Abnormal vital signs
- ✓ Organ system disease
- ✓ Feeding, stools, urine output, extremity movement

Laboratory Studies:

Evidence of Infection

- ✓ Culture from a normally sterile site (blood, CSF, Urine, other)
- ✓ Demonstration of a microorganism in tissue or fluid
- ✓ Molecular detection (blood, urine, CSF) by specific PCR
- ✓ Maternal or neonatal serology (syphilis, toxoplasmosis)

Evidence of Inflammation

- ✓ Leukocytosis, increased immature/total neutrophil count ratio
- ✓ Acute-phase reactants: C-reactive protein, erythrocyte sedimentation rate, procalcitonin
- ✓ Cytokines: interleukin-6, interleukin-B, tumor necrosis factor
- ✓ Pleocytosis in CSF or synovial or pleural fluid
- ✓ Disseminated intravascular coagulation: fibrin degradation products, D-dimer
- ✓ CXR

Evidence of Multiorgan System Disease

- ✓ Metabolic acidosis: pH, PCO₂
- ✓ Pulmonary function: PO₂, PCO₂

- ✓ RBS
- ✓ Renal function: blood urea nitrogen, creatinine
- ✓ Hepatic injury/function: bilirubin, alanine transaminase, aspartate transaminase, ammonia, prothrombin time,
- ✓ partial thromboplastin time, INR
- ✓ Bone marrow function: neutropenia, anemia, thrombocytopenia

Culture-Based and Non–Culture-Based Diagnostics for Neonatal Sepsis (From: Shane AL, Stoll BJ. Recent developments and current issues in the epidemiology, diagnosis, and management of bacterial and fungal neonatal sepsis, *Am J Perinatol* 30(2):131– 141, 2013.)

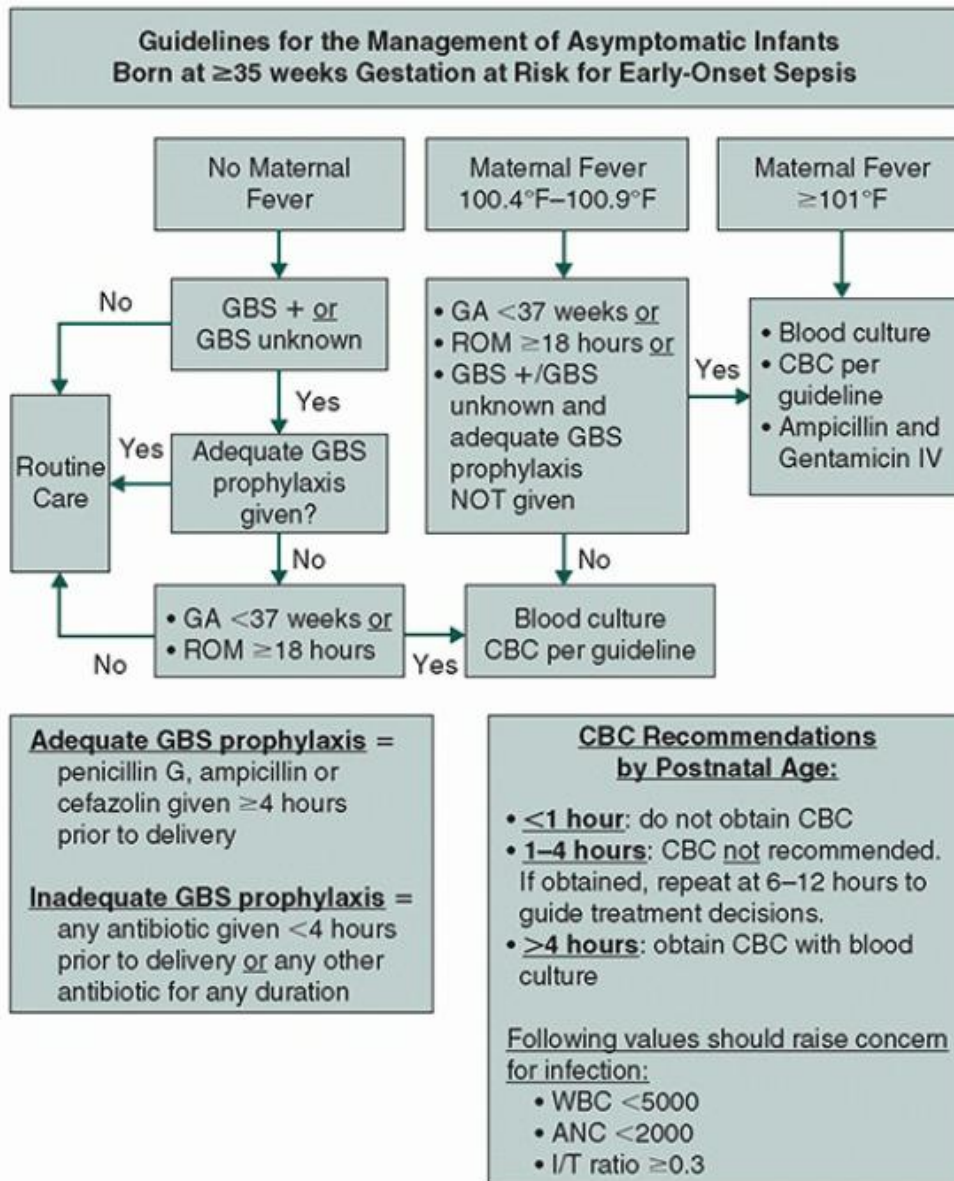
Category Parameter		Optimal Timing, Volume of Specimen, Routine/Investigational	Applicability For Neonatal Sepsis
Culture Based			
Blood	Culture	>1 mL of whole blood, from 2 sites	Gold standard for bacteremia
CSF	Culture	When clinically feasible	Optimize antimicrobial therapy
Urine	Culture	>72 hr of life	Not useful for EOS; potential benefits for LOS
Tracheal aspirate	Culture	Neonates with endotracheal tube in place and signs of progressive respiratory distress	Usually reflects colonization
Non–Culture Based			
Neutrophil indices	Neutropenia Absolute neutrophil count Absolute immature neutrophil count	After 12 hr of life Consider GA, delivery mode, altitude, arterial versus venous sampling, time since birth	Neutropenia better predictor for sepsis than leukocytosis
Platelet count	Thrombocytopenia and thrombocytosis	Late findings; slow to respond	Thrombocytopenia associated with fungal infection
CSF cell count	CSF WBC	Uninfected neonates: mean 10 cells/mm ³ ; range up to 20 cells/mm ³	Does not predict culture-proven meningitis
CSF chemistries	CSF protein CSF glucose	Term <100 mg/dL Preterm higher; 70–80% of serum glucose	Elevated in fungal meningitis Low glucose specific for bacterial meningitis
Acute phase reactants	CRP Procalcitonin	8–24 hr after infection 2–12 hr after infection	Good NPV Better sensitivity but less specificity than CRP
Sepsis panels/scores		After 24 hr of life Investigational	Most useful for NPV and discontinuation of antimicrobial therapy

Management and Prevention of Neonatal Sepsis (From: Nelson Textbook of Pediatrics.)

CONDITION	THERAPY	ADDITIONAL CONSIDERATIONS
EMPIRICAL MANAGEMENT		
Early-onset sepsis	Ampicillin + aminoglycoside 10 days for bacteremia; 14 days for GBS and uncomplicated meningitis; extend to 21-28 days for complicated infections	Consider a third-generation cephalosporin (cefotaxime preferred) or carbapenem for meningitis. Tailor therapy to pathogen. Consider discontinuation of therapy if pathogen not isolated.
Late-onset sepsis	Vancomycin + aminoglycoside Duration dependent on pathogen and site	Alternatives to vancomycin may be considered based on local epidemiology and clinical presentation. Aminoglycoside-based regimen preferred to cephalosporin given reduced risk of resistance. Consider cephalosporin if meningitis suspected. Consider a carbapenem if third generation cephalosporin recently received. Consider amphotericin for fungal etiologies. Tailor therapy to pathogen. Consider discontinuation of therapy if pathogen not isolated.
NONANTIMICROBIAL TREATMENT STRATEGIES		
Recombinant G-CSF Recombinant GM-CSF	Enhance neutrophil number and function, but no reduction in infection when administered as prophylaxis or improvement in survival when administered as therapy.	Insufficient evidence to support the clinical use of G-CSF or GM-CSF either as treatment or prophylaxis to prevent systemic infections.
IVIG	Augments antibody-dependent cytotoxicity and improves neutrophilic function, <i>but</i> no evidence that IVIG in suspected or proven sepsis reduces death.	Insufficient evidence from 10 RCTs or quasi-RCTs to support use in neonates with confirmed or suspected sepsis.
Double volume Exchange Transfusion		may be performed in a case of deteriorating sepsis with sclerema, DIC provided the general condition of the baby allows the procedure. Use fresh whole blood compatible with baby and mother.
PREVENTION STRATEGIES		
IAP	Administration of penicillin or ampicillin 4 hr before parturition	Successfully reduces rates of EOS caused by GBS. No effect on LOS GBS
Fluconazole prophylaxis	Administration of weight-based dosing to neonates <1,500 g	Most beneficial in NICUs with high baseline rates of invasive candidiasis

EOS, early-onset sepsis; GBS, group B streptococcus; G-CSF, granulocyte colony-stimulating factor; GM-CSF, granulocyte-macrophage colony stimulating factor; IAP, intrapartum antimicrobial prophylaxis; IVIG, intravenous immune globulin; RCTs, randomized controlled trials.

Algorithm for sepsis evaluations in at-risk asymptomatic infants ≥ 35 weeks' gestation
(From: Eichenwald. E.C., Hansen. AR., Martin. CR., Stark. AR., Cloherty and Starks Manual of Neonatal Care; 8th edition, 2016.)



ADDITIONAL NOTES

1. Chorioamnionitis is an obstetrical clinical diagnosis made on the basis of clinical findings, laboratory data and fever. If obstetrical staff diagnose chorioamnionitis, the infant should be evaluated for sepsis and receive empiric antibiotic treatment.
2. Maternal fever that occurs within one hour of delivery should be treated like intrapartum fever.
3. Women with a previous infant with GBS disease should receive intrapartum GBS prophylaxis.
4. Blood cultures should consist of aerobic and anaerobic bottles with minimum 1 mL blood in each bottle.

Management of Neonatal Sepsis:

Supportive care — Supportive care is initially focused on ensuring:

- ✓ Maintaining adequate oxygenation and perfusion
- ✓ Prevention of hypoglycemia and metabolic acidosis
- ✓ Maintenance of normal fluid and electrolyte status
- ✓ Maintenance of a thermoneutral environment.
- ✓ Early removal of catheters that may be the foci of bacterial infection
- ✓ Sodium bicarbonate for metabolic acidosis
- ✓ Anticonvulsants for seizures.

Severely ill patients may require ventilatory, volume, and/or vasopressor support to maintain adequate oxygenation and perfusion.

Suggested antimicrobial regimens in the management of neonatal sepsis in term and late preterm infants (From: UPTODATE 2018.)

	Antibiotic regimen
Empiric therapy	
Early onset (<7 days)	Ampicillin AND gentamicin
Late onset (≥7 days): Admitted from the community	Ampicillin AND gentamicin
Late onset (≥7 days): Hospitalized since birth	Gentamicin AND vancomycin
Special circumstances:	
Suspected meningitis - early onset	Ampicillin AND gentamicin*
Suspected meningitis - late onset, admitted from the community	Ampicillin, gentamicin, AND cefotaxime [¶]
Suspected meningitis - late onset, hospitalized since birth	Gentamicin, vancomycin, AND cefotaxime [¶]
Suspected pneumonia	Ampicillin AND gentamicin Alternatives: ▪ Ampicillin AND cefotaxime[¶], OR ▪ Vancomycin AND cefotaxime[¶], OR ▪ Vancomycin AND gentamicin
Suspected infection of soft tissues, skin, joints, or bones (<i>S. aureus</i> is a likely pathogen)	Vancomycin or vancomycin AND nafcillin
Suspected intravascular catheter-related infection	Vancomycin AND gentamicin
Suspected infection due to organisms found in the gastrointestinal tract (eg, anaerobic bacteria)	Ampicillin, gentamicin, AND clindamycin Alternatives: ▪ Ampicillin, gentamicin, AND metronidazole OR ▪ Piperacillin-tazobactam AND gentamicin

Pathogen-specific therapy	
Group B <i>Streptococcus</i>	Penicillin G
<i>Escherichia coli</i> : Ampicillin-sensitive	Ampicillin
<i>Escherichia coli</i> : Ampicillin-resistant	Cefotaxime [¶] Alternative: ▪ Meropenem
Multidrug-resistant gram-negative bacilli (including ESBL-producing organisms)	Meropenem
<i>Listeria monocytogenes</i>	Ampicillin AND gentamicin
Methicillin-susceptible <i>S. aureus</i> (MSSA)	Nafcillin OR cefazolin
Methicillin-resistant <i>S. aureus</i> (MRSA)	Vancomycin
Coagulase-negative staphylococci	Vancomycin

ESBL: extended-spectrum beta-lactamase.

* Cefotaxime should be added to the empiric regimen for suspected early-onset meningitis if the CSF Gram stain reveals gram-negative bacilli.

¶ If cefotaxime is not available, alternative agents include ceftazidime, ceftriaxone (not to be used in infants with clinically significant hyperbilirubinemia or those receiving concurrent IV calcium [including parenteral nutrition]), or meropenem. Meropenem is preferred if there is concern for infection due to a multidrug-resistant gram-negative organism.

Ampicillin and gentamicin are preferred over ampicillin and a third-generation cephalosporin (eg, cefotaxime), based upon the following:

- ✓ The regimen of ampicillin and a third-generation cephalosporin is not more effective than the combination of ampicillin and gentamicin.
- ✓ The emergence of cephalosporin-resistant gram-negative organisms (eg, *Enterobacter cloacae*, *Klebsiella*, and *Serratia* species) can occur when cefotaxime is used routinely.
- ✓ Ampicillin and gentamicin are synergistic in treating infections caused by GBS and *Listeria monocytogenes*. Cephalosporins are **not** active against *L. monocytogenes*.
- ✓ Ceftriaxone is highly bound to albumin and appears to displace bilirubin.

Of note, neonates should not receive ceftriaxone intravenously if they also are receiving, or are expected to receive, IV calcium in any form, including parenteral nutrition. The combination of this therapy is associated with adverse cardiopulmonary events.

Because of the emergence of cephalosporin-resistant organisms, especially *Enterobacter*, *Klebsiella*, and *Serratia* species, the routine use of a third-generation cephalosporin to treat neonatal sepsis is not recommended, except for patients with suspected gram-negative bacterial meningitis because of its excellent penetration in the cerebrospinal fluid.

Management (EOS):

- Every newborn at risk for sepsis should be evaluated (clinically and by laboratory tests) for sepsis.
- If the baby is symptomatic with risk factors – Collect blood culture and start on antibiotics.
- If the baby is asymptomatic with risk factors – Collect blood culture at birth, monitor the baby and perform septic screen at 6-12 hours. If the septic screen is positive, start on antibiotics. If septic screen is negative, continue to monitor the baby and repeat septic screen after 24 hours.
- If the baby is symptomatic without risk factors and CXR is suggestive of pneumonia OR have no alternate reasons for the symptoms must be started on antibiotics after blood culture.

Neonates with extreme risk factors must be started on empirical antibiotics

- ✓ Very prolonged rupture of membranes (≥ 72 hours),
- ✓ Very prolonged labor (≥ 24 hours),
- ✓ Foul smelling liquor,
- ✓ Unclean per vaginal examinations,
- ✓ Maternal septicemia or other systemic infections

Management (LOS):

- If the neonate is symptomatic after 72 hours of life, evaluate for LOS.
- The risk factors for LOS are: VLBW, Use of antibiotics > 5 days, Multiple interventions, Use of H2 blockers, steroids, Presence of central venous catheters (umbilical, percutaneous, and tunneled), Delayed enteral feeding, Inability to feed with EBM, Prolonged hyperalimentation or TPN, Mechanical ventilation, Complications of prematurity (Patent ductus arteriosus, Bronchopulmonary dysplasia, Necrotizing enterocolitis).
- Categorize the neonates into low probability or high probability of sepsis.
- If low probability for sepsis, perform a septic screen. If all the parameters of the screen are negative, antibiotics may not be started and the neonate must be monitored clinically. However Repeat screen after 12-24 hours. A negative repeat screen strongly indicates against starting antibiotics whereas a positive repeat screen with persistence of symptoms may warrant antibiotics.
- If high probability of sepsis, start on antibiotics pending blood culture and septic screen.
- A CSF examination must be performed in all neonates with a high probability of sepsis as well as in those neonates with a low probability of sepsis with a positive sepsis screen.
- Monitor the baby with sepsis for clinical deterioration, respiratory worsening, shock, coagulopathy, cholestasis.

Blood culture:

Under strict aseptic precautions 5-cm in diameter area should be cleansed thoroughly with alcohol, allow to dry, followed by povidone iodine, allow to dry and followed again by alcohol. Povidone-iodine should be applied in concentric circles moving outward from the centre. The skin should be allowed to dry for at least 1 minute before the sample is collected. One-mL sample of blood for a blood culture bottle containing 5-10 ml of culture media. Bactec culture reports should be available in 48 hours.

Sepsis screen:

- ✓ More valuable in a low probability situation to exclude sepsis than to start antibiotics in a suspect sepsis.
- ✓ A sepsis screen is not warranted in neonates with a high probability for sepsis. Instead these neonates should be directly started on antibiotics pending blood culture.
- ✓ If two (or more) parameters are abnormal, it should be considered as a positive screen and the neonate should be started on antibiotics. If the screen is negative but clinical suspicion persists, it should be repeated after 12-24 hours. Two consecutive completely negative screens are suggestive of no sepsis.

1. **Total leukocyte count: Low WBC** (<5000/mm³ is more sensitive).

2. **Absolute neutrophil count: Low ANC (<1000) is more sensitive.**

3. **Immature/total neutrophil** > 0.2. Immature neutrophils (band forms, metamyelocytes, myelocytes) / Mature + immature neutrophils

4. **Micro-ESR** more than 3+ age in days in the first week of life or more than 10 thereafter.

5. **C reactive protein (CRP)** >1 mg/dl Or Positive

Lumbar Puncture (LP):

- In EOS, LP is indicated in (i) infants with positive blood cultures or if the clinical picture is consistent with septicemia. (ii) symptomatic infants with a high risk for EOS whose condition is stable enough to tolerate LP and (iii) infants with negative blood cultures who are treated empirically for the clinical diagnosis of sepsis.
- In Late onset sepsis, LP should be done in all infants prior to starting antibiotics (high probability of sepsis).
- Lumbar puncture could be postponed in a critically sick neonate under cover of antimeningitic dose of antibiotics.
- In a neonate with meningitis not showing clinical recovery after institution of antibiotics, LP should be repeated after 48 hours.
- If the LP is traumatic, the CSF should be sent for gram stain and culture & CSF glucose. Repeat after 48 hours. Pending culture reports, the neonate may be started on meningitic dose of antibiotics. If culture is negative and repeat CSF is normal, neonate should not be labeled or treated for meningitis.
- WBC cell count must be performed within 30 minutes.
- Preterm infants: Treat if CSF WBC count ≥ 10 OR glucose < 24 OR protein > 170 . Do not treat if CSF WBC count < 25 AND glucose ≥ 25 AND protein < 170 . For in-between results, clinical judgment will have to be used, keeping in mind clinical features (seizures, degree of altered sensorium, fullness of fontanelles) and prematurity (the lower the gestation, lower should be the threshold for diagnosis).

- Term infants: Treat if CSF WBC count >8 OR glucose <20 OR protein >120. Do not treat if CSF WBC count <20 AND glucose ≥20 AND protein <120".

Urine culture

In early onset sepsis, urine cultures have a low yield and are not indicated. Urine cultures obtained by suprapubic puncture or bladder catheterization recommended in all cases of LOS.

Neonates at risk for fungal sepsis and very low birth weight infants with poor weight gain should have a urine examination done to exclude urinary tract infection (UTI).

UTI may be diagnosed in the presence of one of the following: (a) >10 WBC/mm³ in a 10 mL centrifuged sample (b) >104 organisms /mL in urine obtained by catheterization and (c) any organism in urine obtained by suprapubic aspiration.

Antibiotic therapy:

Both for out born & inborn

- ✓ 1st line: Ampicillin and Gentamycin –
- ✓ 2nd line: Cefotaxime/ceftazidime/ with ampicillin OR ciprofloxacin
- ✓ 3rd line: Meropenem- single drug

Scenarios after starting 1st line (After a minimum of 24 hours of first line Antibiotics)

1. Worsening clinical condition: add Cefotaxime/ceftazidime (± lab or Xray worsening).
2. Static / no significant improvement in a sick baby: Worsening septic screen-add Cefotaxime/ceftazidime
3. Clinical improvement: continue 1st line till culture reports are available.

Scenarios after starting 2nd line (After a minimum duration of 48 to 72 hours)

1. Worsening clinical condition: Change to 3rd line (± lab or Xray worsening).
2. Static / no significant improvement in a sick baby: Worsening septic screen-replace ceftazidime/meropenem
3. Clinical improvement: continue 2nd line till culture reports are available.

After culture report

- ✓ If the organism is sensitive to an antibiotic with a narrower spectrum, therapy must be changed to such an antibiotic, even if the neonate was improving with the empirical antibiotics and/or the empirical antibiotics are reported sensitive.
- ✓ If the empirical antibiotics are reported sensitive, but the neonate has worsened on these antibiotics, it may be a case of *in vitro* resistance. Antibiotics may be changed to an alternative sensitive antibiotic with the narrowest spectrum and lowest cost.
- ✓ If the empirical antibiotics are reported resistant but the neonate has improved clinically, it may or may not be a case of *in-vivo* sensitivity. In such cases a careful assessment must be made before deciding on continuing with the empirical antibiotics. One must not continue with antibiotics with *in vitro* resistance in case of *Pseudomonas*, *Klebsiella* and MRSA; and in cases of CNS infections and deep-seated infections.

- ✓ If no antibiotic has been reported sensitive, but one or more has been reported 'moderately sensitive', therapy must be changed to such antibiotics at the highest permissible dose.
- ✓ After treatment of UTI, all cases must be started on cephalexin (15 mg/kg HS) oral prophylaxis, till such time that a renal ultrasound, MCU and DMSA scan are performed to exclude VUR or malformation.

For suspect bone and joint infections, cover for Gram + (cloxacillin / vancomycin and gram-negative organism (aminoglycosides)

Suggested Antibiotic Regimens for Sepsis and Meningitis (From: Eichenwald. E.C., Hansen. AR., Martin. CR., Stark. AR., Cloherty and Starks Manual of Neonatal Care; 8th edition, 2016.)

Organism	Antibiotic	Bacteremia	Meningitis
GBS (Streptococcus agalactiae)	Ampicillin or penicillin G	10 days	14-21 days
<i>Escherichia coli</i>	ampicillin and gentamicin, or Cefotaxime	10-14 days	21 days
CONS	Vancomycin	7 days	14 days
<i>Klebsiella</i> , <i>Serratia</i>	Cefotaxime or meropenem and gentamicin	10-14 days	21 days
<i>Enterobacter</i> , <i>Citrobacter</i>	Cefepime or meropenem and gentamicin	10-14 days	21 days
<i>Enterococcus</i>	Ampicillin or vancomycin and gentamicin	10 days	21 days
<i>Listeria</i>	Ampicillin and gentamicin	10-14 days	14-21 days
<i>Pseudomonas</i>	Ceftazidime or piperacillin/tazobactam and gentamicin or tobramycin	14 days	21 days
<i>Staphylococcus aureus</i>	Nafcillin	10-14 days	21 days
MRSA	Vancomycin	10-14 days	21 days

GBS, group B *Streptococcus*; CONS, coagulase-negative staphylococci; MRSA, methicillin resistant *Staphylococcus aureus*.

Many infectious disease specialists recommend repeat lumbar punctures at the completion of therapy for meningitis to ensure eradication of the infection.

Extended-spectrum β -lactamases (ESBL) among enteric pathogens such as *E. coli*, *Klebsiella*, and *Serratia* is an increasing clinical problem. ESBL-containing organisms can be effectively treated with cefepime or meropenem.

The use of additional agents such as linezolid, daptomycin, and rifampin to eradicate persistent *S. aureus* infection or to treat vancomycin-intermediate *Staphylococcus aureus* (VISA) and vancomycin-resistant *Staphylococcus aureus* (VRSA) strains.

Duration of Antibiotics (From: St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)

	Duration of Antibiotics
Culture negative	
Asymptomatic at risk	Stop antibiotics
Suspect screen Negative asymptomatic	Stop antibiotics
Probable EOS or LOS screen +, and the neonate becomes completely asymptomatic by day 5	Antibiotics X 5 days
Suspected/probable EOS or LOS and the neonate improves but does not become asymptomatic	repeat a CRP ✓ If CRP + ve: continue antibiotics ✓ If CRP –ve: stop antibiotics
Suspected/probable EOS or LOS and the neonate have not improved or have worsened	Upgrade
Culture-proven	
CONS sepsis	5-7 days
Sepsis	10 - 14 days
Meningitis	21-day
Pneumonia	7 -10 days
UTI	7-14 days
Bone or joint infections	6 weeks (4 weeksIV, 2 weeks oral)
Ventriculitis	6 weeks

Adjunctive therapies

- **Intravenous immunoglobulins (IVIG):** no evidence. May be considered in sick infants <34 weeks. 1 g/kg
- **Colony stimulating factors):** no evidence. May be considered in sick newborns with neutropenia <1000 / mm³ and not improving with antibiotics. 10 mcg/kg SC BD X 3 days
- **Double volume Exchange Transfusion:** may be performed in a case of deteriorating sepsis with sclerema, DIC provided the general condition of the baby allows the procedure. Use fresh whole blood compatible with baby and mother.

Indications for shifting to isolation:

- ✓ Blood/urine/CSF culture positive other than CONS
- ✓ Open skin lesions / Sclerema
- ✓ Colonization with MDR organism
- ✓ Clinical certainty of sepsis.
- ✓ Clinical+lab deterioration with suspicion of MDR organism
- ✓ After sterilization of cultures, infants from isolation could be shifted out after surface cleaningwith chlorhexidine2 % , and surface and stool culture sterile.

Anaerobic Bacterial Infections.

Neonatal tetanus. This syndrome is caused by the effect of a neurotoxin produced by the anaerobic bacterium *Clostridium tetani*. Infection can occur by invasion of the umbilical cord due to unsanitary childbirth or cord care practices. Treatment consists of the administration of tetanus immunoglobulin (TIG) (500 U intramuscular [IM]) and penicillin G (100,000 U/kg/day divided every 4 to 6 hours for 7 to 10 days) or metronidazole (30 mg/kg/day divided every 6 hours) as well as supportive care with mechanical ventilation, sedatives, and muscle relaxants. IVIG may be given if TIG is not available. Neonatal tetanus does not result in immunity to tetanus, and infants require standard tetanus immunizations after recovery.

It results from unclean delivery and unhygienic management of the umbilical cord in an infant born to a mother who has not been immunized against tetanus. The surveillance case definition of neonatal tetanus requires the ability of a newborn to suck at birth and for the 1st few days of life, followed by an inability to suck. Neonatal tetanus typically occurs in infants 5-7 days after birth (range: 3-24 days), difficulty swallowing, spasms, stiffness, seizures, and death. **Bronchopneumonia**, presumably resulting from aspiration, is a common complication and cause of death.

Perinatal Asphyxia and Hypoxic-Ischemic Encephalopathy

Definitions:

ACOG Criteria for diagnosis of **perinatal asphyxia**:

An arterial cord pH < 7.0(accepted definition of fetal acidosis)

- ✓ Apgar score of < 3 at 5 minutes.
- ✓ Evidence of altered neurological status (altered level of consciousness, seizures, hypotonia, obtundation).
- ✓ Multisystem organ dysfunction (CNS 72%, renal 42%, heart 29%, intestine 29% and lungs 26%).

WHO definition: Failure to initiate and sustain breathing at birth.

Perinatal hypoxia, ischemia, and asphyxia. These pathophysiologic terms describe respectively, decreased oxygen (O₂), blood flow, and gas exchange to the fetus or newborn.

Neonatal encephalopathy is a clinical and not an etiologic term that describes an abnormal neurobehavioral state consisting of an altered level of consciousness (including hyperalert state) and usually other signs of brainstem and/or motordysfunction. It does not imply a specific etiology, nor does it imply irreversible neurologicinjury because it may be caused by such reversible conditions as maternal medications or hypoglycemia.

Hypoxic-ischemic encephalopathy (HIE) is a term that describes clinical evidence of encephalopathy as defined earlier, with objective data to support a hypoxic-ischemic (HI) mechanism as the underlying cause for the encephalopathy.

Incidence. The frequency of perinatal asphyxia is approximately 1.5% of live births in developed countries. It occurs in 0.5% of live-born newborns >36 weeks' gestation and accounts for 20% of perinatal deaths (50% if stillbirths are included).

Etiology:

- ✓ **Maternal factors:** hypertension (acute or chronic), hypotension, infection (including chorioamnionitis), hypoxia from pulmonary or cardiac disorders, diabetes, maternal vascular disease, and in utero exposure to cocaine
- ✓ **Placental factors:** abnormal placentation, abruption, infarction, fibrosis, or hydrops
- ✓ **Uterine rupture**
- ✓ **Umbilical cord accidents:** prolapse, entanglement, true knot, compression
- ✓ **Abnormalities of umbilical vessels**
- ✓ **Fetal factors:** anemia (e.g., from fetal-maternal hemorrhage), infection, cardiomyopathy, hydrops, severe cardiac/circulatory insufficiency
- ✓ **Neonatal factors:** cyanotic congenital heart disease, persistent pulmonary hypertension of the newborn (PPHN), cardiomyopathy, other forms of neonatal cardiogenic and/or septic shock, respiratory failure due to meconium aspiration syndrome, neonatal pneumonia, pneumothorax, or other etiologies

Pathophysiology;

A. Events that occur during the normal course of labor cause most babies to be born with little O₂ reserve.

These include the following:

- ✓ Decreased blood flow to placenta due to uterine contractions, some degree of cord compression, maternal dehydration, and maternal alkalosis due to hyperventilation
- ✓ Decreased O₂ delivery to the fetus from reduced placental blood flow
- ✓ Increased O₂ consumption in both mother and fetus

B. Hypoxia-ischemia causes a number of physiologic and biochemical alterations:

1. With brief asphyxia, there is a transient increase, followed by a decrease in heart rate (HR), mild elevation in blood pressure (BP), an increase in central venous pressure (CVP), and essentially no change in cardiac output (CO). This is accompanied by a redistribution of CO with an increased proportion going to the brain, heart, and adrenal glands (diving reflex).

2. With prolonged asphyxia, there can be a loss of pressure autoregulation and/or CO₂ vasoreactivity. This, in turn, may lead to further disturbances in cerebral perfusion, particularly when there is cardiovascular involvement with hypotension and/or decreased CO. Prolonged asphyxia typically results in diffuse injury to both cortical and subcortical structures.

C. Cellular dysfunction occurs as a result of diminished oxidative phosphorylation and ATP production. Impaired oxidative phosphorylation can occur during the primary HI insult(s) as well as during a secondary energy failure that usually begins approximately 6 to 24 hours after the initiating insult. Cell death can be either immediate or delayed and either necrotic or apoptotic.

- ✓ Immediate neuronal death (necrosis)
- ✓ Delayed neuronal death (apoptosis) occurs secondary to uncontrolled activation of enzymes and second messenger systems within the cell.
- ✓ Reperfusion of previously ischemic tissue may cause further injury because it can promote the formation of excess reactive oxygen.

Topography of Brain Injury in Term Infants with Hypoxic- Ischemic

Encephalopathy and Clinical Correlates (From: Nelson Textbook of Pediatrics 21st Ed.)

AREA OF INJURY	LOCATION OF INJURY	CLINICAL CORRELATES	LONG-TERM SEQUELAE
Selective neuronal necrosis	Entire neuraxis, deep cortical area, brainstem and pontosubicular	Stupor or coma Seizures Hypotonia Oculomotor abnormalities Suck/swallow abnormalities	Cognitive delay Cerebral palsy Dystonia Seizure disorder Ataxia Bulbar and pseudobulbar palsy
Parasagittal injury	Cortex and subcortical white matter Parasagittal regions, especially posterior	Proximal-limb weakness Upper extremities affected > lower extremities	Spastic quadriplegia Cognitive delay Visual and auditory processing difficulty
Focal ischemic necrosis	Cortex and subcortical white matter Vascular injury (usually middle cerebral artery distribution)	Unilateral findings Seizures common and typically focal	Hemiparesis Seizures Cognitive delays
Periventricular injury	Injury to motor tracts, especially lower extremity	Bilateral and symmetric weakness in lower extremities More common in preterm infants	Spastic diplegia

Diagnosis:

A. Perinatal assessment of risk includes awareness of preexisting maternal or fetal problems that may predispose to perinatal asphyxia and of changing placental and fetal conditions ascertained by ultrasonographic examination, biophysical profile, and nonstress tests.

B. Low Apgar scores and need for resuscitation in the delivery room are common but nonspecific findings.

Many features of the Apgar score relate to cardiovascular integrity and not neurologic dysfunction resulting from asphyxia.

1. In addition to perinatal asphyxia, the differential diagnosis for a term newborn with an Apgar score ≤ 3 for ≥ 10 minutes includes depression from maternal anesthesia or analgesia, trauma, infection, cardiac or pulmonary disorders, neuromuscular, and other central nervous system (CNS) disorders or malformations.

2. If the Apgar score is > 6 by 5 minutes, perinatal asphyxia is not likely.

C. Umbilical cord or first blood gas determination. The specific blood gas criteria that define asphyxia causing brain damage are uncertain; however, the pH and base deficit on the cord or first blood gas is helpful for determining which infants have, severe acidosis was defined as pH ≤ 7.0 or base deficit ≥ 16 mmol/L.

D. Clinical presentation and differential diagnosis. HIE should be suspected in encephalopathic newborns with a history of fetal and/or neonatal distress and laboratory evidence of asphyxia. The diagnosis of HIE should not be overlooked in scenarios such as meconium aspiration, pulmonary hypertension, birth trauma, or fetal maternal

hemorrhage, where HIE may be missed because of the severity of pulmonary dysfunction, anemia, or other clinical manifestations. The diagnosis of neonatal encephalopathy includes a number of etiologies in addition to perinatal hypoxia-ischemia.

Asphyxia may be suspected and HIE reasonably included in the differential diagnosis when there is:

- ✓ Prolonged (>1 hour) antenatal acidosis
- ✓ Fetal HR <60 beats per minute
- ✓ Apgar score ≤3 at ≥10 minutes
- ✓ Need for positive pressure ventilation for >1 minute or first cry delayed >5 minutes
- ✓ Seizures within 12 to 24 hours of birth
- ✓ Burst suppression or suppressed background pattern on EEG or amplitude-integrated electroencephalogram(aEEG)

Neurologic Signs. The clinical spectrum of HIE is described as mild, moderate, or severe. EEG is useful to provide objective data to grade the severity of encephalopathy.

Seizures occur in up to 50% of newborns with HIE and usually start within 24 hours after the HI insult.

Seizures indicate that the severity of encephalopathy is moderate or severe, not mild.

Seizures may be subtle, tonic, or clonic. It can sometimes be difficult to differentiate seizures from jitteriness or clonus, although the latter two are usually suppressible with firm hold of the affected limb(s).

Because seizures are often subclinical (electrographic only) and abnormal movements or posture may not be seizure, EEG remains the gold standard for diagnosing neonatal seizures, particularly in HIE.

Increased intracranial pressure (ICP) resulting from diffuse cerebral edema in HIE indicates a poor prognosis. Treatment to reduce ICP does not affect outcome.

Sarnat and Sarnat Stages of Hypoxic-Ischemic Encephalopathy*(from Sarnat HB, Sarnat MS: Neonatal encephalopathy following fetal distress: a clinical and electroencephalographic study, *Arch Neurol* 33:696–705, 1976.)

Stage	Stage 1 (Mild)	Stage 2 (Moderate)	Stage 3 (Severe)
Level of consciousness	Hyperalert; irritable	Lethargic or obtunded	Stuporous, comatose
Neuromuscular control:	Uninhibited, overreactive	Diminished spontaneous movement	Diminished or absent spontaneous movement
Muscle tone	Normal	Hypotonic	Flaccid
Posture	Normal	Flexion	Decerebration
Deep tendon reflexes	Overactive	Overactive	Decreased or absent
Myoclonus	Present	Present	Absent
Complex reflexes:	Normal	Suppressed	Absent
Suck	Weak	Weak or absent	Absent
Moro	Strong	Weak, incomplete	Absent
Autonomic function:	Generalized sympathetic	Generalized parasympathetic	Both systems depressed
Pupils	Mydriasis	Miosis	Midposition, often unequal; poor light reflex
Respirations	Spontaneous	Spontaneous; occasional apnea	Periodic; apnea
Heart rate	Tachycardia	Bradycardia	Variable
Bronchial and salivary secretions	Sparse	Profuse	Variable
Gastrointestinal motility	Normal or decreased	Increased, diarrhea	Variable
Seizures	None	Common focal or multifocal (6-24 hours of age)	Uncommon (excluding decerebration)
Electroencephalographic findings	Normal (awake)	Early: generalized low voltage, slowing (continuous delta and theta) Later: periodic pattern (awake); seizures focal or multifocal; 1.0-1.5 Hz spike and wave	Early: periodic pattern with isopotential phases Later: totally isopotential
Duration of symptoms	<24 hours	2-14 days	Hours to weeks
Outcome	About 100% normal	80% normal; abnormal if symptoms more than 5-7 day	About 50% die; remainder with severe sequelae

*The stages in this table are a continuum reflecting the spectrum of clinical states of newborns over 36weeks' gestational age.

Multiorgan Systemic Effects of Asphyxia (From: Nelson Textbook of Pediatrics 21st Ed.)

SYSTEM	EFFECTS
Central nervous	Hypoxic-ischemic encephalopathy, infarction, intracranial hemorrhage, seizures, cerebral edema, hypotonia, hypertonia
Cardiovascular	Myocardial ischemia, poor contractility, cardiac stunning, tricuspid insufficiency, hypotension
Pulmonary	Pulmonary hypertension, pulmonary hemorrhage, respiratory distress syndrome
Renal	Acute tubular or cortical necrosis
Adrenal	Adrenal hemorrhage
Gastrointestinal	Perforation, ulceration with hemorrhage, necrosis
Metabolic	Inappropriate secretion of antidiuretic hormone, hyponatremia, hypoglycemia, hypocalcemia, myoglobinuria
Integumentary	Subcutaneous fat necrosis
Hematologic	Disseminated intravascular coagulation

The kidney is the most common organ to be affected in the setting of perinatal asphyxia. The proximal tubule of the kidney is especially affected by decreased perfusion, leading to acute tubular necrosis (ATN) with oliguria and a rise in serum creatinine (Cr).

Laboratory Evaluation of Asphyxia:

A. Cardiac evaluation. An elevation of serum creatine kinase myocardial bound (CK-MB) fraction of >5% to 10% may indicate myocardial injury.

B. Neurologic markers of brain injury: Serum CK-BB may be increased in asphyxiated newborns within 12 hours of the insult but has not been correlated with long-term neurodevelopmental outcome.

C. Renal evaluation

1. Blood urea nitrogen (BUN) and serum Cr may be elevated in perinatal asphyxia. Typically, elevation is noted 2 to 4 days after the insult.

2. Fractional excretion of Na⁺ (FENa) or renal failure index may help confirm renal insult.

3. Urine levels of β_2 -microglobulin have been used as an indicator of proximal tubular dysfunction, although not routinely. This low molecular weight protein is freely filtered through the glomerulus and reabsorbed almost completely in the proximal tubule.

Brain Imaging:

A. Cranial sonographic examination can demonstrate edema as loss of gray white differentiation and small ventricles when severe but is generally insensitive for the detection of HI brain injury, particularly in the first days after birth.

B. CT may be used to detect cerebral edema, hemorrhage, and eventually HI brain injury.

C. MRI. Conventional T1- and T2-weighted MRI sequences are the best modality for determining the severity and extent of irreversible HI brain injury, but the injury is not apparent on these sequences in the first days after the HI insult, unless the injury is older than suspected or very severe. These conventional sequences are best

for the detection of brain injury at least 7 to 10 days, and a scan as late as 14 days or older may be needed to show the full extent of the injury,

EEG is used to both detect and monitor seizure activity and also to define abnormal background patterns such as discontinuous, burst suppression, low voltage, or isoelectric patterns.

Pathologic Findings of Brain Injury:

A. Specific neuropathology may be seen after moderate or severe HIE.

1. Selective neuronal necrosis is the most common type of injury seen following perinatal asphyxia.

2. A watershed pattern of ischemic injury occurs in boundary zones between cerebral arteries, particularly following severe hypotension.

3. Focal or multifocal cortical necrosis affecting all cellular elements can result in cystic encephalomalacia and/or dysplasia (injury to cortex in depths of sulci) due to loss of perfusion in one or more vascular beds.

B. Neuropathology may reflect the type of asphyxial episode, although the precise pattern is not predictable.

1. Prolonged partial episodes of asphyxia tend to cause diffuse cerebral (especially cortical) necrosis, although there is often involvement of subcortical ± brainstem structures as well.

2. Acute total asphyxia, when relatively brief, affects primarily the brainstem, thalamus, and basal ganglia and tends to spare the cortex in large part, except for the periorlandic cortex.

Treatment:

A. Perinatal management of high-risk pregnancies

1. Fetal HR abnormalities may provide supporting evidence of asphyxia, especially if accompanied by presence of thick meconium. However, they provide few data concerning duration or severity of an asphyxia event.
2. Measurement of fetal scalp pH is a better determinant of fetal oxygenation than PO₂. With intermittent hypoxia ischemia, PO₂ may improve transiently whereas the pH progressively falls. Fetal scalp blood lactate has been suggested as easier and more reliable than pH but has not gained wide acceptance.
3. Close monitoring of progress of labor with awareness of other signs of in utero distress is important.

B. Delivery room management.

C. Postnatal management of neurologic effects of asphyxia

1. Ventilation. CO₂ should be maintained in the normal range. Hypercapnia can cause cerebral acidosis and cerebral vasodilation. This may result in more flow to uninjured areas and relative ischemia to damaged areas ("steal phenomenon"). Excessive hypocapnia (CO₂ <25 mm Hg) decreases cerebral perfusion so should also be avoided.
2. Oxygenation. O₂ levels should be maintained in the normal range, although poor peripheral perfusion may limit the accuracy of continuous noninvasive monitoring. Hypoxemia should be treated with supplemental O₂ and/or mechanical ventilation. Hyperoxia may cause decreased CBF or exacerbate free radical damage so should be avoided.
3. Temperature. Passive cooling by turning off warming lights is an effective way to initiate therapeutic hypothermia as soon as possible after the HI insult. Hyperthermia should always be avoided.
4. Perfusion. Cardiovascular stability and adequate mean systemic arterial BP are important in order to maintain adequate cerebral perfusion pressure.

5. Maintain physiologic metabolic state

- a. Hypocalcemia is a common metabolic alteration after neonatal asphyxia. It is important to maintain calcium in the normal range because hypocalcemia can compromise cardiac contractility and may cause or exacerbate.
- b. Hypoglycemia is often seen in asphyxiated newborns. Blood glucose level should be maintained in the normal range for term newborns. Hypoglycemia may increase CBF, exacerbate the energy deficit, and cause or exacerbate seizures. Hyperglycemia may lead to increased brain lactate, damage to cellular integrity, cerebral edema, or further disturbance in vascular autoregulation.

6. Judicious fluid management is needed, and both fluid overload and inadequate circulating volume should be avoided. Two processes predispose to fluid overload in asphyxiated newborns:

- a. ATN can result from the “diving reflex” and result in oliguria followed by polyuria.
- b. Syndrome of inappropriate antidiuretic hormone (SIADH) secretion is often seen 3 to 4 days after the HI event. It is manifested by hyponatremia and hypo-osmolality in combination with low urine output and inappropriately concentrated urine (elevated urine specific gravity, osmolality, and Na⁺).

Fluid restriction may aid in minimizing cerebral edema, although the effect of fluid restriction on long-term outcome in newborns who are not in renal failure is not known.

7. Control of seizures.

Seizures: Correct hypoglycemia / hypocalcemia. If seizures persist load 20 mg/ Kg

Phenobarbitone; If seizures recur again 2 additional doses of 10 mg/kg (up to 40 mg/kg total) may be needed. followed by 5 mg/ kg/ day of maintenance. If seizures recur again 2 repeat doses of phenobarbitone of 10 mg/Kg each may be given. If seizures recur despite this, there is a high degree of variability regarding choice of 2nd agent. Historically, phenytoin (20 mg/kg loading dose) or lorazepam (0.1

mg/kg) have been preferred, currently the use of levetiracetam is preferred (at

times even as a first-line agent) as the most used second-line agent. In addition to levetiracetam and phenytoin, other second- or third-line agents commonly used include midazolam, topiramate, and lidocaine. Pyridoxine should also be attempted, particularly in ongoing refractory seizures with highly abnormal EEG background.

Long-term anticonvulsant management. Anticonvulsants can be weaned when the clinical exam and EEG indicate that the newborn is no longer having seizures. If a newborn is receiving more than one anticonvulsant, weaning should be in the reverse order of initiation, with phenobarbital being weaned last, unless there is strong evidence that a particular drug was more effective. There is controversy regarding when phenobarbital should be discontinued, with some favoring discontinuation shortly before discharge and some favoring continued treatment for 1 to 6 months or more. Newborns who have a higher risk of developing epilepsy in infancy or childhood are those with a large area of HI brain injury and those with a persistently epileptiform EEG.

Management of other target organ injury

a. **Cardiac dysfunction** should be managed with correction of hypoxemia, acidosis, hypocalcemia and hypoglycemia, and avoidance of volume depletion or overload. Diuretics may be less effective if concomitant renal impairment is present. Newborns will require continuous monitoring of systemic mean arterial BP, CVP (if available), and urine output. Newborns with cardiovascular compromise may require inotropic drugs such as dopamine and may need afterload reduction (e.g., dobutamine or milrinone) to maintain BP and perfusion.

b. **Renal dysfunction** should be monitored by measuring urine output, with serum electrolytes, paired urine/serum osmolarity, urinalysis, and urine specific gravity.

i. In the presence of oliguria or anuria, avoid fluid overload by limiting free water administration to replacement of insensible losses and urine output (~60 mL/kg/day).

ii. Volume status should be evaluated before instituting strict fluid restriction. If there is no or low urine output, a 10 to 20 mL/kg fluid challenge followed by a loop diuretic such as furosemide may be helpful.

c. **GI effects.** Feeding should be withheld until BP is stable, active bowel sounds are audible, and stools are negative for blood.

d. **Hematologic abnormalities.** Coagulation profile should be monitored with partial thromboplastin time (PTT) and prothrombin time (PT), fibrinogen, and platelets. Abnormalities may need to be corrected with fresh frozen plasma, cryoprecipitate, and/or platelet infusions.

e. **Liver function** should be monitored with measurement of transaminases (ALT, AST), clotting (PT, PTT, fibrinogen), albumin, bilirubin, and ammonia. Levels of drugs that are metabolized or eliminated by the liver must be monitored.

f. **Lung.** Management of the pulmonary effects of asphyxia depends on the specific etiology.

Neuroprotective Strategies:

Therapeutic hypothermia, whether head cooling or systemic cooling (by servo-control to a core rectal or esophageal temperature of 33.5°C [92.3°F] within the 1st 6 hr after birth and maintained for 72 hr), has been shown in various trials to reduce mortality and major neurodevelopmental impairment at 18 mo of age. Infants treated with systemic hypothermia have a lower incidence of cortical neuronal injury on MRI, suggesting systemic hypothermia may result in more uniform cooling of the brain and deeper CNS structures than selective head cooling. The therapeutic effect of hypothermia likely results from decreased secondary neuronal injury achieved by reducing rates of apoptosis and production of mediators known to be neurotoxic, including extracellular glutamate, free radicals, nitric oxide, and lactate. There is also benefit in seizure reduction. Complications of induced hypothermia include thrombocytopenia (usually without bleeding), reduced heart rate, and subcutaneous fat necrosis (sometimes with associated hypercalcemia) as well as the potential for overcooling and the cold injury syndrome. The latter is usually avoided with a servo-controlled cooling system.

Cooling should be started before 6 hours of age; therefore, early recognition is essential. The target core temperature goal during cooling is 33.5°C (33° to 34°C) with acceptable range: 32.5° to 34.5°C.

Whole body cooling for perinatal asphyxia:

Inclusion criteria: Should fulfill 1, 2 and 3.

A) Inborn:

1. GA >35 wks
2. PHYSIOLOGICAL CRITERIA- Any 1 of the following
 - ✓ ABG (UC/1st postnatal hr) pH < 7.0 or ABE > - 12
 - ✓ Apgar score < 5 at 5'
 - ✓ Ventilation required for at least 10'
3. NEUROLOGICAL CRITERIA- Seizures OR Evidence of moderate or severe encephalopathy (3 of 6 criteria in modified Sarnat)

B) Out born:

1. GA >35 wks
2. Physiological Criteria - Babies who did not cry immediately after birth/required resuscitation/APGAR score < 5 at 5' (if available)

3. Neurological Criteria – Seizures or Evidence of encephalopathy

Exclusion criteria:

- ✓ Inability to start cooling the baby by 6 hours of age
- ✓ Chromosomal abnormality
- ✓ Major congenital anomaly
- ✓ FiO₂ requirement >80%
- ✓ Fixed and dilated pupil

Ancillary therapy:

1. Seizures: Correct hypoglycemia / hypocalcemia. If seizures persists use anticonvulsants.
2. Feeding & nutrition: Its preferable to keep the neonate NPO during the 48 hours of cooling due to risk of impaired mesenteric perfusion & risk of NEC. Trophic feeds may be started after 48 hours
3. I/O charting: Strict input output monitoring should be maintained. Intravenous fluid intake & content will be adjusted Q24 H based on the neonate's urine output & lab values. Urine output should be same as fluid intake
4. Access: It's preferable to secure umbilical venous access for all neonates undergoing therapeutic hypothermia.
5. Sedation & pain relief: Minimizing stimulation by reducing light, noise & physical manipulation to prevent increased ICP. Pain may be difficult to assess. Appropriate pharmacological & non-pharmacological treatment to be provided as indicated.
5. Infectious disease: Antibiotics should be started after complete blood count (CBC) and blood culture drawn per routine. If concerns regarding renal function, change from gentamicin to cefotaxime.

Safety monitoring of newborns during 72 hours of therapeutic hypothermia and rewarming:

At the end of 72 hours of induced hypothermia, the newborn is rewarmed at a rate of 0.5°C every 2 hours until patient reaches 36.5°C. This should take approximately 10 to 12 hours. If a patient is discovered to meet an exclusion criterion or undergoes a major adverse event while undergoing hypothermia treatment, rewarm according to the same procedure.

Complications of therapeutic hypothermia:

- ✓ Hypotension
- ✓ Arrhythmia
- ✓ Coagulopathy
- ✓ Hypoglycemia
- ✓ Hyperglycemia
- ✓ Thrombocytopenia
- ✓ PPHN
- ✓ Shivering & discomfort
- ✓ Sepsis.

Outcome in Perinatal Asphyxia:

The overall mortality rate is approximately 20%. The frequency of neurodevelopmental sequelae in surviving newborns is approximately 30%.

The risk of cerebral palsy (CP) in survivors of perinatal asphyxia is 5% to 10% compared to 0.2% in the general population. Most CP is not related to perinatal asphyxia, and most perinatal asphyxia does not cause CP.

Severity of encephalopathy can be ascertained using the Sarnat clinical stages of HIE.

Stage 1 or mild HIE: <1% mortality, 98% to 100% of newborns will have a normal neurologic outcome.

Stage 2 or moderate HIE: 20% to 37% die or have abnormal neurodevelopmental outcomes.

Stage 3 or severe HIE: Death from effects of severe systemic asphyxia is more likely. Survivors are likely to have one or more major neurodevelopmental disability, such as CP, intellectual disability, visual impairment, or epilepsy.

The presence of seizures increases a newborn's risk of CP 50- to 70-fold.

Persistently low voltage activity or isoelectric background by EEG is a prognostic indicator of poor neurologic outcome.

Of note, some maternal medications can transiently alter the neonatal EEG.

Poor Predictive Variables for Death/Disability After Hypoxic-Ischemic Encephalopathy:

- ✓ Infants with initial cord or initial blood pH <6.7
- ✓ Low (0–3) 10 min Apgar score
- ✓ Need for CPR in the delivery room
- ✓ Delayed onset (≥ 20 min) of spontaneous breathing
- ✓ Severe neurologic signs (coma, hypotonia, hypertonia)
- ✓ Seizures onset ≤ 12 hr or difficult to treat
- ✓ Severe, prolonged (~ 7 days) EEG findings including burst suppression pattern
- ✓ Prominent MRI basal ganglia/thalamic lesions
- ✓ Oliguria/anuria >24 hr
- ✓ High base deficit (>20-25 mmol/L),
- ✓ Decerebrate posture,
- ✓ Persistence of abnormal neurologic signs at 2 wk of age.

Follow up:

- ✓ Immediately after discharge within 7 days.
- ✓ At 6 wks, 3 months, 9 months, 12 months, 18 & 24 months.

Polycythemia in Neonates

I. Introduction:

A. Polycythemia is defined as Hct concentration >2 SD above the normal value for gestational and postnatal age. Accordingly, a term infant is considered to be polycythemia if the Hct from a **peripheral venous** sample is >65 percent.

Hematocrit measurements vary greatly with site of sample, and capillary hematocrit may be up to 20% higher than venous. Hematocrit initially rises after birth from placental transfer of RBCs and then decreases to baseline by approximately 24 hours. The mean venous hematocrit of term infants is 53% in cord blood, 60% at 2 hours of age, 57% at 6 hours of age, and 52% at 12 to 18 hours of age.

Plethora: a ruddy, deep red-purple appearance associated with a high hematocrit, is often due to polycythemia, defined as a central hematocrit of 65% or higher.

B. Hyperviscosity is defined as viscosity >2 standard deviations greater than the mean. Blood viscosity, as described by Poiseuille, is the ratio of shear stress to shear rate and is dependent on such factors as the pressure gradient along the vessel, radius, length, and flow. The relationship between hematocrit and viscosity is nearly linear below a hematocrit of 60%, but viscosity increases exponentially at a hematocrit of 70% or greater.

Hyperviscosity determined at constant shear rates (11.5 sec^{-1}) is present when whole blood viscosity is >18 cycles/sec. Hyperviscosity is accentuated because neonatal RBCs have decreased deformability and filterability, which predispose to stasis in the microcirculation.

Newborns have larger, irregularly shaped red blood cells (RBC) with different membrane characteristics than the RBCs of adults. As viscosity increases, there is impairment of tissue oxygenation and decreased glucose in plasma, leading to increased risk of microthrombus formation. If these events occur in the cerebral cortex, kidneys, or adrenal glands, significant damage may result. Hypoxia and acidosis increase viscosity and deformity further. Poor perfusion increases the possibility of thrombosis.

II. Incidence: The incidence of polycythemia is 1% to 5% in term newborns. Polycythemia is increased in babies that have IUGR, SGA, and are born postterm.

III. Causes of Polycythemia:

A. Placental red cell transfusion

1. Delayed cord clamping: When the cord is clamped within 1 minute after birth, the blood volume of the infant is approximately 80 mL/kg. When the cord is clamped 2 minutes after delivery, the blood volume of the infant is 90 mL/kg.

2. Cord stripping (thereby pushing more blood into the infant)

3. Holding the baby below the mother at delivery

4. Maternal-to-fetal transfusion is diagnosed with the Kleihauer-Betke stain.

5. Twin-to-twin transfusion.

6. Forceful uterine contractions before cord clamping

B. Placental insufficiency (increased fetal erythropoiesis secondary to chronic intrauterine hypoxia)

1. SGA and IUGR infants

2. Maternal hypertension syndromes (preeclampsia, renal disease, etc.)

3. Postterm infants

4. Infants born to mothers with chronic hypoxia (heart disease, pulmonary disease).

5. Pregnancy at high altitude

6. Maternal smoking

C. Other conditions

1. Infants of diabetic mothers (increased erythropoiesis)

2. Some large for gestational age (LGA) babies

3. Infants with congenital adrenal hyperplasia, Beckwith-Wiedemann syndrome, neonatal thyrotoxicosis, congenital hypothyroidism, trisomy21, trisomy13, trisomy18

4. Drugs (maternal use of propranolol)

5. Dehydration of infant

6. Sepsis (increase in fibrinogen, lower RBC deformability)

IV. Clinical findings: Most infants with polycythemia are asymptomatic. Clinical symptoms, syndromes, and laboratory abnormalities that have been described in association with polycythemia include the following:

A. Central nervous system (CNS). Poor feeding, lethargy, hypotonia, apnea, tremors, jitteriness, seizures, cerebral venous thrombosis.

B. Cardiorespiratory. Cyanosis, tachypnea, heart murmur, congestive heart failure, cardiomegaly, elevated pulmonary vascular resistance, prominent vascular markings on chest x-ray.

C. Renal. Decreased glomerular filtration, decreased sodium excretion, renal vein thrombosis, hematuria, proteinuria.

D. Other. Other thrombosis, thrombocytopenia, poor feeding, increased jaundice, persistent hypoglycemia, hypocalcemia, testicular infarcts, necrotizing enterocolitis (NEC), priapism, disseminated intravascularcoagulation

V. Screening: Routine screening of well term newborns for polycythemia/hyperviscosity is not recommended by many because there are few data showing that treatment of asymptomatic patients with partial exchange transfusion is beneficial in the long term.

VI. Diagnosis: The capillary blood or peripheral venous hematocrit level should be determined in any babywho appears plethoric, who has any predisposing cause of polycythemia or who is not well for any reason.

- A.** Warming the heel before drawing blood for a capillary hematocrit determination will give a better correlation with the peripheral venous or central hematocrit. If the capillary blood hematocrit is above 65%, the peripheral venous hematocrit should be determined.
- B.** Few hospitals are equipped to measure blood viscosity. If the equipment is available, the test should be done because some infants with venous hematocrits under 65% will have hyperviscous blood.

VII. Management:

- A.** Once other causes of illness have been considered and excluded (e.g., sepsis, pneumonia, hypoglycemia), any child with symptoms that could be due to hyperviscosity should be considered for **partial exchange transfusion if the peripheral venous hematocrit is >65%.**
- B. Asymptomatic infants** with a peripheral venous hematocrit between 60% and 70% **can usually be managed by increasing fluid intake and repeating the hematocrit in 4 to 6 hours.**
- C.** Many neonatologists perform an exchange transfusion when the peripheral venous hematocrit is >70% in the absence of symptoms, but this is a controversial issue.
- D.** The following formula can be used to calculate the exchange with normal saline. In infants with polycythemia, the blood volume varies inversely with the birth weight.

Exchange volume = [(observed Hct – desired Hct) x blood volume] ÷ observed Hct

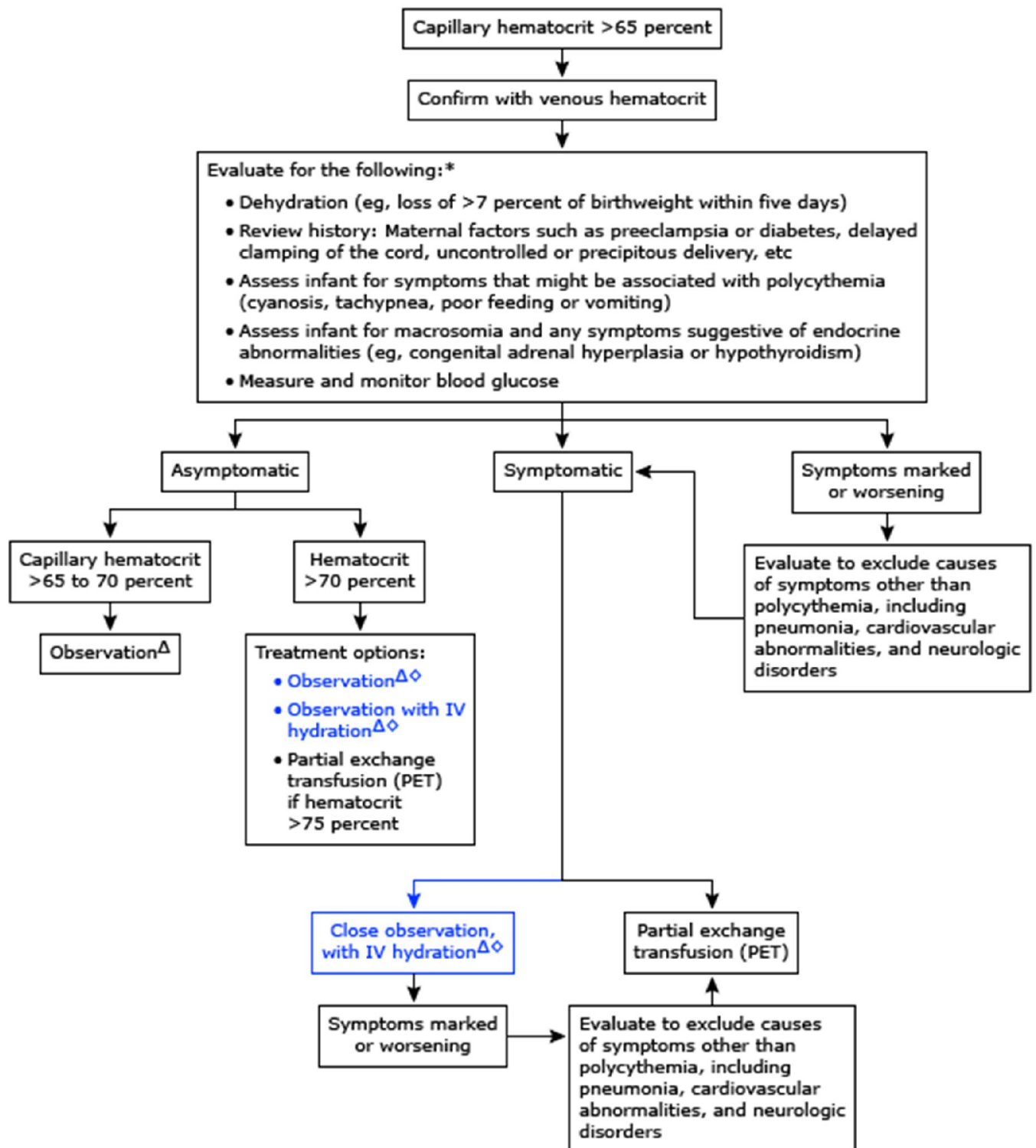
VIII. Outcome:

- A. Infants with polycythemia and hyperviscosity who have decreased cerebral blood flow velocity and increased vascular resistance develop normal cerebral blood flow following partial exchange transfusion.** They also have improvement in systemic blood flow and oxygen transport.
- B. The long-term neurologic outcome** in infants with asymptomatic polycythemia/hyperviscosity, whether treated or untreated, **remains controversial.**

The long-term prognosis of polycythemic infants is unclear. Reported adverse outcomes include speech deficits, abnormal fine motor control, reduced IQ, school problems, and other neurologic abnormalities. The underlying etiology (chronic intrauterine hypoxia) and hyperviscosity is thought to contribute to adverse outcomes.

Partial exchange transfusion (PET): — Isovolumetric PET will lower hematocrit (without causing hypovolemia), decrease viscosity, and reverse many of the physiologic abnormalities associated with polycythemia/hyperviscosity. There is some evidence that PET has beneficial short-term effects on measures of perfusion like: physiologic measures related to viscosity (eg, cerebral blood flow, cardiac index, and oxygen transport); however, PET does not appear to affect long-term outcome in infants with polycythemia.

Algorithm for management of neonatal polycythemia (From: UpToDate 2018.)



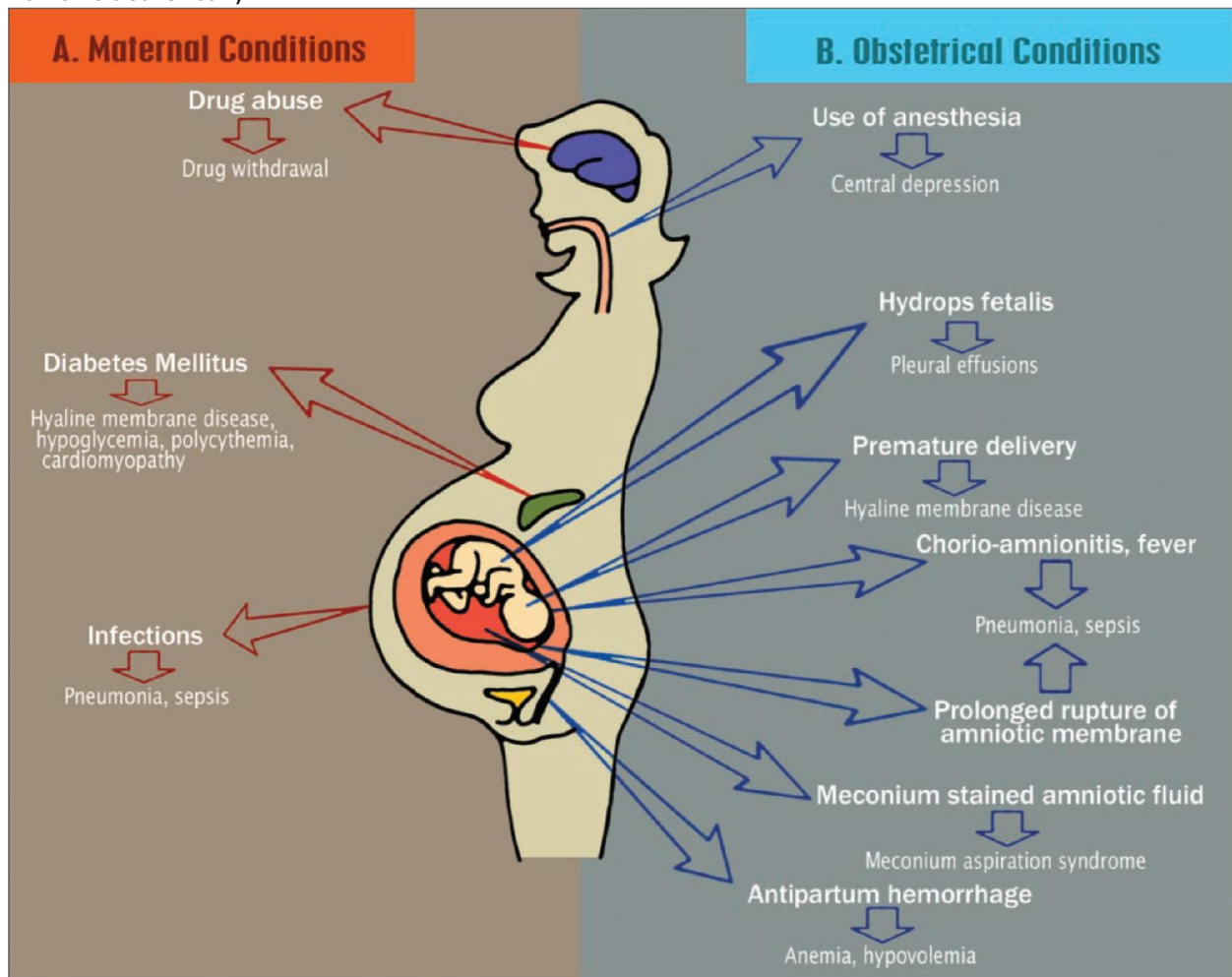
Respiratory Distress Syndrome

Respiratory Distress in Infants can be caused by:

- I. Maternal and obstetric conditions associated with respiratory distress in neonates.
- II. Respiratory diseases
- III. Cardiac diseases
- IV. Neurologic diseases
- V. Other miscellaneous causes

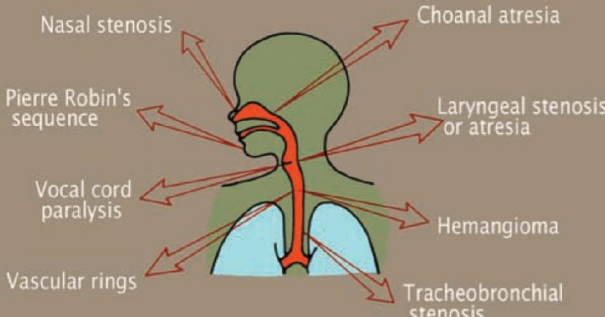
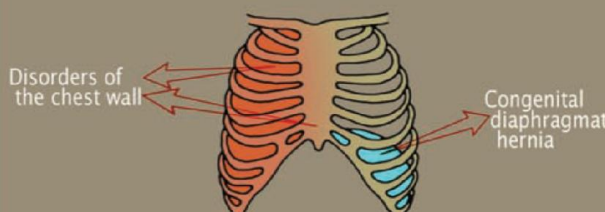
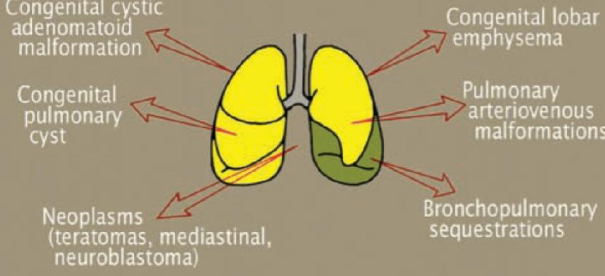
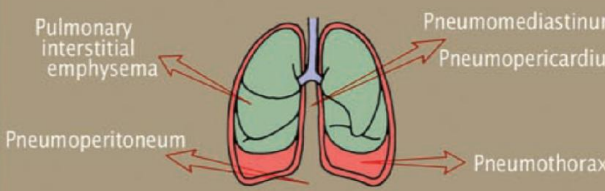
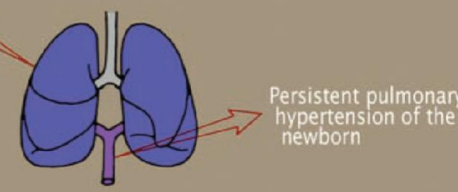
Maternal and obstetric conditions associated with respiratory distress in

neonates (From: Aly H., Respiratory Disorders in the Newborn: Identification and Diagnosis, Pediatrics in Review Vol.25 No.6 June 2004.)



Respiratory and Extra respiratory Causes of Respiratory distress in Infants

(From: Aly H., Respiratory Disorders in the Newborn: Identification and Diagnosis, Pediatrics in Review Vol.25 No.6 June 2004.)

1. Respiratory Diseases	
A. Airway Obstructions 	B. Disorders of the Chest Wall and Diaphragm 
C. Malformation of the Mediastinum and Lung Parenchyma 	D. Air Leak Syndromes 
2. Cardiac Diseases	E. Pulmonary Parenchymal and Vascular Diseases 
A. Cyanotic - Transposition of great arteries - Total anomalous pulmonary venous return - Ebstein's anomaly - Tricuspidal atresia - Pulmonic stenosis - Tetralogy of Fallot - Severe congestive heart failure 	3. Neurological Disorder - Birth trauma - Intraventricular hemorrhage - Meningitis - Hypoxic ischemic encephalopathy - Primary seizure disorder - Obstructed hydrocephalus - Infantile botulism - Spinal cord injury - Muscular diseases (myasthenia gravis, poliomyelitis) 
B. Acyanotic - Hypoplastic left heart syndrome - Interrupted aortic arch - Critical aortic coarctation - Patent ductus arteriosus 	4. Other Miscellaneous Diseases - Sepsis - Anemia or polycythemia - Hypo or hyperthermia - Hypo or hypernatremia - Hypoglycemia - Inborn errors of metabolism - Maternal medication (magnesium sulfate, opiates) or drug abuse

Respiratory Distress Syndrom:

Definition: Respiratory distress syndrome (RDS), formerly known as hyaline membrane disease (HMD), describes a disease typical of preterm infants that is caused by insufficient pulmonary surfactant in alveoli.

A pre-term neonate with HMD:

Classic features of RDS:

- ✓ Respiratory distress having onset within 6 hours of birth
- ✓ Amniotic fluid L/S ratio of <1.5 , or negative gastric aspirate shake test, or
- ✓ CXR showing poor expansion with air bronchogram/ reticulogranular pattern/ ground glass opacity.

Incidence:

It occurs in 60–80% of infants <28 wk GA, in 15–30% of those between 32 and 36 wk GA, and rarely in those >37 wk GA.

Etiology:

Surfactant deficiency (decreased production and secretion) is the primary cause of RDS.

Pathophysiology:

As a consequence of progressive injury to epithelial and endothelial cells from atelectasis (atelectrauma), volutrauma, ischemic injury, and oxygen toxicity, effusion of proteinaceous material and cellular debris into the alveolar spaces (forming the classic hyaline membranes) further impairs oxygenation. Alveolar atelectasis, hyaline membrane formation, and interstitial edema make the lungs less compliant in RDS, so greater pressure is required to expand the alveoli and small airways.

Although surfactant is present in high concentrations in fetal lung homogenates by 20 wk of gestation, it does not reach the surface of the lungs until later. It appears in amniotic fluid between 28 and 32 wk. Mature levels of pulmonary surfactant are present usually after 35 wks. of gestation.

Atelectasis results in perfused but not ventilated alveoli, causing hypoxia. Decreased lung compliance, small tidal volumes, increased physiologic dead space, and insufficient alveolar ventilation eventually result in hypercapnia. The combination of hypercapnia, hypoxia, and acidosis produces pulmonary arterial vasoconstriction with increased right-to-left shunting through the foramen ovale and ductus arteriosus and within the lung itself.

The major constituents of surfactant are dipalmitoyl phosphatidylcholine (lecithin), phosphatidylglycerol, apoproteins (surfactant proteins SP-A, SP-B, SP-C, and SP-D), and cholesterol.

Surfactant recovered from lungs of all mammalian species by alveolar wash procedures contains 70% to 80% phospholipids, about 8% to 10% protein, and about 10% neutral lipids, primarily cholesterol. The phosphatidylcholine species of the phospholipids

represents about ~~60%~~ weight of the surfactant and about 80% of the phospholipids. It is synthesized, packaged, and secreted from alveolar type II cells of the lung.

Risk factors:

The risk for development of RDS increases with:

- ✓ Fetal sex. Male infants are at higher risk for RDS due to the presence of circulating weak fetal androgens that inhibit the production of surfactant phospholipids.
- ✓ Maternal diabetes: due to enhanced production of fetal insulin which inhibits the production of proteins important for surfactant function.
- ✓ Multiple births,
- ✓ Cesarean delivery,
- ✓ Precipitous delivery,
- ✓ Asphyxia, The baby does not get enough oxygen during labor, delivery, or immediately after birth.
- ✓ Cold stress,
- ✓ Maternal history of previously affected infants.
- ✓ Mutations in surfactant-related proteins Mutations, especially in SP-B and SP-C, impair surfactant production or function.

Prematurity (Most important)

The risk of RDS is reduced in

- ✓ Pregnancies with chronic or pregnancy-associated hypertension,
- ✓ Maternal heroin use,
- ✓ Prolonged rupture of membranes,
- ✓ Antenatal corticosteroid prophylaxis.

expose the fetus to chronic intrauterine stress
stimulates production of fetal corticosteroids.

Corticosteroids:
Accelerate maturation of type II pneumocytes.
Increase surfactant synthesis.
Increase surfactant release.
Improve lung compliance.
Promote structural lung development.
Neonatal mortality

They significantly reduce:
Respiratory Distress Syndrome (RDS).
Need for mechanical ventilation.
Intraventricular hemorrhage (IVH).
Necrotizing enterocolitis (NEC).
Neonatal mortality.

Clinical Manifestations:

Signs of RDS usually appear within minutes of birth, although they may not be recognized for several hours in larger premature infants, until rapid, shallow respirations become more obvious. A later onset of tachypnea should suggest other conditions.

Characteristically, tachypnea, prominent (often audible) expiratory grunting, intercostal and subcostal retractions, nasal flaring, and cyanosis are noted. Breath sounds may be normal or diminished with a harsh tubular quality, and on deep inspiration, fine crackles may be heard.

a. **Tachypnea**. Inadequate FRC leads to inadequate tidal volumes. To maintain minute ventilation (the product of tidal volume × respiratory rate), infants with RDS increase respiratory rate.

b. **Retractions**. To maximize negative inspiratory pressure and thus lung inflation, affected infants use accessory muscles of breathing to supplement diaphragmatic contractions. The high negative inspiratory pressure draws in the highly compliant chest wall resulting in suprasternal, intercostal, and subcostal retractions.

Types of retractions
Suprasternal
Intercostal
Subcostal---in sever case

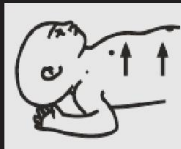
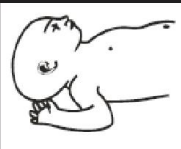
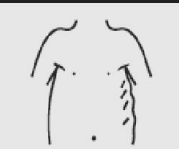
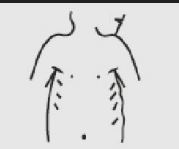
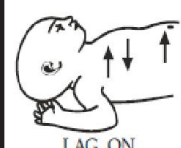
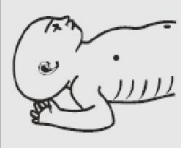
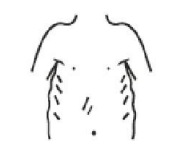
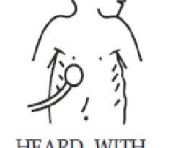
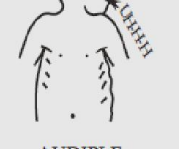
c. **Flaring** of the alae nasi. To maximize air entry into the lungs in babies who are obligate nose breathers, flaring of the alae nasi reduces the resistance to air flow through the upper airways.

d. **Grunting**. Grunting is active exhalation against a partially closed glottis and results in a pressure gradient at the level of the vocal cords that provides expiratory distending pressure to stabilize patent but surfactant-poor alveoli.

Cyanosis is a sign of severe hypoxemia and requires immediate oxygen assessment and management.

Respiratory distress score:

In order to objectively grade the severity, the signs of respiratory distress are assigned a numerical score (0 indicating best score in the category and 2 indicating the worst) and individual scores are combined to produce a final respiratory distress score. The two commonly used respiratory distress scores are the Silverman Anderson score and Downes' Vidyasagar Score.

	UPPER CHEST	LOWER CHEST	XIPHOID RETRACTIONS	NARES DILATATION	EXPIRATORY GRUNT
Grade 0	 SYNCHRONIZED	 NO Retractions	 NONE	 NONE	 NONE
Grade 1	 LAG ON INSPIRATION	 JUST VISIBLE	 JUST VISIBLE	 MINIMAL	 HEARD WITH STETHOSCOPE
Grade 2	 SEE-SAW	 MARKED	 MARKED	 MARKED	 AUDIBLE

SILVERMAN ANDERSON ASSESSMENT OF RESPIRATORY DISTRESS: Scoring:

0 – 3 ----- Mild, 4 – 6 -----Moderate, >6 -----Sever

Signs	0	1	2
Thoraco-abdominal movement	Rhythmic and regular	Immobile thorax movement in abdomen	Thorax and abdomen in up and down
Intercostals retraction	Absent	Discrete	Accentuated and constant
Xiphoid retraction	Absent	Discrete	Very marked
Flaring of alar nasi	Absent	With closed mouth	Very marked with open mouth
Grunting	Absent	Mild and inconstant	Constant and accentuated

(From: 1. Silverman WA, Andersen DH. A controlled clinical trial of effects of water mist on obstructive respiratory signs, death rate and necropsy findings among premature infants. *Pediatrics*. 1956;17 :1- 10.

2. Federal Ministry of Health of Ethiopia, Neonatal Intensive Care Unit (NICU) Training Participants' Manual, November 2014, Addis Ababa.)

Downe's score for grading severity of respiratory distress (From: Downes JJ, Vidyasagar D, Boggs TR, Jr., Morrow GM, 3rd. Respiratory distress syndrome of newborn infants. I. New clinical scoring system (RDS score) with acid—base and blood-gas correlations. *Clin Pediatr (Phila)*. 1970;9 :325-31.)

Feature	Score 0	Score 1	Score 2
Cyanosis	None	In room air	In 40% FiO ₂
Retractions	None	Mild	Severe
Grunting	None	Audible with stethoscope	Audible without stethoscope
Air entry	Normal	Decreased	Barely audible
Respiratory rate	<60	60-80	>80 or apnea

Grades of severity according to the total score: 0 – 3 - mild; 4 – 6 - moderate, >6 severe

(From: 1. Downes JJ, Vidyasagar D, Boggs TR, Jr., Morrow GM, 3rd. Respiratory distress syndrome of newborn infants. I. New clinical scoring system (RDS score) with acid—base and blood-gas correlations. *Clin Pediatr (Phila)*. 1970;9 :325-31.
2. St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)

The natural course of untreated RDS is characterized by:

- ✓ Progressive worsening of cyanosis and dyspnea.
- ✓ If the condition is inadequately treated, blood pressure may fall; cyanosis and pallor increase, and grunting decreases or disappears, as the condition worsens.
- ✓ Apnea and irregular respirations are ominous signs requiring immediate intervention.
- ✓ Untreated patients may also have a mixed respiratory-metabolic acidosis, edema, ileus, and oliguria. Respiratory failure may occur in infants with rapid progression of the disease.
- ✓ In most cases the signs reach a peak within 3 days, after which improvement is gradual. Improvement is often heralded by spontaneous diuresis and improved blood gas values at lower inspired O₂ levels and/or lower ventilator support.

Differential diagnosis of Respiratory Distress Syndrom:

a. Transient tachypnea of the newborn. Excess fetal lung fluid can mimic RDS and can complicate RDS. Signs are indistinguishable from RDS, but TTN often resolves rapidly over the first several hours after birth. Radiographic findings are consistent with retained fetal lung fluid, with characteristic prominent perihilar streaking (sunburst pattern) due to engorgement of periarterial lymphatics that participate in the clearance of alveolar fluid and often fluid retained in the lateral fissure of the right lung.

b. Pneumonia, especially due to group B Streptococcus. Proinflammatory cytokines elaborated in the course of an infection can inactivate surfactant constituents and down regulate surfactant production. Signs and radiographic findings of group B Streptococcus (GBS) sepsis/pneumonia are indistinguishable from RDS; therefore, obtaining blood cultures and initiating antibiotics should be considered.

Evaluation:

Lung maturity testing:

the most readily available tests assess the lecithin (desaturated phosphatidylcholine) component of surfactant. Lecithin is the most abundant surfactant phospholipid, and its production is developmentally regulated. However, it is also present in cell membranes, necessitating correction for the presence of contaminants like blood.

a. The lecithin/sphingomyelin (L/S) ratio corrects for the presence of a neutral lipid in low abundance in surfactant, whereas the TDx-FLM II corrects for the presence of albumin in the amniotic fluid sample. In both cases, samples contaminated significantly by blood or meconium can be difficult to interpret. RDS risk is low when the L/S ratio is >2 , but notable exceptions to this include maternal diabetes, erythroblastosis fetalis, and intrapartum asphyxia. The TDx-FLM II has established gestational age-specific cutoffs but in general is predictive of low RDS risk at >55 mg lecithin per gram albumin.

b. The presence of lamellar bodies in amniotic fluid samples is a rapid and inexpensive test that may be useful in resource-poor settings. Lamellar bodies are the organelles in alveolar type II cells that receive, concentrate, and store surfactant constituents for regulated secretion. $>50,000$ lamellar bodies per microliter of amniotic fluid has been correlated with lung maturity.

- Lung maturity can be assessed in a newborn at risk of RDS by the shake test
- For all preterm <34 weeks of gestation perform shake test on the gastric fluid within 1 hour of age by passing a nasogastric tube into the stomach. Take absolute alcohol 0.5ml and add 0.5 ml of gastric aspirate in a 4ml glass test tube. The capped test tube is vigorously shaken for 15 seconds and allowed to stand for 15 minutes.

Observation	Interpretation	Risk of RDS
Immature	No bubbles	High (60%)
1+	Very small bubbles in meniscus extending one third or less of circumference (magnifying glass may be used)	Intermediate (20%)
2+	Single rim of bubbles extending one third to all the way around the test tube.	Moderate
3+	A rim of bubbles all the way around the test tube with double row in some areas	Low ($<1\%$)
4+	A double row or more of bubbles all the way around the test tube indicate lung maturity	Nil

- Watch for evidence of respiratory distress (≥ 2 of the following)
 - ✓ RR > 60 /min
 - ✓ Working of accessory muscle of respiration
 - ✓ Cyanosis
 - ✓ Grunting
- Monitor HR, RR, CRT, and NIBP 2 hourly

- Monitor SaO₂ and FiO₂ every ½ hourly
 - Take X-ray chest
 - ✓ Low volume poorly inflated lungs
 - ✓ Homogeneous micro atelectasis that has the appearance of ground glass or fine homogeneous reticulogranular pattern (grade 1)
 - ✓ widespread air bronchogram highlighted by the surrounding micro atelectasis become visible (grade 2)
 - ✓ Cardio pulmonary differentiation difficult due to alveolar shadowing (grade 3)
 - ✓ Complete white-out of the lung fields (grade 4)
 - Record intake/output chart strictly
 - Watch for apnea, pallor, air entry, liver size, murmur
- Blood gas measurement will demonstrate hypoxemia and hypercarbia.

Prevention

Avoidance of unnecessary or poorly timed early cesarean delivery or induction of labor, appropriate management of high-risk pregnancy and labor (including administration of antenatal corticosteroids), and prediction of pulmonary immaturity with possible in utero acceleration of maturation are important preventive strategies.

Administration of antenatal corticosteroids to women before 37 wk gestation significantly reduces the incidence and mortality of RDS as well as overall neonatal mortality.

Antenatal steroids also reduce:

- ✓ Overall mortality,
- ✓ Admission to the neonatal intensive care unit (NICU) and need for/duration of ventilatory support, and
- ✓ Incidence of severe IVH,
- ✓ Necrotizing enterocolitis (NEC), and
- ✓ Neurodevelopmental impairment.

For all women between 24 and 36 wk gestation who present in preterm labor and are likely to deliver a fetus within 1 wk, antenatal corticosteroid administration should be considered.

Practical points in Prevention:

- ✓ Indicated for 24-34 weeks gestation age.
- ✓ Betamethasone or Dexamethasone.
- ✓ Dose: 2 doses of betamethasone are administered 12mg IM at an interval of 24 hours or 4 doses of dexamethasone, 6mg each at an interval of 12 hours Effect starts 24 hours (possibly as early as 4 hours) after 1st dose and continues for 7 days.
- ✓ A repeat course of steroids is to be considered if it was taken > 7 days ago and gestation is < 32 weeks and delivery is likely.

Management:

The basic defect requiring treatment in RDS is inadequate pulmonary O₂ -CO₂ exchange.

- ✓ Basic supportive care (thermoregulatory, circulatory, fluid, electrolyte, and respiratory) is essential while FRC is established and maintained.
- ✓ Careful and frequent monitoring of heart and respiratory rates, SaO₂, PaO₂, PaCO₂, pH, electrolytes, glucose, hematocrit, blood pressure, and temperature are essential.
- ✓ Give antibiotic if risk factors are present or if septic screen is positive.

The hallmark of RDS is surfactant deficiency and collapse of the alveoli.

Basic pathophysiology:

- Compliance: ↓↓↓
- Resistance: N/↑
- Time constant: ↓
- FRC: ↓↓↓, Less commonly N or ↓
- Ventilation: Heterogeneous lung disease
- Perfusion: normal to reduced

CPAP:

- Early CPAP is the corner stone of RDS management. If the baby has respiratory distress / labored breathing, initiate CPAP at 5 cm H₂O in the labor room. It is preferred to use blended oxygen. Target SaO₂ 90-94 %.
 - Start with CPAP of 5 cm H₂O, flow of 5 l/ min and FiO₂ – 0.5.
 - Bubble CPAP and Ventilator CPAP are available.
 - In Bubble CPAP the pressure is increased by increasing the depth of the underwater seal. Ensure bubbling.
 - Interface: should be comfortable, atraumatic to nose, easily cleanable and deliver CPAP effectively. The interface should be fixed well with head gear.
 - ✓ Short binasal prongs are preferred. Hudson prongs, dragger prongs and mask are used.
 - ✓ Masks are used only if there is columellar injury.
 - ✓ Ensure there is no blanching on the nose. There should be a small column of air
 - ✓ between the prongs and the columella.
 - Titrate CPAP pressures based on retractions. It is preferred to increase CPAP rather than FiO₂.
- Ensure that the stomach is decompressed by keeping the infant feeding tube open.
- Increase CPAP to a maximum of 7 cm H₂O especially if the lungs are low volume. based on saturations. If the baby requires > 0.4 FiO₂, and CXR is suggestive of RDS, give surfactant.
 - INSURE - Intubate Surfactant Rapid extubation is preferred.

- Contraindications to CPAP – hemodynamic instability is a relative contraindication CPAP is not possible in cleft palate, TEF, bilateral choanal atresia, recurrent apnea not responding to CPAP or SIPPV

- After surfactant is given, the pressures should be reduced over the next 1 hours. Volume guarantee is a good mode post surfactant to avoid volutrauma.
- Prophylactic surfactant is considered for extreme preterms < 29 weeks) especially if the mother has not received steroids.
- Rescue therapy is considered for all preterms with respiratory distress and $FiO_2 > 0.4$. Early rescue is within 2 hours. Late rescue is > 2 hours.
- Repeat dose: If the baby is still on ventilator and MAP is > 8 and FiO_2 is > 0,4, repeat dose of surfactant may be administered. Repeat dose after 6 hours in case of survanta and after 12 hours in case of curosurf

Surfactant is administered to achieve a phospholipid dose that is at least 100mg/kg. The standard dosing interval is every 12 hours for a maximum of three doses. However, in the absence of a therapeutic response or if the initial response is waning, the next dose can occur as soon as 6 hours later, rather than waiting to complete the 12-hour dosing interval.

- **Precautions:** Adjust the ventilator settings appropriately after administration, as the compliance improves rapidly and may lead to barotrauma.
- Mechanical blockage of the tube leading to hypoxemia.
- Hemorrhagic pulmonary edema.

Ventilatory management

- Mechanical Ventilation: Start with patient triggered ventilation.
 - Low PIP - 10-20 cm H₂O
 - Moderate PEEP – 5 – 6 cm H₂O
 - Relatively rapid rate (40-60/min)
- If retractions persist, increase PEEP to a maximum of 7 cm H₂O.

Target SaO₂:

- ✓ pH 7.25-7.35
- ✓ PaO₂ 50-70 mm Hg
- ✓ PaCO₂ 45-55 mm Hg

Further management:

- An uncomplicated RDS should be out of the ventilator by 3 days if surfactant is given.
 - PDA, sepsis, apnea, air leaks, CCF, IVH are early onset and features of chronic lung disease
- Watch for onset of diuresis
- Watch for clinical and ABG improvement and gradually wean off the baby.
 - Give maintenance fluids and start feeds gradu

In the past, intratracheal surfactant replacement for symptomatic premature infants immediately after birth (prophylactic) or during the 1st few hr of life (early rescue) showed reduced air leak and mortality from RDS. However, substantial evidence supports the feasibility and efficacy of prophylactic Nasal Continuous Positive Airway Pressure (nCPAP) as the primary means of respiratory support for preterm infants with RDS. CPAP started at birth is as effective as prophylactic or early surfactant and is associated with a reduction in BPD. Prophylactic nCPAP is therefore the approach of choice for the delivery room management of a preterm neonate at risk for RDS.

Therefore, surfactant should be considered in preterm infants with signs of RDS in the immediate perinatal period after a failed trial of CPAP or in infants for whom CPAP is contraindicated, i.e., apnea, air leak.

Outcomes. In the presurfactant era, RDS in late preterm infants typically resolved at 2 to 4 days of age, often preceded by spontaneous diuresis. RDS in infants born at ≥ 32 weeks' gestational age and without other complications typically resolves fully with no long-term pulmonary sequelae. Infants < 32 weeks' gestational age are at risk for BPD; risk increases with decreasing gestational age.

Resuscitation in the Delivery Room

I. Introduction:

The critical period for the physiologic changes is during the first several breaths, which result in lung expansion and elevation of the partial pressure of oxygen (PO₂) in both the alveoli and the arterial circulation.

Elevation of the PO₂ from the fetal level of approximately 25 mm Hg to values of 50 to 70 mm Hg is associated with:

- ✓ Decrease in pulmonary vascular resistance,
- ✓ Decrease in right-to-left shunting through the ductus arteriosus,
- ✓ Increase in venous return to the left atrium,
- ✓ Rise in left atrial pressure, and
- ✓ Cessation of right-to-left shunt through the foramen ovale.

The end result is conversion from fetal to transitional to neonatal circulatory pattern. Adequate systemic arterial oxygenation results from perfusion of well-expanded, well-ventilated lungs and adequate circulation.

Conditions at delivery may compromise the fetus's ability to make the necessary transitions. Alterations in tissue perfusion and oxygenation ultimately result in depression of cardiac function, but human fetuses initially respond to hypoxia by becoming apneic. Even a relatively brief period of oxygen deprivation may result in this **primary apnea**. Rapid recovery from this state is generally accomplished with appropriate stimulation and oxygen exposure. If the period of hypoxia continues, the fetus will irregularly gasp and lapse into **secondary apnea**.

This state may occur remote from birth or in the peripartum period. Infants born during this period require resuscitation with assisted ventilation and oxygen.

II. Preparation:

Anticipation is key to ensuring that adequate preparations have been made for a neonate likely to require resuscitation at birth. It is estimated that as many as 10% of neonates require some assistance at birth for normal transition, whereas <1% require extensive resuscitative measures.

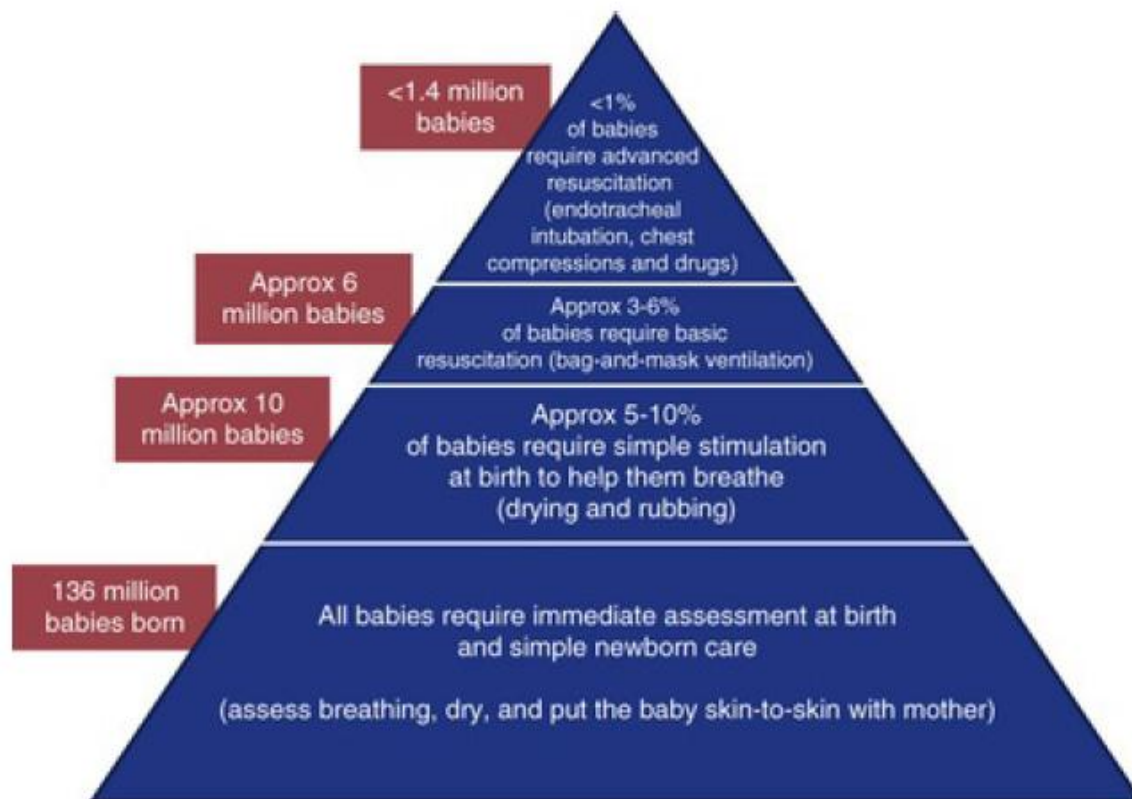
A. Identifying factors associated with a high-risk delivery (From: UPTODATE 2018)

I. Antepartum
Maternal conditions
Age (>40 years, <16 years)
Poor socioeconomic status (poverty, malnutrition)
Detrimental habits (smoking, drug and/or alcohol abuse)
Medical conditions
• Diabetes mellitus
• Hypertension
• Chronic heart and/or lung diseases
• Kidney diseases/urinary tract infections
• Blood disorders (thrombocytopenia, anemia, blood group incompatibilities)
Obstetric conditions
• Prior stillbirth/fetal loss/early neonatal death
• Prior birth of a high-risk infant
• Antepartum hemorrhage
• Premature rupture of membranes
• Serious infection during pregnancy
• Placental anomalies: praevia, polyhydramnios or oligohydramnios, pregnancy-induced hypertension, group B streptococcus carrier
Fetal conditions
Prematurity
Postmaturity
Intrauterine growth retardation
Macrosomia
Multiple gestation
Congenital anomalies
Hydrops
II. During birth
Complications of maternal medical disease
Preterm labor
Prolapsed cord
Utero-placental bleeding
Abnormalities of presentation (transverse lie, breech, etc)
Chorioamnionitis or systemic maternal infection
Foul-smelling or meconium-stained amniotic fluid
Abnormal fetal heart rate patterns
Instrumented delivery (forceps or vacuum deliveries)
Cesarean delivery for maternal or fetal compromise
Narcotic administered to mother within four hours of birth

Goals of resuscitation are the following:

- ✓ **Minimizing immediate heat loss** by drying and providing warmth, thereby decreasing oxygen consumption by the neonate
- ✓ **Establishing normal respiration and lung expansion** by clearing the upper airway and using positive pressure ventilation if necessary
- ✓ **Increasing arterial PO₂** by providing adequate alveolar ventilation. The **routine** use of added oxygen is not warranted, but this therapy may be necessary in some situations
- ✓ **Supporting adequate cardiac output**

Estimates of infants requiring resuscitation at birth (From: Wall SN, Lee ACC, Niermeyer S, et al: Neonatal resuscitation in low-resource settings: what, who, and how to overcome challenges to scale up? Int J Gynaecol Obstet 107: S47–S64, 2009.)



Estimates of global numbers of babies undergoing resuscitation at birth. Source: Estimates based on references.

B. Necessary equipment that must be present in the delivery room

(From: UPTODATE 2018)

I. Suction equipment
Bulb syringe
Mechanical suction, tubing, and catheters
Meconium aspirator
8F feeding tube and 20 cc syringe
II. Intubation equipment
Laryngoscope with straight blades: Make sure the light is bright and has an appropriate blade for the anticipated baby (no. 1 for full-term neonates, no. 0 for premature neonates, no. 00 for extremely low birth weight neonates).
Face masks (preterm and term infant sizes)
Oxygen source with flowmeter
For Positive pressure ventilation equipment: Functioning self-inflating bag – with term and preterm mask, the bag must be appropriately sized for neonates (generally about 750 mL) and capable of delivering 100% oxygen.; T piece resuscitator (if preterm delivery).
Uniform diameter ET tubes (2.5-, 3.0-, and 3.5-mm internal diameters), two of eachSet out an appropriate ET tube for the expected birth weight (3.5 mm for full-term infants, 3.0 mm for premature infants >1,250 g, and 2.5 mm for smaller infants). The NRP recommends a 4.0-mm tube for larger babies, but this is rarely necessary. For all babies, the tube should be 13 cm long. An intubation stylet may be used if the tip is kept at least 0.5 cm from the distal end of the ET tube.
III. Medications
Dextrose solution 10 percent
Epinephrine (0.1 mg/mL)
Isotonic solution
Naloxone hydrochloride (0.4 mg/mL)
Needles
Syringes
Umbilical vessel catheterization supplies
Sterile gloves, scalpel, antiseptic prep solution, umbilical catheter, tape, three-way stopcock
IV. Miscellaneous
Radiant warmer (heated for 20 min prior), 2 sterile warm towels, Cling wrap (GA< 32 wk)
Warm towels
Cardiac monitor
Pulse oximeter and probe
Oropharyngeal airways
V. Additional equipment for delivery of preterm infants
Compressed air source
Oxygen blender
Plastic wrap
Transport incubator
Stethoscope with infant- or premature-sized head

III. During Delivery. The team should be aware of the type and duration of anesthesia, extent of maternal bleeding, and newly recognized problems such as a nuchal cord or meconium in the amniotic fluid.

A. Immediately following delivery, begin a process of evaluation, decision, and action (resuscitation)

Assess if the baby cries immediately after birth If yes – provide Routine Care:

- ✓ Provide routine care on the mother's abdomen
- ✓ If secretions are present, turn head to one side and wipe off secretions gently. Deep pharyngeal stimulation with a suction catheter may cause arrhythmias.
- ✓ Dry and discard wet cloth. Cover the head
- ✓ Cut cord after 1 minute in both term and preterm.
- ✓ Initiate breastfeeding as soon as baby is ready.
- ✓ Continue skin to skin care for at least 1 hour.
- ✓ All routine care such as Vit K, applying identification band, eye care can be done in skin to skin contact. Routines such as birth weight check, complete examination may be delayed. However, if the baby is separated for these routines, the baby should be provided SSC as soon as possible.

Sequence of intervention. Although Apgar scores are assigned at 1 and 5 minutes, resuscitative efforts should begin during the initial neonatal stabilization period. The NRP recommends that at the time of birth, the baby should be assessed by posing three basic questions: (i) Is it a term gestation? (ii) Does the baby have good muscle tone? (iii) Is the baby crying or breathing? If the answer to any of these questions is “no,” the initial steps of resuscitation should commence. In the newly born infant, essentially all resuscitation problems within the initial postnatal period occur as a result of inadequate respiratory effort or some obstruction to the airway. Therefore, the initial focus must be on ensuring an adequate airway and adequate breathing.

Apgar Scores. Evaluation and decisions regarding resuscitation measures should be guided by assessment of respiration, heart rate, and color/oxygen saturation. This said, Apgar scores are conventionally assigned after birth and recorded in the newborn's chart. The Apgar score consists of the total points assigned to five objective signs in the newborn. Each sign is evaluated and given a score of 0, 1, or 2. Total scores at 1 and 5 minutes after birth are usually noted.

If the 5-minute score is 6 or less, the score is then noted at successive 5-minute intervals until it is >6. A score of 10 indicates an infant in perfect condition; this is quite unusual because most babies have some degree of acrocyanosis. The scoring, if done properly, yields the following information:

One-minute Apgar score. This score generally correlates with umbilical cord blood pH and is an index of intrapartum depression. It does not correlate with outcome. Babies with a score of 0 to 4 have been shown to have a significantly lower pH, higher partial pressure of carbon dioxide (PaCO₂), and lower buffer base.

Apgar scores beyond 1 minute are reflective of the infant's changing condition and the adequacy of resuscitative efforts.

Apgar Scoring System (From: Apgar V. A proposal for a new method of evaluation of the newborn infant. Curr Res Anesth Analg 1953; 32:260-267.)

Sign	Score		
	0	1	2
Heart rate	Absent	<100 bpm	>100 bpm
Respiratory effort	Absent	Slow (irregular)	Good crying
Muscle tone	Limp	Some flexion of extremities	Active motion
Reflex irritability	No response	Grimace	Cough or sneeze
Color	Blue, pale	Pink body, blue extremities	All pink

B. Initial steps in resuscitation (30 seconds)

The steps in neonatal resuscitation follow the **ABCs**:

A, anticipate and establish a patent airway by positioning the baby with the head slightly extended, sniffing position, and suctioning if secretions are blocking the airway;
B, initiate breathing first by using tactile stimulation, followed by positive pressure ventilation (PPV) with a bag-mask device and ETT insertion should the baby remain apneic or PPV is not achieving effective ventilation; and
C, maintain the circulation with chest compressions and medications.

If baby does not cry immediately after birth, then the initial steps of resuscitation are provided. The initial steps in resuscitation consist of

- ✓ Preventing heat loss and providing warmth.
- ✓ Positioning and clearing the airway.
- ✓ Drying and removing the wet linen
- ✓ Initiating breathing by gentle stimulation (gently if required).
- ✓ Evaluation- HR, respiration and color

Evaluation of the newborn is based on

- ✓ Respiratory effort
- ✓ Heart rate

Some newborns do not immediately establish spontaneous respiration but will rapidly respond to tactile stimulation, including vigorous flicking of the soles of the feet or

rubbing the back (e.g., cases of primary apnea). More vigorous or other techniques of stimulation have no therapeutic value and are potentially harmful.

If breathing does not start after two attempts at tactile stimulation, the baby should be considered to be in secondary apnea, and respiratory support should be initiated. It is better to overdiagnose secondary apnea in this situation than to continue attempts at stimulation that are not successful.

C. Assessment of the need for supplemental oxygen.

Positive pressure ventilation

Indications

- ✓ Apnea or gasping respiration
- ✓ Heart rate less than 100 per minute

Steps

- ✓ Check the equipment to ensure that there are no tears or leaks.
- ✓ Position the infant correctly with slight extension at the neck
- ✓ Position the mask correctly and apply a firm seal. Provide 5 breaths and check for rising heart rate, if absent, adequate chest movement.
- ✓ If there is no adequate chest rise, take ventilatory corrective steps (MRSOPA).

Once effective ventilation starts, continue for 30 seconds at 40-60 breaths / minute.

Initiate PPV with room air (21% on blender; 21- 30% in preterm).

Connect to pulse oximeter. Titrate as per minute wise saturation targets. In preterm use 21% to 30% oxygen. If blender is not available, a self-inflating bag (with O₂ and without reservoir) provides 40%.

Pressures required to inflate the lungs:

- ✓ Normal: 15 – 20 cm H₂O
- ✓ Diseased: 20 – 40 cm H₂O

Evaluate the heart rate after 30 seconds of bag and mask ventilation

- ✓ If heart rate is above 100 and infant has spontaneous respiration, discontinue ventilation
- ✓ Heart rate between 60 & 100 – continue ventilation.
- ✓ Heart rate < 60 beats per minute – ensure ventilation with 100% oxygen and initiate chest compression.

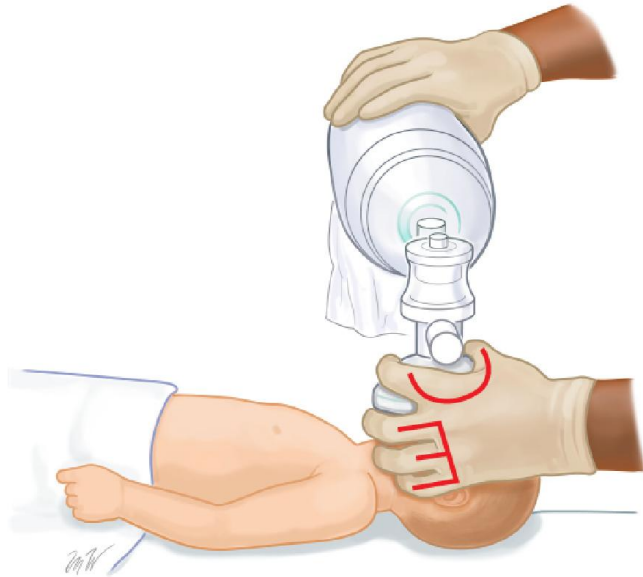
Pass an orogastric tube if bag and mask ventilation is continued for more than 2 minutes.

Contraindications

- ✓ Diaphragmatic hernia

The 6 ventilation corrective steps can be remembered with the mnemonic **MRSOPA**: **m**ask readjustment, **r**eposition the head, **s**uction mouth and nose, **o**pen the mouth, **p**ressure increase, and **a**lternative airway).

c. **Bag and Mask: E-C clamp technique for a neonate** (From: UPTODATE 2018)



D. Circulation. If, after intubation and 30 seconds of ventilation with 100% oxygen, the heart rate remains <60 bpm, cardiac massage should be instituted.

Chest compressions

Indication:

If heart rate less than 60/min despite good assisted ventilation for 30 sec.

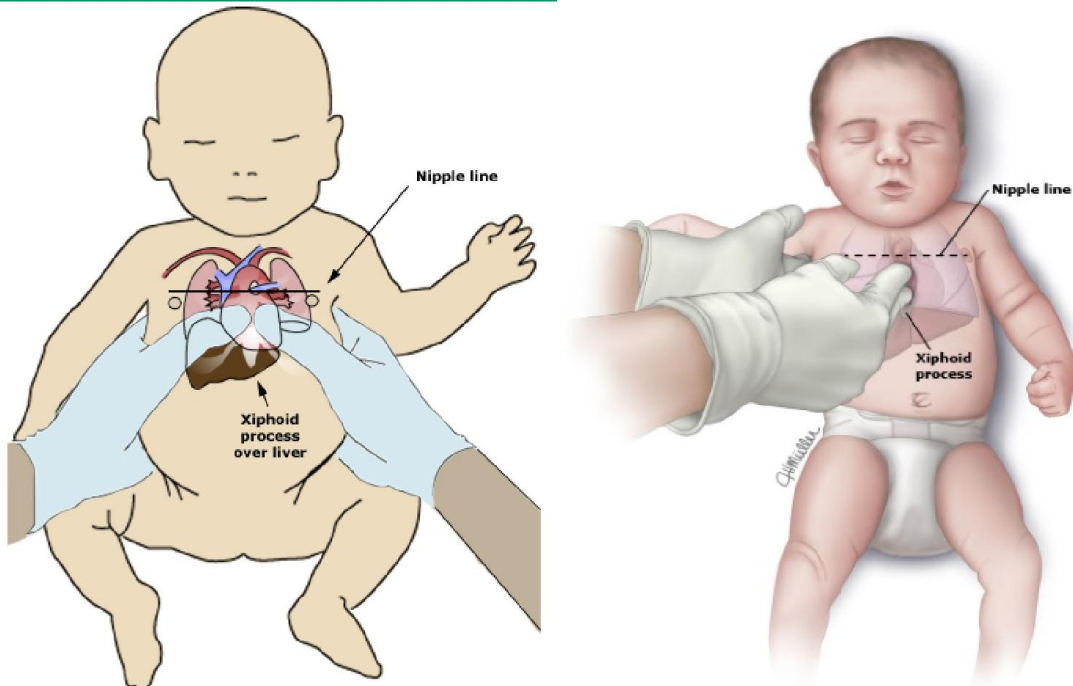
Methods:

- Two thumb method.

Steps

- ✓ Intubate. Use 100% oxygen. Call for help if needed
- ✓ Location: lower third of sternum strictly avoiding applying pressure on the xiphoid.
- ✓ Depth: Compress the sternum approximately 1/3 of the A-P diameter of the chest.
- ✓ Rate: 90 times per minute and ventilation 30 times per minute with a ratio of 3: 1.
- ✓ Evaluation: After 45-60 seconds of chest compression, check the heart rate.
 - If HR < 60/min- Continue compression and ventilation, Initiate medications.
 - If HR > 60/min - Discontinue compression. Continue ventilation till HR > 100/min and good respiratory efforts.

D. Chest compression for infant resuscitation: Two thumb and Two finger techniques respectively: (From: UPTODATE 2018)



E. Medications.

If, despite adequate ventilation with 100% oxygen and chest compressions, a heart rate of >60 bpm has not been achieved by 1 to 2 minutes after delivery, medications such as chronotropic and inotropic agents should be given to support the myocardium, ensure adequate fluid status, and in some situations to correct acidosis. Medications provide substrate and stimulation for the heart so that it can support circulation of oxygen and nutrients to the brain.

Epinephrine:

Indications: HR < 60/min despite adequate ventilation and chest compression for 45-60 seconds. (at approximately 2 min of life).

Dose: 0.1 – 0.3 ml/kg IV or 0.5 - 1 ml/kg Et 1: 10,000 of epinephrine solution, may be repeated every 3-5 min if required. Endotracheal route is easier, but IV is preferred. Initiate UV cannulation as ET dose is given.

Volume expanders

Normal saline, plasma expanders, whole blood can be used.

Indication: Shock, acute bleeding, poor response to resuscitative efforts.

Dose: 10 ml/kg IV over 5 –10 min.

Endotracheal intubation

Indication

- ✓ Bag and mask is ineffective or prolonged.
- ✓ Before chest compression
- ✓ Tracheal administration of medications is desired.
- ✓ Congenital diaphragmatic hernia
- ✓ ELBW for administering surfactant.

TUBE SIZE (ID mm)	B.WEIGHT(g)	GEST. AGE Wks)
2.5	<1000	<28
3.0	1000-2000	28-34
3.5	2000-3000	35-38
4.0	>3000	>38

(From: 1. St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)
2. UPTODATE 2018.)

Placental Transfusion:

Delayed cord clamping for all babies, both term and preterm who do not require resuscitation.

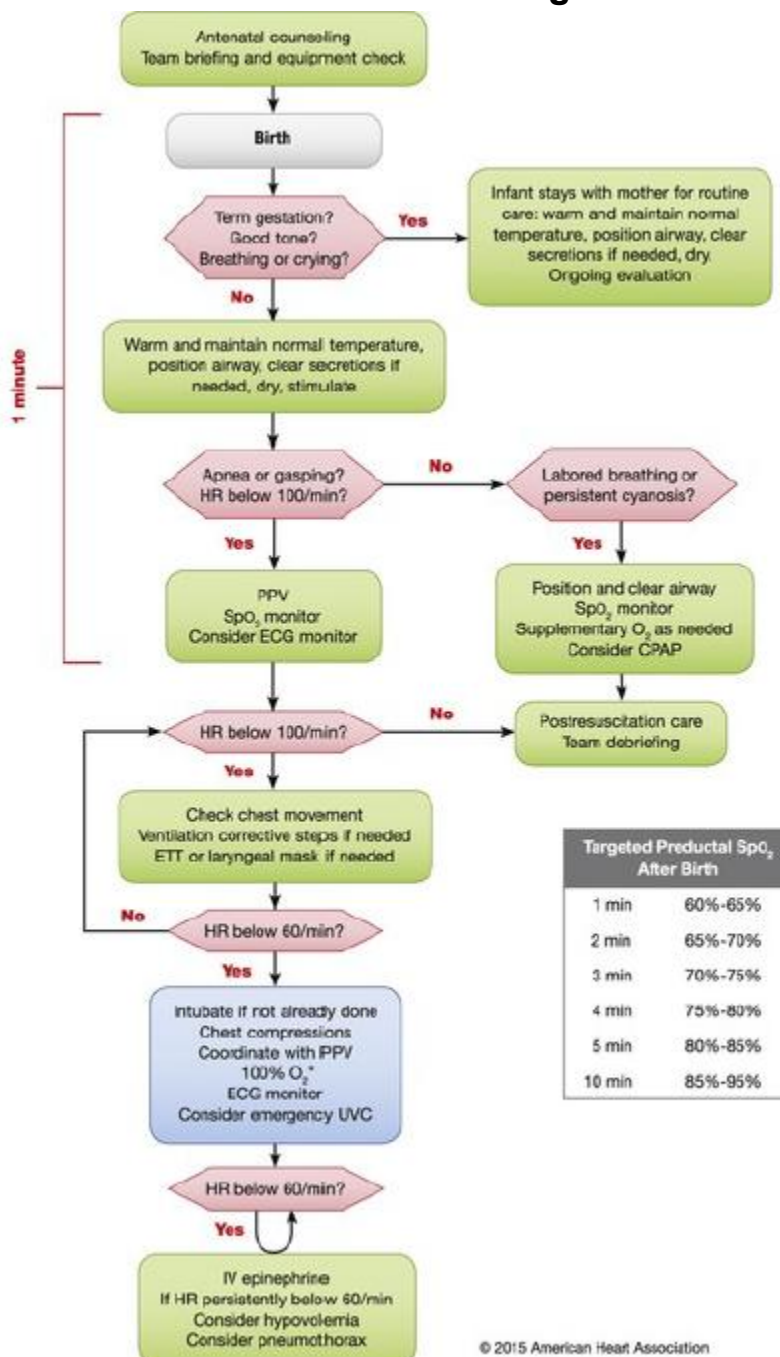
Cord should be clamped at the end of 1 minute. This applies to both normal delivery and cesarean section.

For babies requiring resuscitation, a 25 cm long segment of cord is secured with the help of cord clamping forceps. While initial steps are being done, the blood in the cord should be milked towards the baby.

Temperature Monitoring during Resuscitation:

- After the 60 seconds of PPV, consider switching to servo mode and monitor temperature.
- Check axillary temperature prior to shifting of infant.
- Pre-warm sheets prior to transfer of infant

Newborn resuscitation algorithm:



Newborn resuscitation algorithm. CPAP, Continuous positive airway pressure; ETT, endotracheal tube; HR, heart rate; IV, intravenous; PPV, positive pressure ventilation; UVC, umbilical venous catheter.

(From: American Heart Association guidelines update for cardiopulmonary resuscitation and emergency cardiovascular care, *Circulation* 132(Suppl 2): S542–S560, 2015.)

Shock in Neonates

I. Definition: Shock is defined as acute circulatory dysfunction resulting in insufficient oxygen and nutrient delivery to the tissues relative to their metabolic demand. Cellular hypoxia results in a switch to anaerobic metabolism and accumulation of lactic acid. Increasing levels of lactic acid causes metabolic acidosis, which interferes with cell and organ function and, if not addressed, cell death.

Clinical Features:

- ✓ Tachycardia- one of the earliest manifestations of shock
- ✓ Tachypnea/ apnea
- ✓ Capillary refill time (CRT) is prolonged (> 3 sec).
- ✓ Cold extremities
- ✓ Blood pressure- Hypotension - late sign
- ✓ Oliguria
- ✓ ABG changes: Shows the following changes in the same sequence as shock progresses:
 - Respiratory alkalosis (as a compensation to overcome hypoxia)
 - Metabolic acidosis (due to anaerobic metabolism leading to lactic acidosis).
 - Mixed acidosis (late stage due to respiratory suppression).

Organ specific indicators of shock (From: St. Johns Medical College Hospital, Clinical Protocols in Neonatology; 1st edition, 2018, Bengaluru, India.)

Organ	Effect of shock
Brain	Irritability, lethargy, seizures
Heart	Tachycardia, thready pulse, cardiomegaly
Lungs	Shock lung (In shock, the development of edema, impaired perfusion, and reduction in alveolar space so that the alveoli collapse. A condition in which changed pulmonary compliance and oxygenating capacity causes an ARDS-like picture with defective aeration)
GIT	Mucosal ulceration
Kidney	Acute tubular necrosis
Hematological	DIC
Liver	Jaundice, elevated liver enzymes

The following parameters are used to assess circulation and cardiovascular status:

Cardiac output (CO) is the volume of blood ejected by the heart each minute. It is the product of heart rate (HR) and stroke volume (SV). SV is the volume of blood pumped out during each cardiac contraction and is dependent on preload, afterload, and ventricular contractility: $CO = HR \times SV$

Oxygen content in the arterial blood (CaO₂) depends on the hemoglobin concentration (Hgb), oxygen saturation (SaO₂), and, to a much lesser degree, the arterial oxygen partial pressure (PaO₂): $CaO_2 = 1.36 \times Hgb \times SaO_2 + (0.0031 \times PaO_2)$

Oxygen delivery (DO₂) is the product of CO and the CaO₂: $DO_2 = CO \times CaO_2$

Systemic vascular resistance (SVR) is the resistance that the left ventricle must overcome to pump blood through the systemic circulation. It is affected by the vessel diameter (constriction or dilation), vessel length, and blood viscosity. Changes in SVR will alter the afterload and thereby have an impact on SV.

Blood pressure (BP) is the pressure exerted against the inner walls of the blood vessels and depends on CO and SVR: $BP = CO \times SVR$.

II. Etiology of neonatal shock based on pathogenesis (From: UpToDate 2018.)

Classification	Disorder
Hypovolemic	
Hemorrhage	Fetomaternal hemorrhage Severe bleeding (eg, subgaleal hemorrhage, umbilical cord rupture, internal bleeding), Placental hemorrhage as in abruptio placentae or placenta previaDIC, Twin-twin transfusion,
Non-hemorrhage	Third spacing from acute intestinal injury (eg, volvulus, necrotizing enterocolitis, Sepsis) Gastrointestinal fluid loss Polyuria due to congenital diabetes insipidus
Distributive	
Sepsis	
Non-sepsis	Adrenal insufficiency Hydrops fetalis Neonatal toxic shock syndrome Neurologic injury
Cardiogenic	
Cardiomyopathic	Myocardial ischemia/hypoxemia Myocarditis, Congenital cardiomyopathy
Arrhythmogenic	Congenital complete heart block Tachyarrhythmia (eg, SVT, VT)
Structural	Congenital heart disease: ▪ Hypoplastic left heart syndrome ▪ Critical aortic stenosis ▪ Critical coarctation of the aorta ▪ Obstructed total anomalous pulmonary venous connection
Obstructive	
Pulmonary vascular	Severe pulmonary hypertension Pulmonary embolus
Mechanical	Tension pneumothorax Pericardial tamponade Constrictive pericarditis

III. DIAGNOSIS:

compensated shock: During compensated shock, clinical findings may be subtle and include increased systemic vascular resistance (SVR) presenting as decreased peripheral perfusion (cold, pale skin with delayed capillary refill), tachycardia to maintain cardiac output, weak peripheral pulses and narrow pulse pressure (raised diastolic BP), ileus (decreased splanchnic circulation), and oliguria (decreased renal perfusion).

Uncompensated shock: refers to the phase of shock when the patient develops hypotension, and its clinical presentation will reflect decreased perfusion to vital organs. Lack of perfusion to the brain may cause changes in consciousness and lethargy. Lack of coronary perfusion increases the risk of cardiac arrest.

IV. INVESTIGATIONS.

- ✓ CBC: Infection, Anemia
- ✓ Prolonged PT/INR and PTT tests can occur with consumptive coagulopathy
- ✓ Glucose: may be elevated or decreased during neonatal shock.
- ✓ Hyperkalemia is a result of release of intracellular potassium stores due to tissue injury and cell death, and decreased excretion with renal function impairment.
- ✓ Serum bilirubin levels and liver enzymes may be elevated due to hepatic injury and dysfunction.
- ✓ Serum creatinine and BUN may be elevated due to renal injury and dysfunction.
- ✓ Periodic serum lactate measurements, to see metabolic acidosis often indicates inadequate circulation.
- ✓ Echocardiogram to assess cardiovascular anatomy and function.
- ✓ Electrocardiogram (ECG)
- ✓ Arterial blood gas – The arterial blood gas measures the degree of acidosis and hypoxemia.
- ✓ Blood cultures

V. Treatments: Fluids, inotropes, vasopressors, and hydrocortisone replacement are used to treat shock in the neonate.

Therapeutic Endpoints:

- ✓ capillary refill time of < 2 seconds
- ✓ normal pulses without differential between peripheral and central pulses
- ✓ warm extremities
- ✓ urine output greater than 1ml/kg/hr, low serum lactate, and
- ✓ mixed venous saturation of >70%
- ✓ CVP at 5 to 8 mm Hg

ABC of life:

Maintain airway & breathing. Optimise perfusion. Start oxygen, if saturation (<90%); low threshold for invasive ventilation

Establish iv access (central line preferred if requiring inotropes)

A. Fluid therapy: Fluid resuscitation aims to improve preload, thereby increasing stroke volume (SV) and cardiac output (CO).

Intravascular volume is increased by using crystalloids (Normal saline / Ringer lactate) 10cc / kg given over 20 minutes in term; 1 hr in ELBW. Consider repeat fluid bolus of 10 ml/kg after 20 minutes in septic or hypovolemic shock; (Maximum 60 ml/kg in term) & can be repeated until CVP is normal.

However, aggressive fluid resuscitation may be harmful for neonates with cardiogenic shock or those who have compensated shock with certain comorbidities (eg, prematurity, acute renal failure, intrinsic respiratory diseases). Patients with obstructive shock may also require volume expanders to improve cardiac preload, but therapy should focus on emergent correction of the underlying cause (eg, needle or chest tube thoracostomy for tension pneumothorax or pericardiocentesis for cardiac tamponade).

In acute blood loss

- Whole blood 15- 20ml/kg less than 7 days old over one hour.
- If blood is not available, volume expanders like 0.9% N/S can be used till blood is prepared.

B. Medications

1. Inotropes are used for augmentation of Myocardial contractility.

- ✓ Dopamine (5 - 15 µg/kg/min) - first line of inotrope in septic shock. Maximum 20µg/kg/min. In septic shock choice of initial treatment- Upto 60 ml/kg of NS as 20 ml/kg aliquots if no signs of fluid overload followed by **dopamine** start at 5 mcg/kg/min and titrate.
- ✓ Dobutamine start at 3mcg/kg/min. First line of inotrope in cardiac dysfunction, perinatal asphyxia or unexplained shock in ELBW. When baby is on dopamine esp > 10 µg/kg/min, to consider adding dobutamine if blood pressure is normal but perfusion is poor. Dopamine and dobutamine can be increased upto a maximum dose of 20 mcg/kg/min. It is choice of initial treatment in cardiogenic shock- dobutamine start at 3 mcg/kg/min and titrate.
- ✓ Epinephrine (0.1-1.5 µg/kg/min) – if no adequate response to Dopamine and Dobutamine. Reduce and wean off dobutamine after starting epinephrine. Epinephrine is an effective adjunct therapy to dopamine because cardiac norepinephrine stores are readily depleted with prolonged and high-rate dopamine infusions.
- ✓ Titrate inotropes fast – remember half-life is only 2 minutes; While tapering- taper the one which is least likely to be effective; Avoid combination of 3 or more inotropes- difficult to titrate.
- ✓ Other inotropes- milrinone; It may be useful for some causes of cardiogenic shock,

2. Other agents:

Hydrocortisone replacement. Corticosteroids may be useful in infants with hypotension refractory to volume expansion and vasopressors, especially among premature infants and also is indicated in Congenital adrenal hyperplasia (Drug of choice).

Hydrocortisone stabilizes BP through multiple mechanisms. It induces the expression of the cardiovascular adrenergic receptors that are downregulated by prolonged use of sympathomimetic agents and also inhibits catecholamine metabolism.

The BP response is evident as early as 2 hours after hydrocortisone treatment.

Dose – Hydrocortisone 1 mg / kg every 12 hourly for 2-3 days. Hydrocortisone is indicated if requiring >1 inotrope in very preterm and >2 in >32 wk; 1–2 mg/kg every 8 hrs till normal cortisol report or resolving shock.

C. Supportive treatment.

- ✓ Correction of negative inotropic factors such as hypoxia, acidosis, hypoglycemia, and other metabolic derangements will improve cardiac output.
- ✓ Maintain normal blood glucose, electrolytes, calcium, pH, and Temperature.
- ✓ Supportive therapy for multi organ dysfunction: Monitor renal function, UOP
- ✓ Correction of a significant coagulopathy and anemia (PCV < 36%).
- ✓ NPO if > 1 inotrope or dopamine >10
- ✓ Treatment of underlying cause. Consider starting or changing antibiotics: Because neonates are at-risk for sepsis, intravenous (IV) empiric antibiotics are administered pending results from blood cultures. The combination of ampicillin and gentamicin or ampicillin and cefotaxime are potential regimens that provide empiric coverage until culture results are available.

VI. Management of Neonatal Dehydration

Mild, moderate, and severe degrees of dehydration in infants correspond roughly to a loss of 5% (50 mL/kg), 10% (100 mL/kg), and 15% (150 mL/kg) of body weight, respectively.

If signs of hypovolemia are present (eg, tachycardia, hypotension), a rapid infusion of fluid (“bolus”) is indicated.

By convention, dehydration often is expressed as percent of body weight, and bolus replacement is considered in milliliters per kilogram of body weight. Recognizing that 20 mL/kg is equivalent to 2% of body weight permits the clinician to set expectations for the impact of a bolus of fluid.

Parenteral Rehydration of a Moderately Dehydrated Infant:

An initial bolus of 10 mL/kg is provided to restore normal hemodynamics; adequacy of the bolus can be assessed by resolution of hypovolemia-induced tachycardia, and this bolus can be repeated based on the response.

After the bolus, there are two options.

1. A common approach is to provide 50% of the remainder of the deficit in the next 8 hours along with 8 hours maintenance which is one third of the 24 maintenance fluid; the remainder of the deficit is administered in

the subsequent 16 hours, along with the usual hourly maintenance requirement which is two third of the remaining 24-hour maintenance.

2. An easier alternative is to recognize that the bolus of 10 mL/kg, if given twice, represents 2% of body weight. If the initial deficit was estimated to be 10%, the remainder after the two boluses is 8%. This can be replaced conveniently over 8 hours at an hourly rate of 10 mL/kg (1% body weight per hour), a number that is easy to calculate mentally. The day's worth of maintenance fluid then is provided in 16 hours. Administering 24 hours worth in 16 hours requires a rate of 1.5 times the usual hourly maintenance rate ($24 / 16 = 1.5$).

The two Methods of Rehydration (e.g.: A 5-kg infant who has moderate dehydration (10%, 500 mL deficit) following a 10 mL/kg bolus given twice (100 mL). Remaining deficit is 400 mL; maintenance is 500 mL/d.)

	Usual methode: Combined Deficit/ Maintenance	Robert's method: Sequential Deficit/ Maintenance
First 8 hours after bolus	$\frac{1}{2}$ remaining deficit + $\frac{1}{3}$ daily maintenance = 367 mL	Remaining deficit: 400 mL
Next 16 hours	$\frac{1}{2}$ remaining deficit + $\frac{2}{3}$ daily maintenance = 533 mL	Daily maintenance: 500 mL
Total	900 mL	900 mL

(From: 1. Roberts KB., Fluid and Electrolytes: Parenteral Fluid Therapy, Pediatrics in Review Vol.22 No.11 November 2001.

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